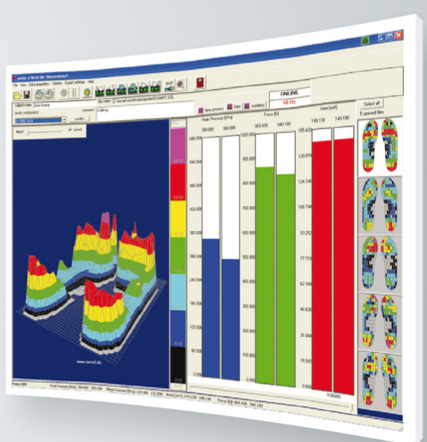
art science

pedar-x system

manual

v.20







novel.de

Welcome to novel pedar-x system

This manual provides instructions for the installation and operation of the pedar-x system. Also included are tips/guidelines for the proper use of the system to obtain the most accurate data results.

The pedar-x is an in-shoe dynamic pressure distribution measuring system which utilizes capacitive sensors. The system is designed for a variety of applications including medical, ergonomic and biomechanical testing scenarios.

The names of software and hardware components as well as proprietary names of companies are in general protected by laws of trademarks and brands or patent law.

All information in this manual was carefully compiled and written; however, errors may be present. novelgmbh cannot be held responsible for consequences of incorrect information in the manual.

If you find any errors when using the manual, please notify us at one of the following locations:

novelgmbh
Ismaninger Straße 51
81675 München
Germany

Tel.: 089 / 41 77 67 – 0
Fax: 089 / 41 77 67 99
email: novel@novel.de
http://www.novel.de

novelelectronics inc.
964 Grand Avenue
St. Paul, MN 55105
USA

651-221-0505
651-221-0404
novelinc@novelusa.com
http://www.novelusa.com

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1 PRECAUTIONS

- The pedar-x system should not be used in extremely high temperatures or high humidity. The recommended operating conditions are 0-40°C (32-100°F) and 20-80% humidity.

- The flexible insole sensors should be handled with care. Pressure ranges outside the designated loading should be avoided as well as use of sharp objects on the sensing device.

- Folding or wrinkling of the insole sensors is not advised and could damage the sensors.

- Please review the section on battery care. The batteries supplied with the pedar-x system must be charged/discharged in the manner instructed to maintain the longest possible lifetime.

- Any attempt to service the pedar-x system or components by unauthorized personnel will void the system warranty.

#####

2 SETUP OF THE pedar-x SYSTEM

2.1 Installation of the System Drivers

IMPORTANT

Administrator privileges are necessary to install drivers and software.

If novel has loaded the software on your computer, skip this section.

1. Insert the CD containing the novel software and drivers into the CD/DVD drive of your computer.
2. If the Auto-run-menu opens automatically:
   a. Install the following drivers and components:
      i. the hardlock driver,
      ii. the Fiber Optic Cable Drivers,
      iii. Firebird (required for novel database software)
      iv. Video filter (only if novel player with video is being used)
3. If the menu does not open automatically:
   a. Select the following setup files from the CD using Windows Explorer:
      i. HASPUserSetup.exe
      ii. nvl_ftdi_setup.exe
      iii. Firebird-1.5.x.exe
      iv. Avisynth_258.exe (only if novel player with video is being used)

2.2 Loading the novel Software

IMPORTANT

The pedar-x system comes with a Hardlock dongle that connects to the USB port of the computer. The Hardlock must be connected to use any of the novel software programs.

If novel has loaded the software on your computer, skip this section.

2.2.1 Loading Software that did not previously exist on the Computer

1. Insert the CD containing the novel software and drivers into the CD/DVD drive of your computer.
2. If the Auto-run-menu opens automatically:
   a. select the pedar-x software option and follow the setup wizard. The installation wizard will recommend to install the novel software on the computer's default system drive (typically "C:") in the subfolder "\novel\". It is highly recommended to follow this recommendation because it helps the novel staff to streamline your support. . The following sections refer to the C:\novel\ path.

      If the software is installed in a different location please replace the above mentioned location by the path which you have chosen.

3. If the Auto-run-menu does not open automatically:
   a. Select setup.exe (will be located in a folder called pedar-x with the version number) file from the CD using Windows Explorer.
   b. Double click on setup file to run the setup wizard.
4. The factory default calibration and configuration files will install automatically into the appropriate folders.

After installation of the software, a program group called pedar-x will be located in the **Start** menu under **Programs** and on the desktop in the novel shortcut folder.

2.2.2 Upgrading pre-existing Software

IMPORTANT

Always make sure the calibration and data files are backed up prior to uninstalling software.

Please note the following procedures are described for a PC with Windows XP (SP3). On Windows Vista or Windows 7 the locations for starting specific tasks or the displays on the screen may look slightly different.

1. First uninstall the old software.
   a. Go to **Start menu | Control Panel | Add/Remove Programs.**
   b. Select the pedar-x program list.
   c. Click Uninstall.
2. Install the new software following the directions in **Section 2.2.1**.
3. All files should still be located in the same directories as previously.
   a. Please note that after version 14 of pedar-x software the calibration and configuration file locations have changed:
      i. The old directory for the configuration files was C:\novel\NOVSYST\pedarx,

while the new directory is
C:\novel\NOVSYST\pedarx\mfg

ii. The old directory for the calibration files was
C:\novel\NOVSYST\pedarx\cab
while the new directory is
C:\novel\NOVSYST\pedarx\cab\pedar

b. It may be necessary to manually move these files to the appropriate directory.

4. Connect the novel pedar hardlock to the PC and test all programs to make sure they open and run and ensure that all previously collected data and configuration/calibration files are accessible.

If any problems occur, please contact your novel representative.

2.3 Calibration and Configuration Files

The factory default calibration and configuration files will automatically be copied during the installation of the software on the computer. It is also possible that a separate CD containing calibration and configuration files will be provided. If so, these files should be copied into the following directories on the computer:

1. Configuration Files – extension .mfg. All files should be copied to C:\novel\NOVSYST\pedarx\mfg.
2. Calibration Files – there are seven associated files with each calibration with the following extensions: *.cab, *.set, *.tbl, *.def, .*.geo, *.chl and *.zpt. All files should be copied to the directory C:\novel\NOVSYST\pedarx\cab\pedar.

For further information on calibration and configuration files please see **Sections 4.3 and 4.4**.

2.4 Connecting the pedar-x system

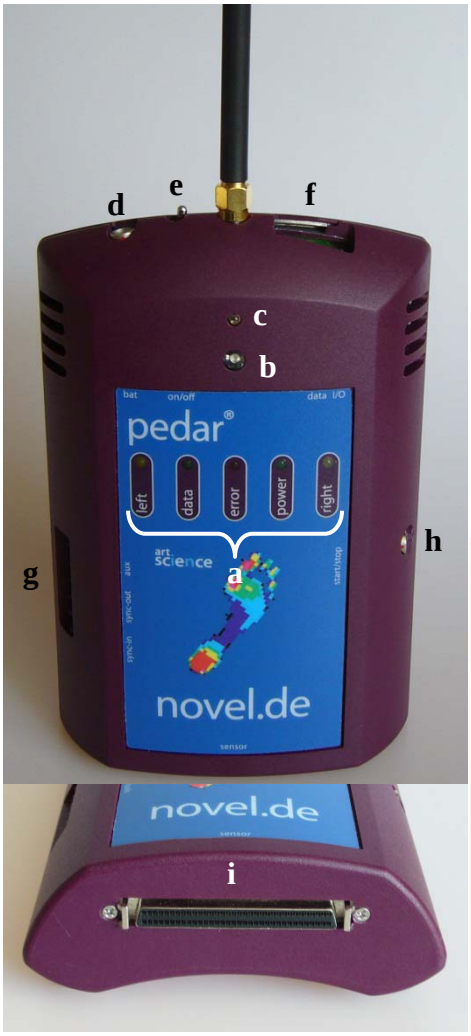
2.4.1 The pedar-x system components



The pedar-x system components include:

- Insoles (A)
- pedar-x double insole cable (B)
- pedar-x box (C)
- pedar-x fiber optic cable with fiber optic/USB adapter (D)
- USB cable (E)
- novel belt (F)
- pedar-x battery (G)
- pedar-x battery cable (H)
- pedar-x battery charger (I)
- start/stop trigger (J)
- Bluetooth dongle (K)
- Velcro straps (L)

2.4.2 The pedar-x Box Components and Connections





- Front Panel:
  five LEDs (a):

left	is used during the zero measurement of the left insole
data	indicates transmission of data between computer and pedar-x box
error	indicates an error during the measurement
power	indicates if the system is powered on
right	is used during the zero measurement of the right insole
top green “flash LED” (b)	During booting the pedar-x system: This LED flashes when the pedar-x box is turned on and when the internal boot process of the pedar x box is finished If the pedar x is in the measuring mode:

	It flashes when a measurement is started.
top blue BT LED (c)	if it is on, it indicates an active Bluetooth connection between the pedar-x and the PC

- Top Panel:

4pin lemo connector (d)	connects the pedar-x box with the battery
on/off-switch (e)	switch to turn the pedar-x box on or off
2 pin fiber optic cable Connector (f)	connects the pedar-x box via fiber optic cable with the PC

- Left side Panel (g):

3 pin fiber optic connector	connects the pedar-x box with optional synchronization unit

- Right side Panel (h):

3pin lemo connector	connects the pedar-x box with the start/ stop trigger

- Bottom Panel (i):

insole cable connector	connects the pedar-x box to the double insole cable

- Back Panel (j)

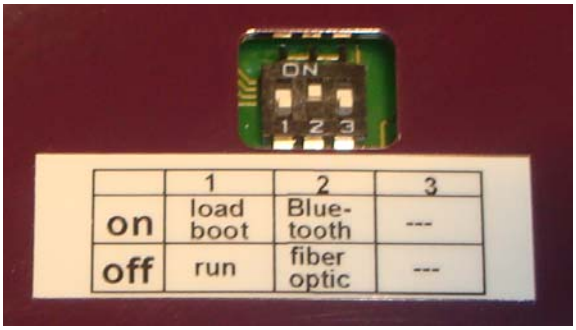
dip switches	Open the panel to locate the pedar serial number, and DIP switches used to select between the Bluetooth telemetry mode and fiber optic cable data transmission mode.

2.4.3 The pedar-x Data Collection Modes

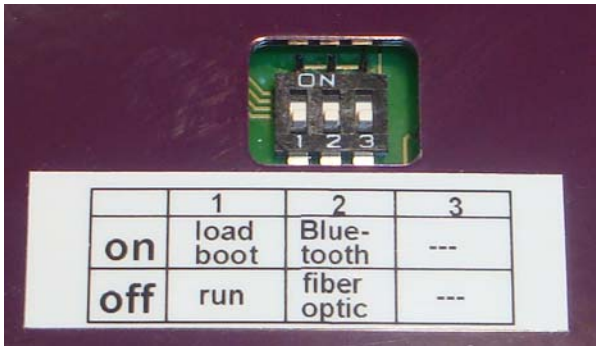
There are three modes of data collection with the pedar-x system:

1) **Online –** in this mode data is transmitted real-time and displayed to the computer. The computer controls the data collection of the device. There are two possible methods to connect the pedar-x box to the computer:

   a) Bluetooth telemetry - This setup utilizes Bluetooth to transfer data via wireless communication to the computer. When using this mode the three dip switches on the back side of the pedar-x box have to be set to 1=“OFF”, 2=”ON”, 3=“OFF”.



b) **Fiber Optic Cable** – With this method a fiber optic cable tethers the pedar –x box to the computer. When using this mode all the three dip switches on the back side of the pedar-x box must be set to the "OFF" position.



2) Online + Flash – in this mode data is transmitted real-time and displayed to the computer and is also stored simultaneously to the flash memory of the device. The computer still controls the pedar-x system within this mode. This mode of data collection can be helpful when the subject reaches the limits of the Bluetooth data transmission and some data might be not received from the PC. Please see **Section 8** for additional information on using this mode with an unreliable BT transmission.

3) **Offline measurement (Monitor Mode)** In this mode data transmission is controlled solely by the start/stop trigger switch connected to the pedar-x box and the data is stored to the internal flash memory of the pedar-x box.

If the computer receives data from the pedar-x box, the pedar-x online program will display the data using the Monitor mode. The computer does not control the data acquisition.

2.4.4 Connect the pedar-x system components

Please refer to labels within the diagrams in **Sections 2.4.1 and 2.4.2**.

1. Connect the pedar-x box to the novel belt as shown below. Please note that a version of the belt in which the box sits horizontally is also available. Please contact your novel representative to obtain a belt:



Thread the two lower horizontal straps through the plastic loops of the device, each from inside to outside.

Close the two upper horizontal straps. The device should now be secure and all LEDs should be visible.

Close the two lower horizontal straps, then move the device to the upmost position and secure the two lower vertical straps.

2. Connect the double insole (B) cable to the pedar box (C) via the insole connector (i). Ensure that the clips on the cable are fully attached to the box. Make sure to go straight in and straight out with this connection and do not torque the connector when attaching it or removing it from the box. It can be helpful to coil the pedar cable and attach it to the Velcro strap provided to the left of the box.
3. Insert the battery (G) into one of the sleeves on the pedar belt. Connect the battery via the battery cable (H) to the pedar box (C) via connector (d). Ensure that the red dots align on the cable and connector before pushing the cable in. Always ensure to pull on the release neck to release the cable without damage.
4. Connect the insoles (A) to the double insole cable (B). Connect the Right or Left insole to the Right or Left cable connector as shown on the connector. The insoles can only be connected one way to properly transmit data - i.e. Match the "R" (Right) on the insole plug to the "R" on the cable and the "L" (Left) on the insole plug to the "L" on the cable. Please note that if you plan to instrument a subject, you may skip this step until later. Please see **Section 2.4.7**.
5. Switch the pedar-x box on.

2.4.5 Fiber optic cable (FOC) Setup (valid for all pedar-x systems)

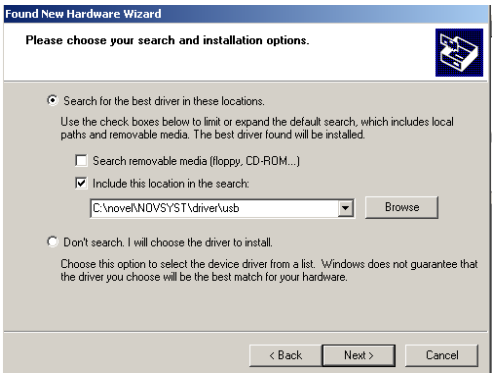
2.4.5.1 Installation of drivers (occurs with initial connection to computer)

Please refer to labels within the diagrams in **Sections 2.4.1 and 2.4.2**.

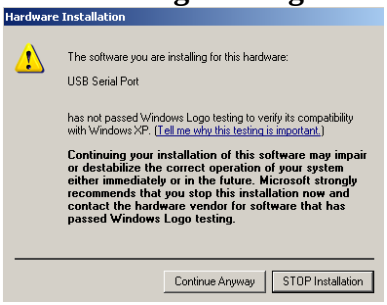
1. Connect the USB cable (E) to the fiber optic/USB adapter (D) and connect the USB cable to the computer.
2. The **Found New Hardware Wizard** will appear
   - NOTE: If the **Found New Hardware Wizard** does not appear and Windows automatically installs drivers, or the connection is not made within the pedar online software, please check that the proper drivers have installed following the instructions in **Section 13**.
3. Choose **Install the software specific location (advanced)**



4. Choose the checkbox **Include file location in the search:** and refer to the directory c:/novel/novsyst/driver/usb



5. Click **Next**
6. Confirm the message for the **Windows Logo testing** with **Continue anyway**.



7. The process will repeat itself for the second required driver.

2.4.5.2 Making the FOC connections

1. Connect the fiber optic cable (D) to the pedar-x box (C) via connector (f).
2. Connect the other side of the fiber optic cable to the fiber optic/USB adapter (D).
3. Ensure that the dip switches are set appropriately (see **Section 2.4.3 – 1a**)
4. Switch on the pedar-x device

The pedar-x system is now ready for online measurements and the pedar-x box should be switched on.

2.4.6 Bluetooth Setup (valid for pedar-x systems with Bluetooth)

2.4.6.1 Installation of Bluetooth-USB-Adapter under Windows XP Service Pack 2 or Service Pack 3 or Windows Vista

Please refer to labels within the diagrams in **Sections 2.4.1 and 2.4.2**.

> **IMPORTANT**
>
> Administrator rights may be necessary to setup the Bluetooth connection.

With Windows XP Service Pack 2 and later versions of Windows Microsoft has included its own Bluetooth stack. novel recommends using these drivers in place of those drivers which come with the Bluetooth dongles. Different Bluetooth dongles have been provided by novel, but each should work with the Windows BT drivers. It may be possible that your computer has an internal Bluetooth connection. This software may interfere with the Windows software. It may be necessary to disable the internal Bluetooth and uninstall the drivers provided with the system if the Windows drivers are not available. For ease of use and connection novel recommends at least Service Pack 2 or 3 be installed on all computers running the pedar system.

To determine if Windows XP Service Pack 2 is installed on your computer either go to the control panel and click on **System**, or right click on the **My Computer** icon on the desktop. The following will show the system information:



1) After connecting the Bluetooth dongle (K) to the computer Windows will display a blue and white “B”, as circled in red in the Windows task bar:

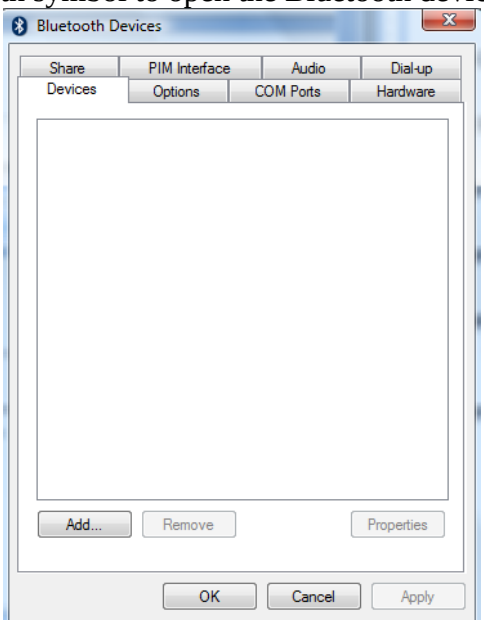


If an additional Bluetooth symbol is present in the task bar, it may be necessary to disable the internal Bluetooth of the computer and/or uninstall the additional Bluetooth software which is on the PC:



Please note that by default Windows will eventually hide the Bluetooth icon. If the icon is not found within the task bar, the Bluetooth device menu can be accessed from the same symbol in the Control Panel. novel recommends creating a shortcut from this item on the desktop for easy access to the Bluetooth devices.

2) Double click on the Bluetooth symbol to open the Bluetooth devices window

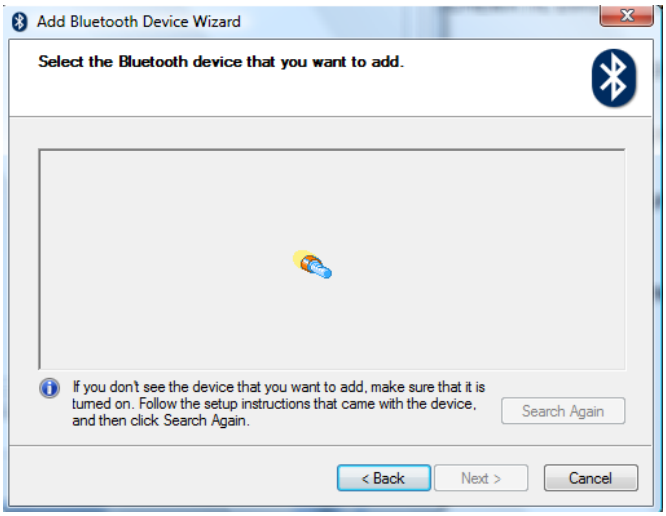


Click **Add** to add the pedar device.

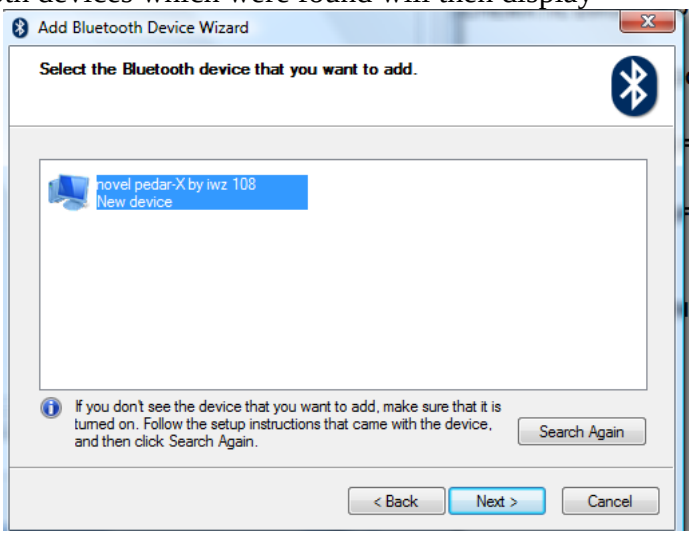
3) The **Add Bluetooth Device Wizard** will begin. Choose **My device is set up and ready to be found** and then click **Next**



4) The computer will begin to search for Bluetooth devices



A list of the Bluetooth devices which were found will then display

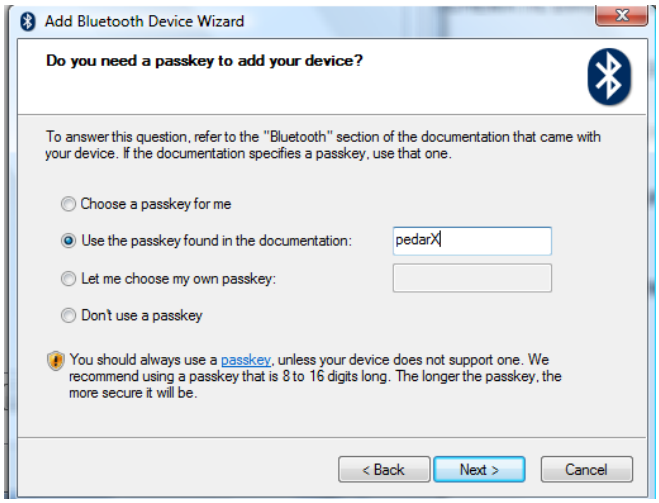


Choose the pedar-x device and click **Next.**

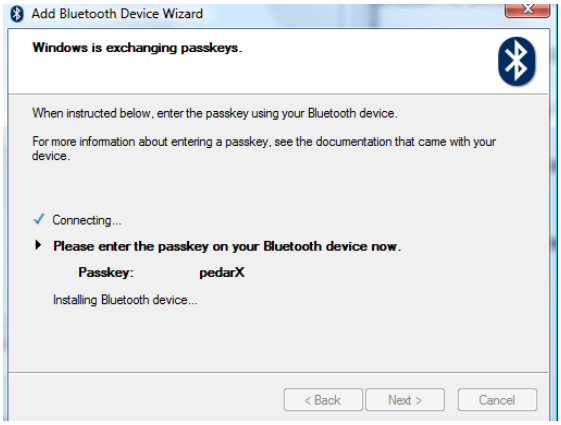
If the pedar –x box is not listed please choose **Search Again** to initiate the search once again. If the box is not found please ensure that the dip switches are set properly (See **Section 2.4.3 – 1b**). Please refer to **Section 8** for additional Bluetooth troubleshooting information.

Please note that if a large number of devices are found within the area, this can interfere with data transmission. If devices such as cell phones, computers, etc are present, and it is determined later that the Bluetooth is not transmitting all frames, it is recommended to turn off the Bluetooth transmission on these devices, if possible.

5) Choose the radial button **Use the passkey found in the documentation** and enter the phrase **pedarX** as the passkey. Click **Next**.

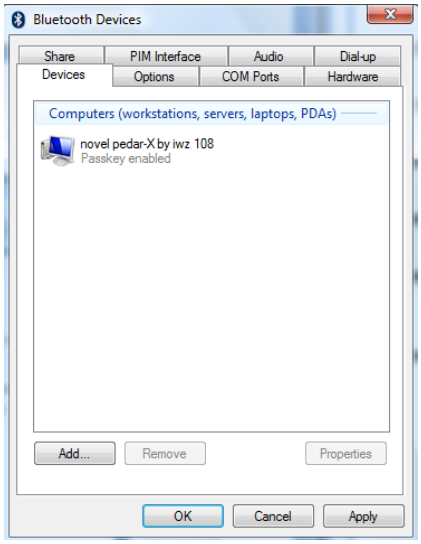


6) The Bluetooth connection will be established and the virtual COM ports will be created. Make note of the **Outgoing COM port**, as this will be needed within the pedar-x online software. Click **Finish**.

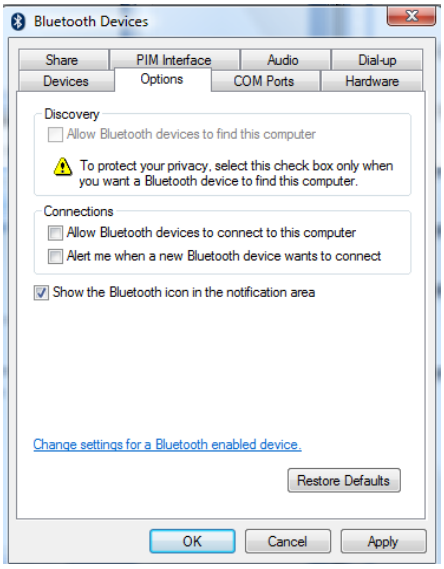




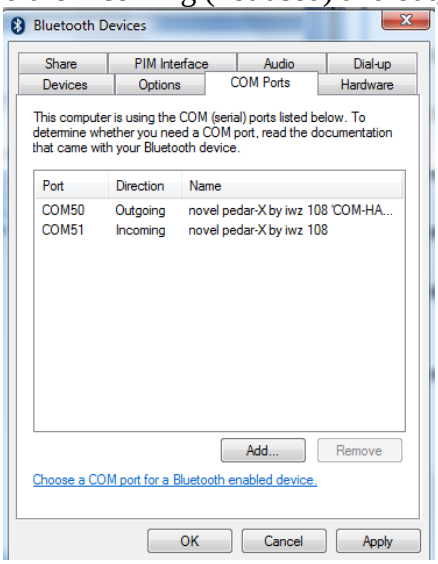
7) The pedar device will now be listed within the Bluetooth Devices window:



To ensure proper data transmission between the pedar and the computer the ability of other Bluetooth devices should be turned off. Click on the **Options** tab and uncheck the options **Allow Bluetooth devices to connect to the computer** and **Alert me when a new Bluetooth device wants to connect**.



8) If later it becomes necessary to determine which Bluetooth port is the Outgoing port you can choose the **COM Ports** tab and the incoming (not used) and outgoing ports will be listed:



9) Please refer to **Section 8** for additional information on the Bluetooth connection including troubleshooting a connection, manually creating COM ports and what to do if Bluetooth transmission is not reliable.

2.4.6.2 Installation of Bluetooth-USB-Adapter under Windows 7

Please refer to the previous section for additional information. Please refer to labels within the diagrams in **Sections 2.4.1 and 2.4.2**.

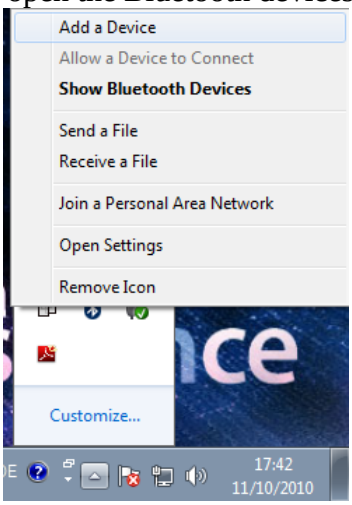
1) After connecting the Bluetooth dongle (K) to the computer Windows will display a blue and white “B”, as circled in red in the Windows task bar:



If an additional Bluetooth symbol is present in the task bar, it may be necessary to disable the internal Bluetooth of the computer and/or uninstall the additional Bluetooth software which is on the PC:

Please note that by default Windows will eventually hide the Bluetooth icon. If the icon is not found within the task bar, the Bluetooth device menu can be accessed from the same symbol in the Control Panel. novel recommends creating a shortcut from this item for easy access to the Bluetooth devices.

2) Click on the Bluetooth symbol to open the Bluetooth devices window



Click **Add a device** to add the pedar device.

3) The **Add a Device Wizard** will begin. The computer will begin to search for Bluetooth devices. A list of the Bluetooth devices which were found will then display

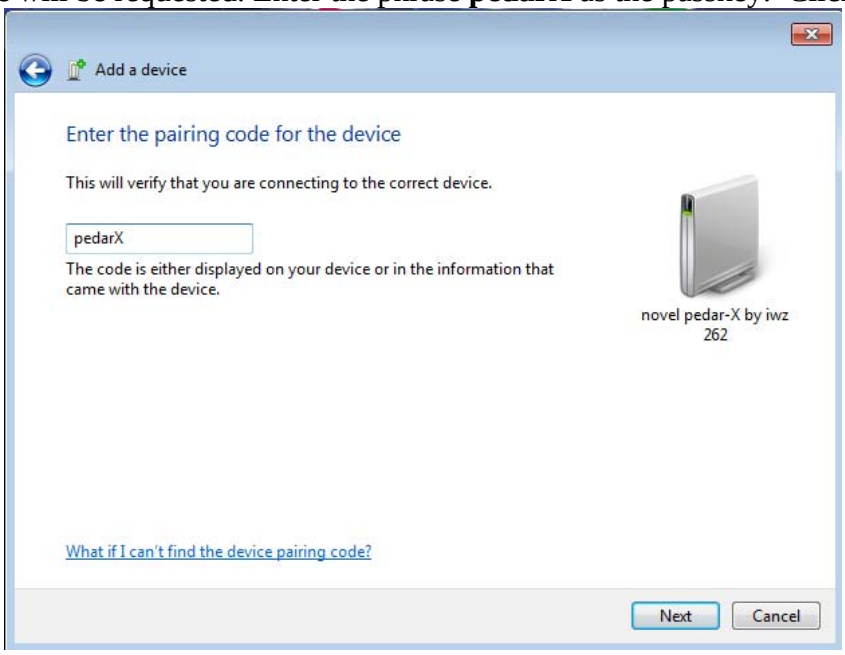


Choose the novel pedar-x by iwz device and click **Next.**

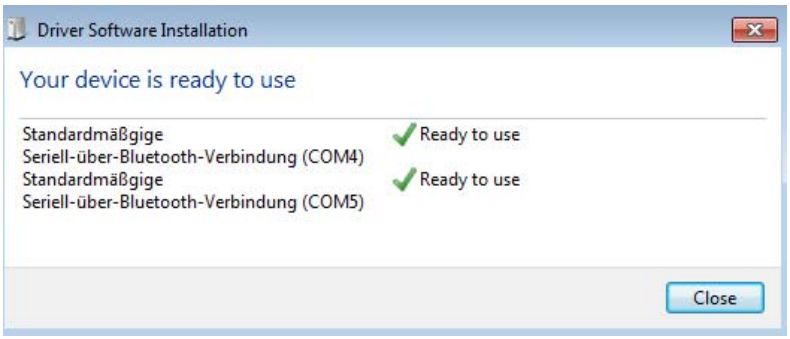
If the pedar –x box is not listed please choose **Search Again** to initiate the search once again. If the box is not found please ensure that the dip switches are set properly (See **Section 2.4.3 – 1b**). Please refer to **Section 8** for additional Bluetooth troubleshooting information.

Please note that if a large number of devices are found within the area, this can interfere with data transmission. If devices such as cell phones, computers, etc are present, and it is determined later that the Bluetooth is not transmitting all frames, it is recommended to turn off the Bluetooth transmission on these devices, if possible.

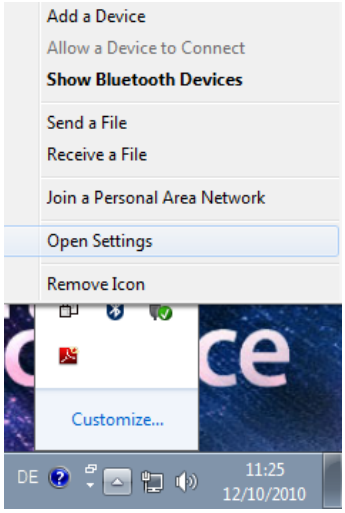
4) A pairing code will be requested. Enter the phrase **pedarX** as the passkey. Click **Next**.



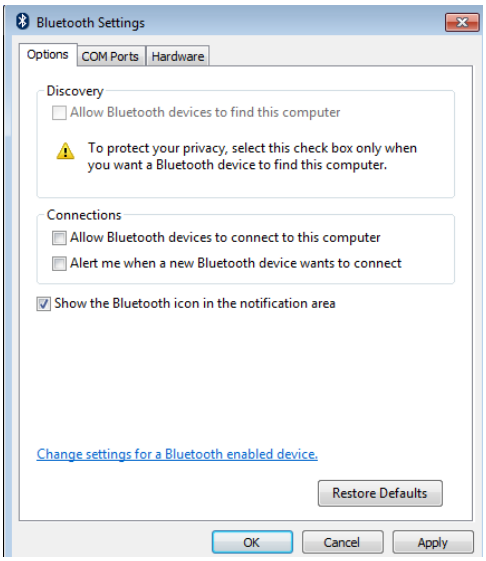
5) The Bluetooth connection will be established and the virtual COM ports will be created. Make note of the **Outgoing COM port**, as this will be needed within the pedar-x online software. Click **Close**.



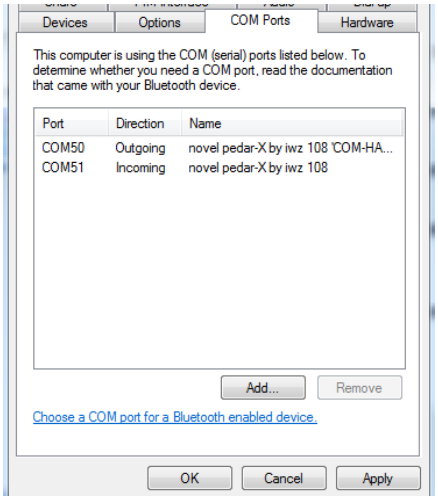
6) The pedar device will now be listed as a Bluetooth Device.
7) To ensure proper data transmission between the pedar and the computer the ability of other Bluetooth devices should be turned off. Choose Bluetooth menu again from the task bar and choose the **Open settings** option.



8) An options menu will appear. Please uncheck the option **Allow Bluetooth devices to connect to the computer** and **Alert me when a new Bluetooth device wants to connect.**



9) If later it becomes necessary to determine which Bluetooth port is the Outgoing port you can choose the **COM Ports** from the **Open Settings** menu item and the incoming (not used) and outgoing ports will be listed:



10) Please refer to **Section 8** for additional information on the Bluetooth connection including troubleshooting a connection, manually creating COM ports and what to do if Bluetooth transmission is not reliable.

2.4.6.3 Making the Bluetooth Connection

Once the Bluetooth connection has been setup, each time the pedar-x device is turned on the connection will automatically be made between the pedar device and the computer. Please note that that the Bluetooth connection is specific to the USB port on the computer, so ensure to connect the Bluetooth adapter to the same port with each data collection.

2.4.7 Preparing the Subject

1. Connect the pedar-x system as described in **Section 2.4.4**.
2. Attach the pedar-x belt to the subject's back and ask them to sit down on a chair. If the chair has a back, ensure that they do not lean back. A backless chair is preferred.
3. Detach the insole cable from the Velcro on the belt and place the Left cable on the left-hand-side of the subject and the Right cable on the right hand side.
4. Place the insoles within the subject's shoes.
    a. Ensure that the left insole is in the left shoe and right insole in the right shoe.
    b. Ensure that the insole fits within the shoe properly, i.e. that the insole is not too big or too small for the shoe. An insole which is too big can be damaged and an insole which is too small may underestimate pressures/forces.
5. Attach the insole to the insole connector. Ensure that the "L" and the "R" on the insoles and connectors match. NOTE: If the insole is connected backwards, you may only see half of the insole when data collection begins. If this occurs, please check the insole attachment.
6. Place the Velcro straps around the subject's limbs to keep the insole cable in place.
    a. Appropriate connection at the ankle is vital to insole longevity. You must ensure that the insole lead has enough slack against the ankle so that that the lead is not being pulled during plantar and dorsiflexion. A good strap fixation is shown on the left; a bad fixation is shown on the right.





    b. Attach the calf and thigh straps to keep the insole cable from moving around, or getting caught on objects.
    c. Attach any extra cable length to the Velcro straps on the belt.
    d. An instrumented subject is shown below:





7. Instruct the subject to remain sitting until the unload process has been completed within the software. See **Section 3.2.1.4 and 4.1.1**.

2.5 Battery care

2.5.1 General info

The batteries provided with the pedar-x system are NiMH batteries.



Usage times:

Standard NiMH battery – approximately 4 hours
Lightweight NiMH battery – approximately 10 minutes

These batteries have effect called “lazy battery effect”. Partial charging and discharging them

often will cause the capacity to decrease. The battery should be completely charged before use. It is recommended once every 3 months (or 50-60 charge cycles) to completely discharge the battery before charging it using the Ansmann charger provided. If this guideline is not followed, batteries may not charge to their full capacity on subsequent use. Thus, battery use time and the lifetime of the battery may be diminished.

If the batteries will not be used for a long length of time it is important to maintain the charge of the batteries. Deep discharge of the batteries can damage them. Batteries with model numbers 2300 – 2700 should be recharged at least once/month. Batteries with model numbers 2000 should be recharged at least twice/year. Please follow the instructions below.

If this process is used for each charge/discharge cycle the batteries will last for 3-5 years (1,000-1,500 charge cycles). Subsequently, the cells inside of the battery may need replaced.

2.5.2 Charging the battery

The following is the process of charging the batteries with the Ansmann charger:
If battery charge is low (as indicated by the LEDs on the battery), perform the following:

1. Connect the battery to the charger and the charge process will begin automatically.
2. Completely charge the battery. The charge LED will turn off and the green "ready" LED on the Ansmann charger will blink. It may be possible that 5 LEDs are not received on the battery; however the battery will be fully charged. If this process occurs regularly, and it is hard to know the actual status of the battery's charge. Please contact your novel representative for further information.

As described above, once every three months (or after 50-60 charge cycles) the battery should be completely discharged.

1. Completely discharge the battery using the Ansmann charger (even if the battery was completely discharged while running the pedar device). Press and hold down the yellow "discharge" button on the Ansmann charger until the orange discharge LED illuminates.
2. The charger will discharge the battery and then immediately begin to recharge the battery. NOTE: The green "ready" LED may blink during the discharge process. Ignore this.

If the battery does not last long enough to perform measurements after charging your battery to full (as indicated on the Ansmann charger), please contact your novel representative for further technical assistance.

Note: If the battery is low during usage with the pedar-x the first LED on the battery will blink.

#####

3 DATA COLLECTION WITH novel database AND pedar

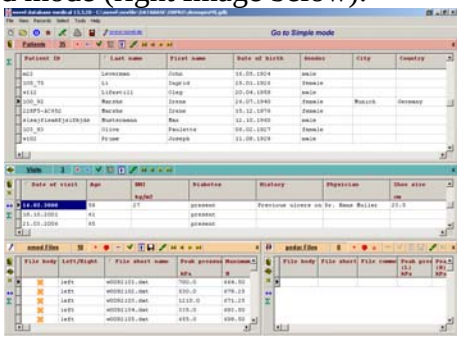
3.1 Database Software

The database novel database organizes patient information with pedar data files. It is possible to calculate biophysical parameters and to create user specified reports.

1. You can begin the novel database software by clicking on the 

icon located in the novel folder on the desktop. The software will begin in one of two viewing modes, the simple (left image below) or advanced mode (right Image below).





   a. To begin entering patient information in the simple mode, simply click on the 

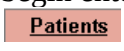
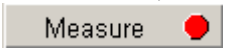
button.
   b. To begin entering patient information in the advanced mode, click the  icon in the 

table.
2. Enter the patient information (a patient ID is required) and then click the 

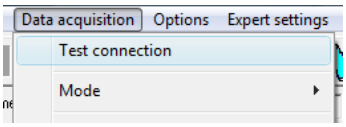
button.
3. Within the visit window the current date will be added. You may also add in any clinical data within this window. For a complete description of the information within this window please see the database medical manual.
4. Once the patient information has been entered, choose the  (pedar) tab.
5. To begin data collection click the 

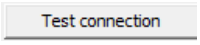
button and the pedar collection software will begin.

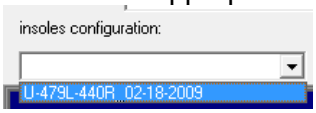
For a complete explanation of all database options, please refer to the novel database manual.

3.2 pedar online Data Collection Software

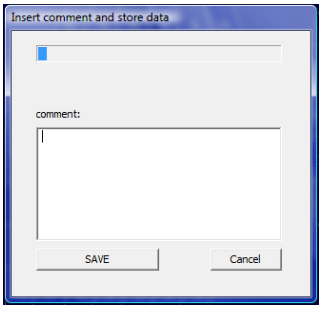
1. The pedar data collection software will open and it is first recommended to test the connection between the pedar box and the software.

2. Go to 

and choose the appropriate port from the drop down menu and click 

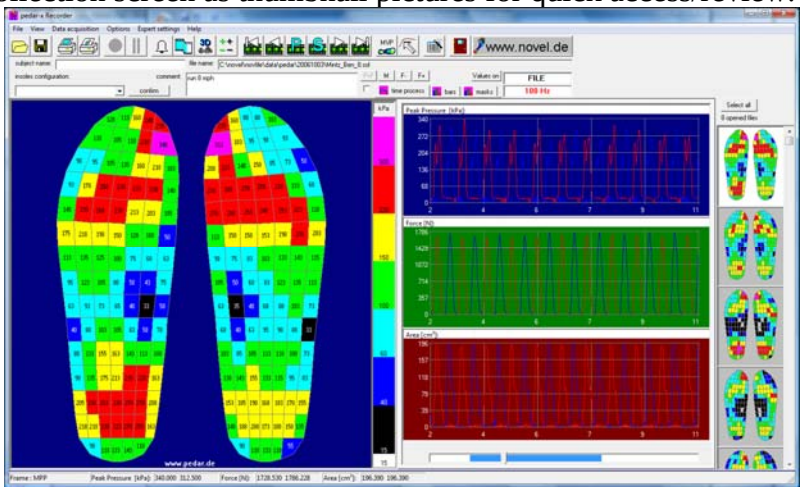
(NOTE: older version of software - enter the appropriate Bluetooth or Fiber Optic Cable port manually. See Section 2.4.6.2 to locate the Port number). You should receive a response in the "flash configuration" Window (a "connection OK" message will appear in older software versions).
3. Instrument the subject according to the defined protocol for your study while sitting in a chair. Please see Section 2.4.7 for details on instrumenting the subject.
4. Choose the appropriate insole configuration from the drop down menu 

and click the 

button.
5. Unload both the right and left insoles by having the participant raise the appropriate foot off the ground.
6. Instruct the patient to tie their shoes
7. Begin recording by clicking on the  button. Ensure to record the appropriate number of steps/repetitions for your activity and click the  button.
8. Enter a comment within the following window and click Save:


9. Record a second trial by clicking the  button again.
For more specific collection options, please refer to Section 4.

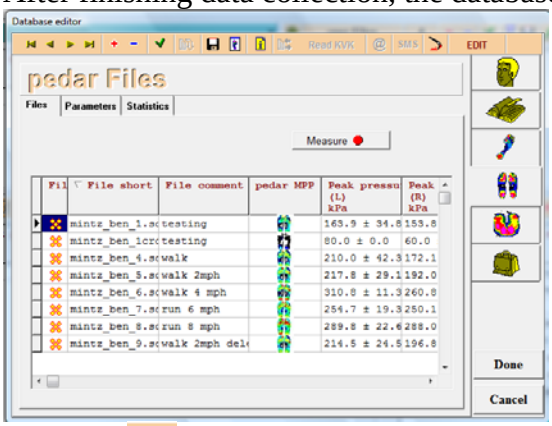
Once data collection has stopped all collected trials will be located in the right hand section of the collection screen as thumbnail pictures for quick access/review.



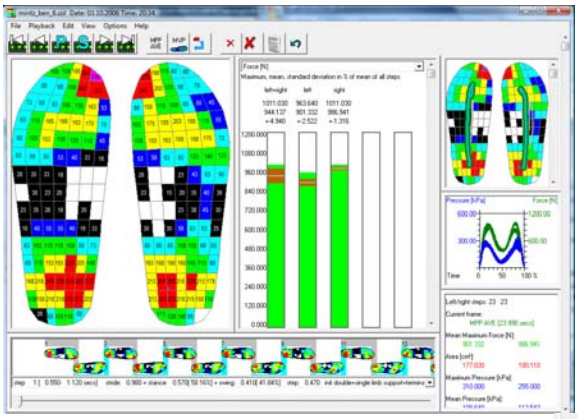
10. To play back an activity, choose the thumbnail on the right hand side of the screen and click the button.
    a. To move forward or backward one frame, use the and buttons, or move to the first or last frame with the and buttons
11. You can choose how to view the pressure picture by using the 2D, 3D, or (isobar) buttons.
    a. You can zoom in and look at the individual pressure values of each sensor by clicking on the pressure picture in the 2D mode. Right click to zoom back out.
12. The values of pressure, force, and area can be viewed in the graphs on the right hand side of the screen
    a. You can zoom in on these graphs by clicking on the button.
    b. You can track (move) the cursor along the graph by clicking the button.
    c. Alternatively you can view the values of pressure, force, and area in bar graphs by clicking the Bars tab.
13. Once finished collecting trials, exit the pedar software by clicking the or X button.

3.3 Post-processing the Steps within the step analysis program

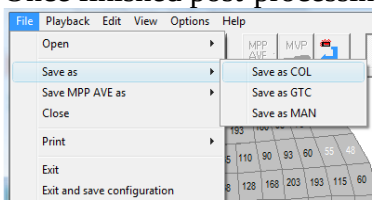
1. Sometimes unwanted steps, such as turn-around steps, are present within the trial. These may be post-processed and removed from the trial. If the trial only contains steps which should be analyzed, post-processing may not be necessary.
2. After finishing data collection, the database editor will display like so:



3. Click the button to select all the trials to post-process and then click the Run Application button and then the pedar button and then click the button to begin the step analysis program.
4. Review the steps by looking at the timing information, the force and pressure curves, the COP lines, etc, as in the following screen:



5. Delete the unwanted steps by selecting the  button and then clicking on the unwanted steps such as  and click the  button to accept changes. Typically at least 10 steps for each foot should be analyzed.
6. Once finished post-processing save the trial as a .col file by going to:



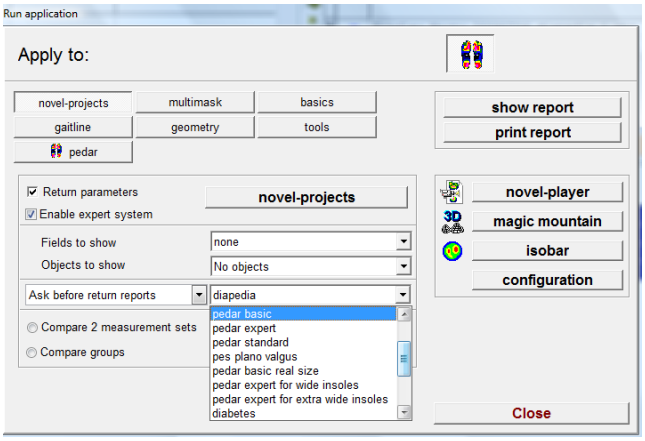
and save the trial under the appropriate patient and visit folder. It will be automatically returned to the database medical.

For more specific collection options, please refer to Section 4.

3.4 Creating Reports and analyzing data with novel projects

To create a pedar Report either the novel projects or pedar Report software must be installed. This software is available in pedar Basic, Standard, Expert level.

1. A report can be created once the .sol or .col file is created.
2. Choose the appropriate file in which you would like to create a report and then click the Run Application  button and you may choose a predefined report from the dropdown menu.



3. You may then choose to either **show report** or **print report**.

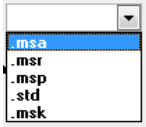
NOTE: To use the novel projects program you must have this optional software installed. For example, this program is included in the multiprojects –ip software package.

i) If you would like to process the data through novel projects and return this processed data to the database, or to create a new report, click the **novel-projects** button to open the novel projects software program.

(1) Select the parameters of interest by clicking the **Parameters** button. Select the parameters from the list of available parameters and use the button to add the parameters to the Selected Parameters list. Once finished, click "OK".

(2) Apply the appropriate mask (if desired) by either:

(a) Applying one mask to all files by choosing the **Many files - one mask** tab and then opening then selecting the mask by clicking the **Select mask** button. Ensure to include the option to use the Mask Editor by selecting **Masks Editor**.

(b) Creating a new type of mask for each file by choosing the **Many files - many masks** tab and then choosing the type of mask to apply from the drop down menu 

Ensure to include the option to use the Mask Editor by selecting the **Masks Editor** button.

(3) If you would like to create a report for these parameters, click the **Edit report** button and select pictures, tables and graphs to place on the report. For a complete explanation on creating reports, please refer to the novel scientific manual.

(4) To calculate the parameters and return the parameters back to the novel database, click the Calculate button. Review or change the mask within the Masks Editor window and click the Continue button.

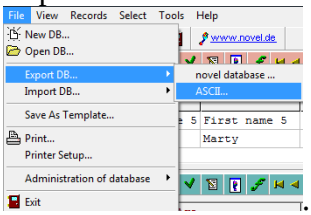
(5) To create the report, click either Show report or Print report to calculate the parameters, return them to the database and create the report. Review or change the mask within the Masks Editor window and click the Continue button.

(6) Once processing is complete, close novel projects by clicking "Close". Save the project, if required.

For a complete explanation on the novel projects program, please refer to the novel projects manual.

3.5 Exporting data from the novel database medical

The pedar software most often includes the novel database light. The following chapter describes features of the more advanced database, novel database medical. This database is available as a standalone program or is also included in the multiprojects –ip software package.

1. Once data analysis is complete all parameters can be exported from the database to an ASCII format, which can be opened in Microsoft Excel (or the like).
   a. It is important to view what parameters you would like to export within the novel database medical. Select the parameters of interest within each table (Patients, Visits, pedar Files, etc) by using the  button in each table.
   b. You may save this exported list by using the  button and choosing Save list to DB.
2. Export the database using the menu item:


. Choose a location to save the database to and then click Export.
3. Files will be saved for each table: .avs – contains information from the Patients and Visits Table, .adt – form the emed table, .apd – from the pedar table, .apl – from the pliance table, .aur – from the user table. Open these in Excel (or the like) to view them in table delimited format.

For a complete explanation of all database options, please refer to the database medical manual.

######

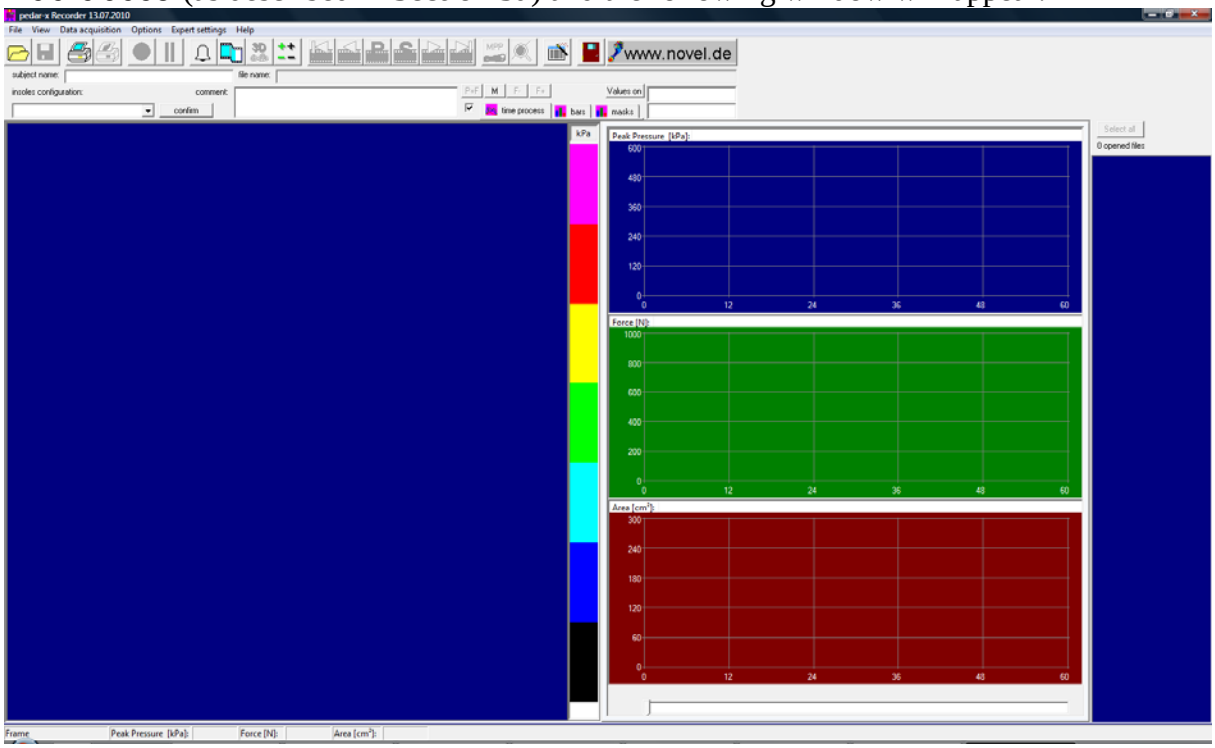
4 THE PEDAR-X ONLINE PROGRAM

The pedar online program collects and displays the data captured with the pedar-x measuring insoles. The program also provides the ability to calibrate the insole sensors (purchase of the trublu calibration device is required for this option). Many different collection and display options are available within the program and will be reviewed in the following sections.

4.1 Data Acquisition

4.1.1 Collecting data

1. Opening the pedar –x online program either from the desktop shortcut or from the novel database (as described in **Section 3a**) and the following window will appear.



2. Test the connection with the pedar-x system by going to **Data Acquisition | Test Connection**. Please see **Section 4.2.3 –a** for additional information.
3. Select insole configuration file from the **insoles configuration** pull down menu.
4. Instrument the subject, as described in **Section 2.4.7**. The subject should be seated with their shoes untied.
5. Click the **confirm** button. This process tells the pedar box which insole configuration you have chosen.

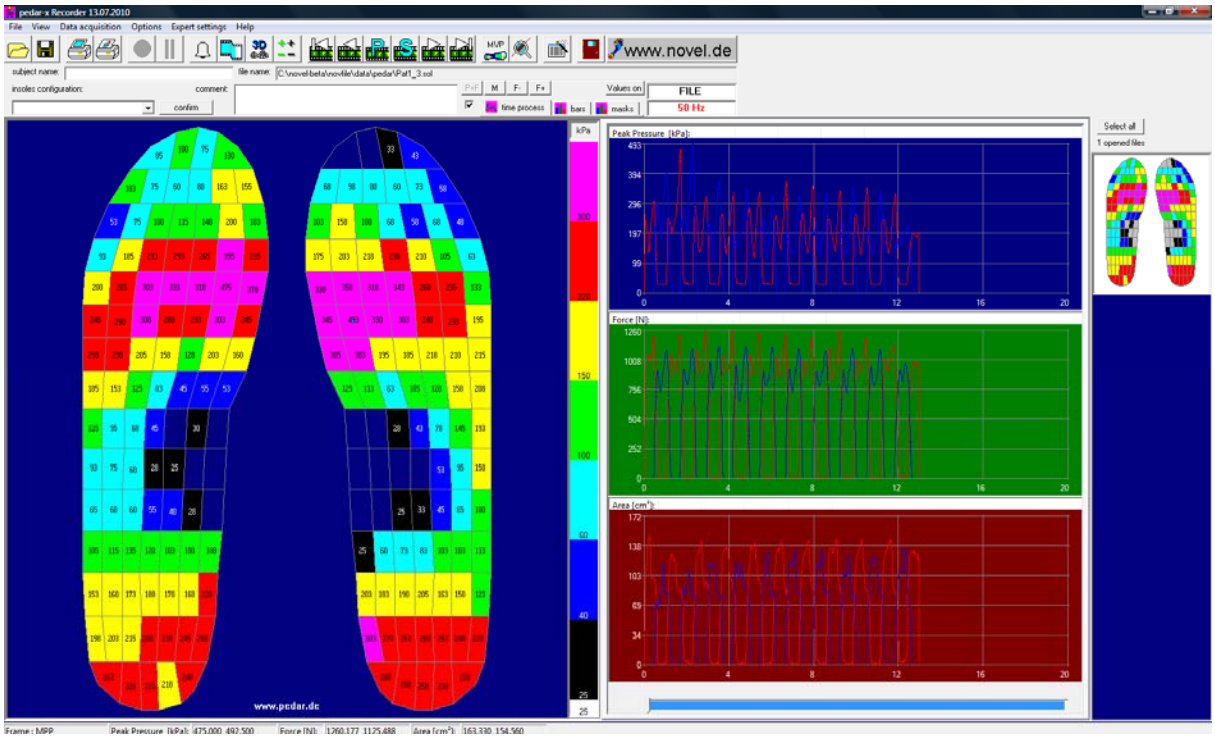
6. A message will appear to unload the left insole. Unload the left insole (i.e. have the subject lift their left foot slightly off the ground. Again, the subject should still be seated to avoid any balance related injuries from standing on one foot) as prompted and then click **OK**. A message will then appear to unload the right insole; repeat the process of unloading.

   NOTE: The zeroing process is done with no load on the insoles. This includes the preload that may occur from (over) tight shoes. Most typical shoes will not provide a pre-load. If a pre-load does exist which you would like to ignore, a trial should be captured with this preload and used in analysis later.

   The insoles should be re-zeroed each time the shoes are removed (i.e. between shoe conditions). This can be done by going to Data Acquisition | Zero Settings | with zero. To use the last zero measurement that was performed (i.e. the box was turned off), go to the menu Data acquisition | Zero settings | previous zero. Please see **Section 4.2.3-d** for additional zeroing information.

7. To begin recording, click the record icon. 
8. To stop the measurement, click the stop and save icon. 
9. By default the automatic save option is enabled within the pedar online software. The measurement is therefore saved as soon as the collection is stopped. By default the data is saved to the c:\novel\novfile\data\pedar\ directory. If this option is not selected, click the save icon  and follow normal Microsoft Windows saving procedures. Once saved, a MPP picture of the data file will be shown on the right side of the screen as seen in the following picture.

   NOTE: If the data collection was begun through one of the novel databases, the subject's name will automatically be assigned to the file name and will display under the **Subject Name** field within the software. The measurement will save by default to the c:\novel\novfile\data\pedar\Lastname_Firstname\DateofVisit\ directory as Lastname_Firstname_#.sol, where # is the incremental measurement number. If the database is not being used a subject name may be manually entered into the **Subject Name** field.

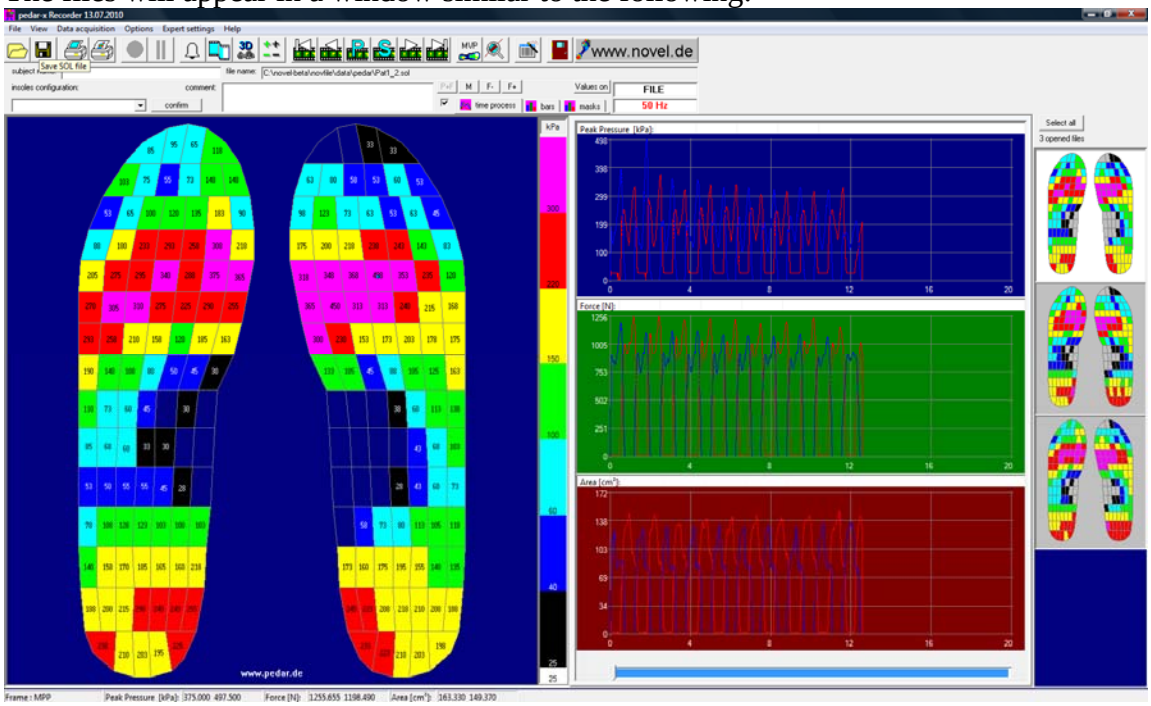


10. Also by default, the option to enter a Comment for the trial will appear. This step is very useful to make note of what occurred within the trial. This can be helpful in post-processing later.
11. To continue collecting data, follow steps 7-10.
12. It is possible to view the data files that were collected during this session simply by clicking on the MPP thumbnail located on the right of the screen. The selected file will appear in the pressure picture window.
13. To close the program, click the exit icon. 

4.1.2 Viewing data

1. Open the pedar –x online program and select the open file icon.  A standard Microsoft Windows open file box will appear. Select the file(s) to open and click **open**.
   Alternatively
   Select the data you would like to view from the novel database and open the pedar-x program from the database (refer to the novel database manual for specific instructions)

2. The files will appear in a window similar to the following:



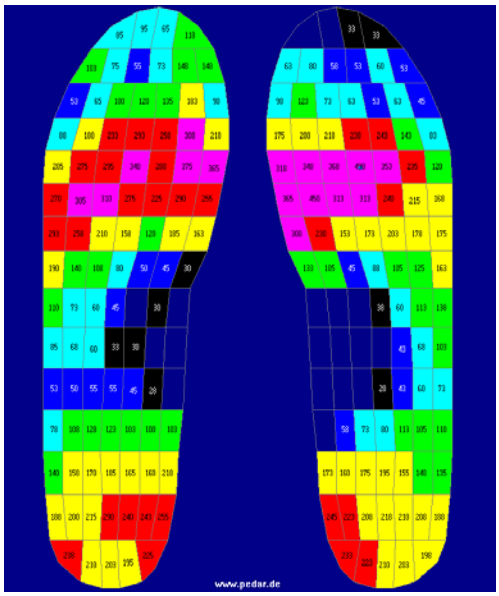
3. Use the toolbar and menu options, described in **Section 3.2.1.9**, to view the files.
4. Close the program. 

4.1.3 Window descriptions

All windows are synchronized to describe the same point in time.

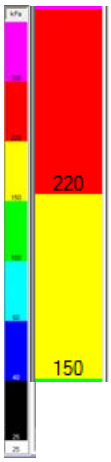
Right foot strikes are represented in blue, the left in red.

1. **Pressure Picture** – displays the pressure distribution for each foot. Located on the left portion of the screen.

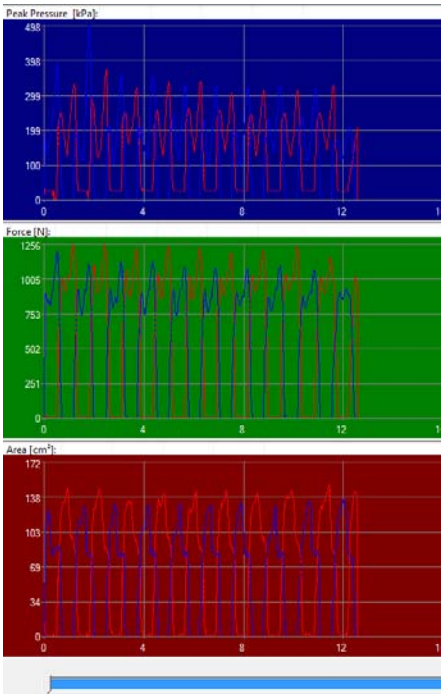


a. Pressure values/colors for each sensor of the insole are represented here in the 2D mode.
b. The left foot is displayed on the left of the screen, the right foot on the right.

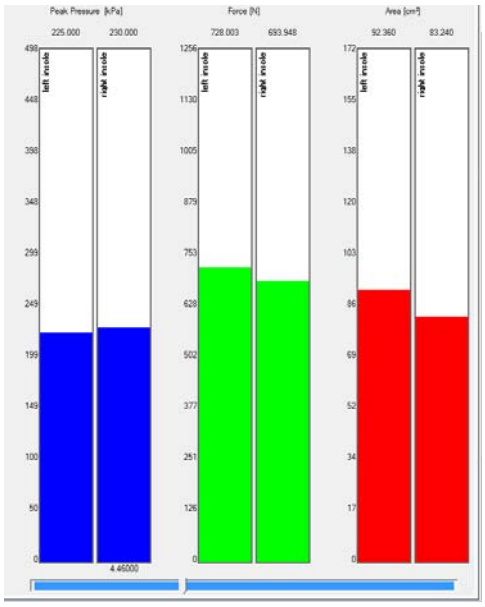
2. **Color Code** – displays the color code relating the pressure values to a color. The units of pressure and the cutoff are also displayed here. The value given for each color is the lowest value for the color level. For example, all pressure from 150 up until 220 kPa will display yellow.



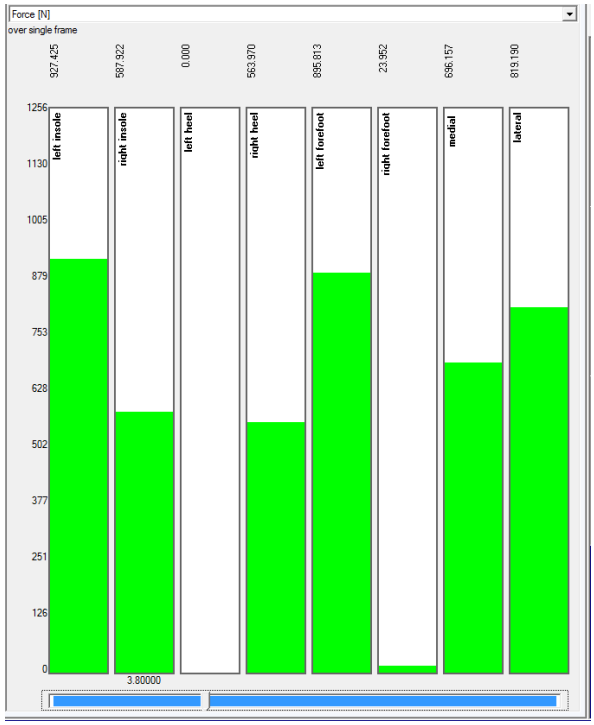
3. **Time process tab** - displays force, pressure and area versus time curves for each foot. The values at the time frame are displayed at the top of each graph.



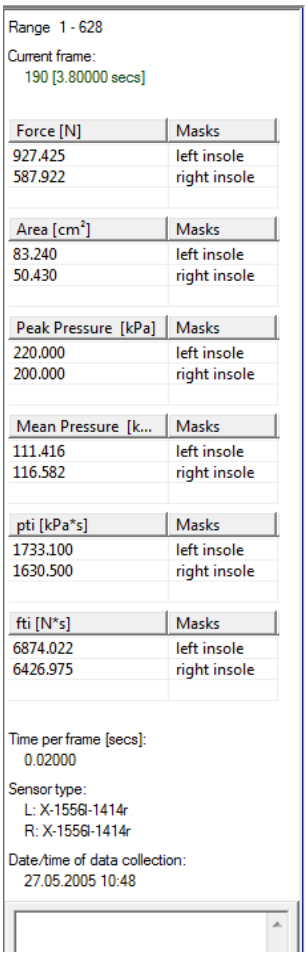
4. **Bars tab** – displays force, pressure and area values for each frame, left and right foot respectively in a bar graph format.



5. **Masks** - When a mask is applied the values for each mask are displayed in this window. Parameters can be calculated and displayed for each mask. For further information on creating these masks, please see **Section 4.2.3-f**. The parameters are Force (N), Peak pressure (kPa), Mean pressure (kPa), Contact Area (cm^2), Pressure-time integral (kPa*s), Force-time integral (N*s) and can be selected from the pull down menu at the top of the window.



6. **Values On** – displays information relative to each frame of data.
    a. **Range:** Shows the number of data frames which are being viewed in the time process window.
    b. **Current frame**: Displays the current frame number and the corresponding time.



c. **Force (N):** Displays total force for the left and right insole.
d. **Area (cm^2):** Displays the total contact area for the left and right insole.
e. **Peak Pressure (kPa):** Shows the maximum pressure that occurred in one sensor element for the displayed frame for the left and right insole.
f. **Mean Pressure (kPa):** Shows the average pressure for the displayed frame for the left and right insole.
g. **PTI (kPa*s):** Shows the pressure-time integral value for the selected time period for the left and right insole.
h. **FTI (N*s):** Shows the force-time integral value for the selected time period for the left and right insole.
i. **Time per frame (sec):** Shows the time duration of each frame. This value is dependent on the sampling rate.
j. **Insole type:** Shows the calibration file used to collect the data file.
k. **Date/time of data collection:** Shows the date (in DD.MM.YYYY format) and time of data collection.
l. **Comment Box:** Will display any comments which were saved with the measurement.

7. Selected Trial Menu

a. Close: Select to close the selected trial

b. Erase from database: Select to exclude the selected trial from the novel database. This may be necessary if a trial should be repeated and should not be included in analysis later. Please note that the trial will still exist on the computer hard drive. Only the copy within the novel database will be removed.

c. Open file in separate window: Opens the selected file in another window of the pedar online software.

d. Add mask difference: Applies a mask to the selected trial and subtracts the difference between the left and right insole.

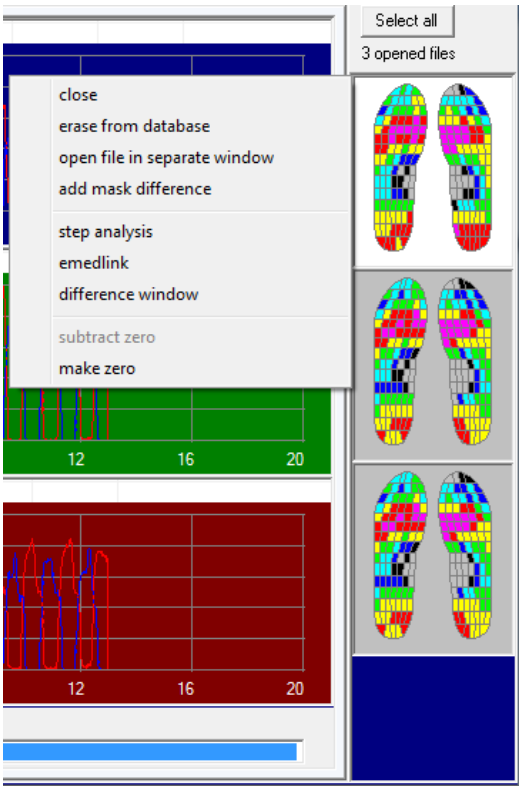
e. Step analysis: Opens the step analysis program. Please see **Section 5.**

f. Emedlink: Opens the emedlink program. Please see **Section 6.**

g. Difference Window: Opens the difference program. Two files can be subtracted from each other. Please see the novel tools manual for additional information.

h. Make Zero: Select a file to make as a zero, or offset for the data. All additional trials can be subtracted from this trial. This can be helpful when an offset is present and would like to be ignored.

i. Subtract Zero: Select to subtract from a file which was made zero. Each sensor element's reading in the zero trial will be subtracted from the selected trial.


Select all
3 opened files
close
erase from database
open file in separate window
add mask difference
step analysis
emedlink
difference window
subtract zero
make zero
12
16
20
12
16
20

4.2 Menu and Toolbar Options

4.2.1 File Menu

a.  Open

Use to open data files. This will display a typical Windows open screen. If using the novel database, data files may be opened directly into the pedar online software.

b. Open Desktop Files

Opens desktop (.dsk) files. Desktop files are files which are created when digital video is captured within the pedar online software. This file contains information on the synchronization of the video (.avi) and pedar (.sol) file.

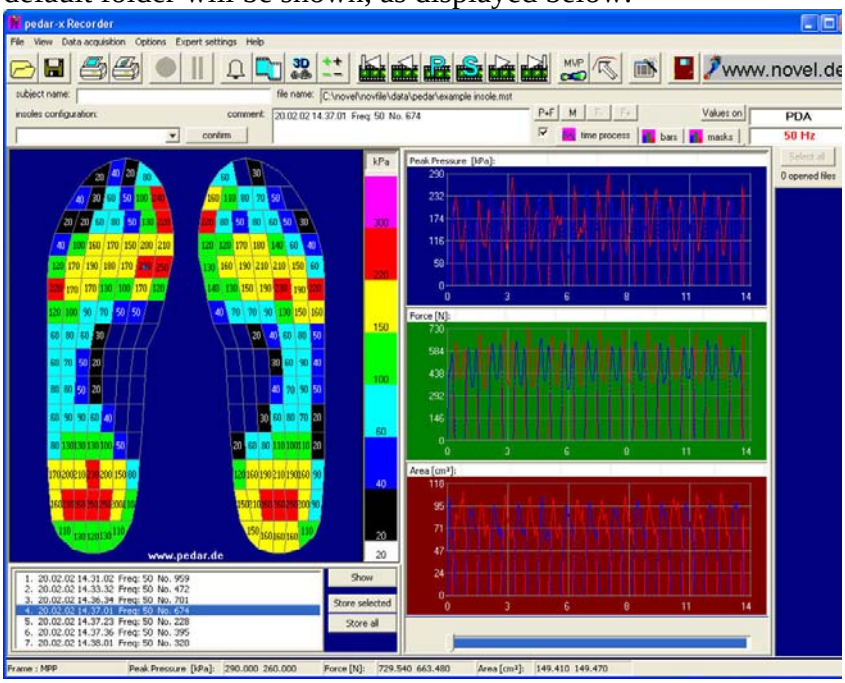
c. Open FAR Files

Opens an archived flash file. The FAR file may contain a set of .sol files which were downloaded from the pedar's flash memory.

d. Open PDA File

The pedar-x box can send data directly to a Windows mobile PDA device. For further information on supported devices and software for these devices, please contact your novel representative.

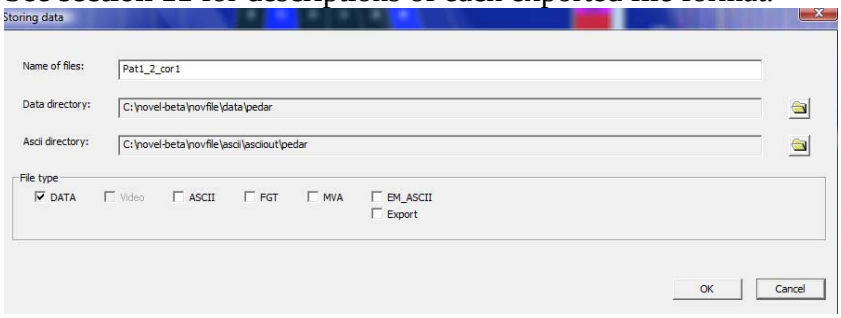
This menu item opens data collected by the PDA device. The files collected will have the extension .mst and may be transferred to the computer either manual or automatically using Windows Sync Center. The user may set the default folder for import of these files. Once the menu item is selected, a listing of the trials collected on the PDA and stored in the default folder will be shown, as displayed below:



The user may choose to **show** the file within the pedar software,

store all files located in the directory, or **store selected** files only.

e.  Save — Use to save data files. Can save in the following formats: sol, asc, fgt, mva, em_ascii. More than one file format can be selected and saved at one time. Also, the default folder may be changed. Please note that an original .sol file may not be overwritten. A new name must be assigned for this file.
See **section 11** for descriptions of each exported file format.



f. Close — Use to close single data file.

g. Close All — Use to close all data files.

h.  Print — Use to print data file in real size. Just the left or right can be selected at the same time and can be printed in black and white or color.

i.  Print screen — Use to print the entire computer screen.

j.  Exit — Use to close the program.

k. Exit with storing changes — Close the program and also save any changes to the display options. Please see **Section 4.2.4** for additional information.

4.2.2 View Menu

a.  First frame — Move to first frame of data file.

b.  Previous frame — Move one frame back in data file.

c.  Next frame — Move one frame forward in data file.

d.  Last frame — Move to last frame in data file.

e. 

MPP/MVP — Toggle between the Maximum Pressure Picture (MPP) and Mean Value Picture (MVP) of the file.

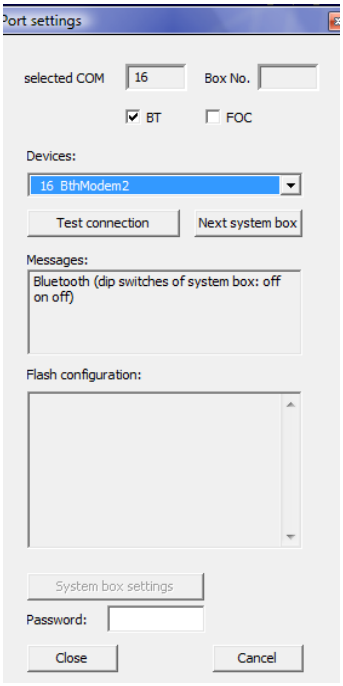
f.  Play — Play back the data file.

g.  Stop — Stop playing the data file.

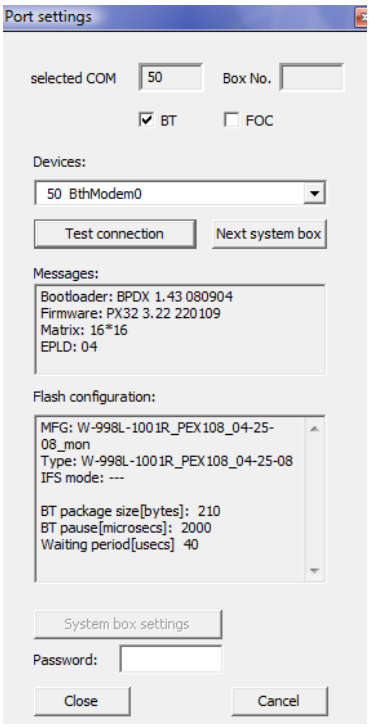
h.  Zoom — Toggle button to switch between:
1. being able to select a frame by clicking on a time curve and
2. being able to zoom on the time process graphs

4.2.3 Data acquisition Menu

a. Test Connection — Select this option to test the connection between the computer and the pedar-x box. The connection should be tested prior to data collection. This window will automatically appear if the connection is lost, or the box was turned off. A window similar to the following will appear:



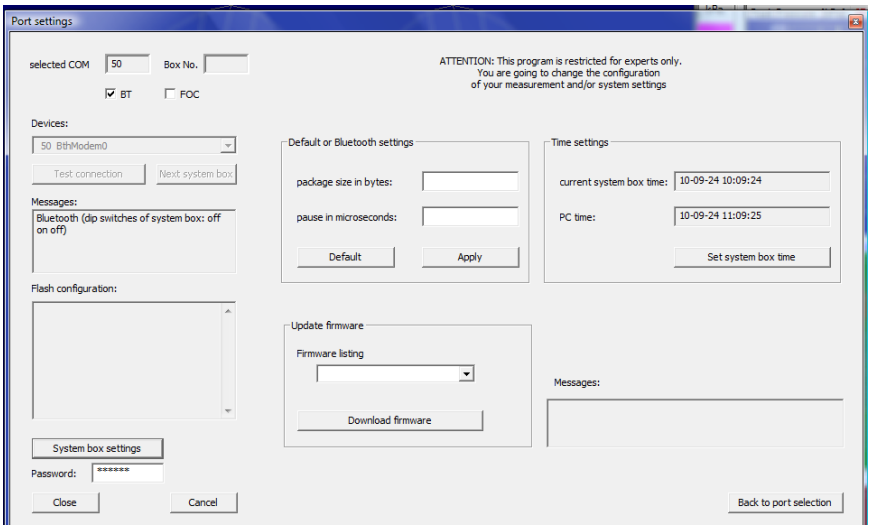
1. Select how the pedar-x is connected to the computer: select BT if using Bluetooth or FOC if using the Fiber Optic Cable. A message will appear at the bottom specifying how the three DIP switches on the pedar-x box should be set. Please refer to **Sections 2.5.4 and 2.5.5** for information on making the initial FOC and BT connection.
2. Choose the appropriate port from the **Devices** menu. With a BT Connection, this port will be the Outgoing COM port listed when the port was created and will contain the statement “BthModem”. For a FOC Connection, the statement “VCP”.
3. Select the test connection button. A message will appear indicating that the connection was found or if a problem exists.



A response in which the information shown above is listed indicates that the connection has been made. This information may be helpful when discussing or troubleshooting with your novel representative.

A message such as **serial port locked**, or **box may be in menu mode** indicates that the virtual COM port (either BT or FOC) is not set up properly. Please refer to **Sections 2.5.4 and 2.5.5** to troubleshoot this connection. If using the FOC, please also refer to **section 13** to ensure that the proper drivers are installed.

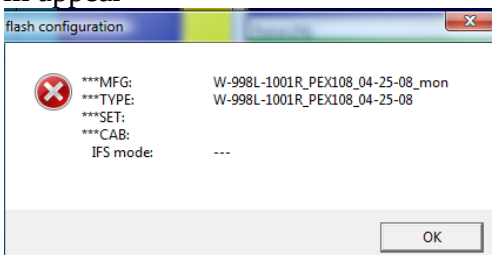
4. Once the connection has been established click **Close.**
5. Additional system box settings may be set with the help of your novel representative. After a connection has been made, enter the password **system** into the dialog box and click **System box settings**. A window similar to the following will appear:



The only option that can be altered without the help of your novel representative are the Time Settings. This allows the user to set the pedar box time to match the system box time. This can be useful when making offline measurements. All other options should not be changed unless instructed to do so.

b. Offline/ Online Measure-ment Settings

This option will display the measurement configuration settings within the flash memory of the pedar device. A window similar to the following will appear



c. Mode

Different modes of operation can be selected depending on the task to be performed. See **Section 2.4.3** for details on these data collection modes:
Online
Online + flash
Monitor + Test Connection
Monitor

d. Zero settings

This menu item allows the user to choose different options for zeroing the system. Please refer to **Section 4.1.1 #6** for information on zeroing philosophy:
With: select to make a zero measurement before collecting data. This option is useful if a footwear change has been made. The system should be re-zeroed
Without: select to collect data without using a zero measurement.
Previous: select to collect data using the last zero measurement made. This option is typically selected if the box was turned off and back on. The user must re-confirm the connection and the software will automatically prompt for the unload to occur. Click **Cancel** and

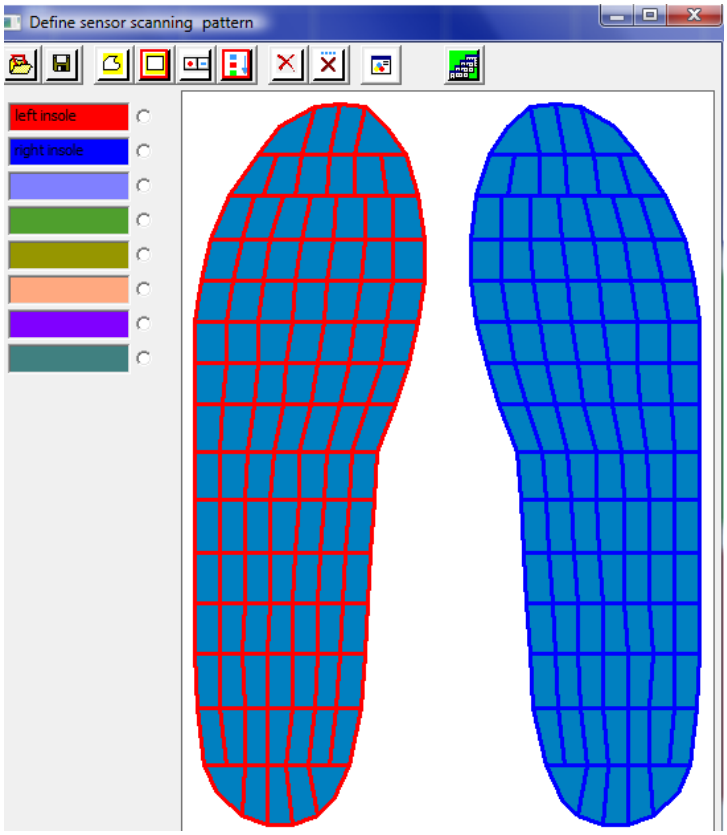
then select this menu option to use the previous zero.

e. Stop System — Stops communication with the pedar box.

f. Define Sensors to be Scanned — This menu allows the user to both turn off sensors which will be scanned and also define masks for display during data collection.

Sensors are commonly turned off to increase the sampling rate of the pedar system. For example, the maximum scanning rate of the pedar system with both insoles is 101 Hz (typically we say 100 Hz). If one of the insoles is turned off, you can capture at 200 Hz. It is also possible to turn off half of the insole if you are only interested in the forefoot or heel.

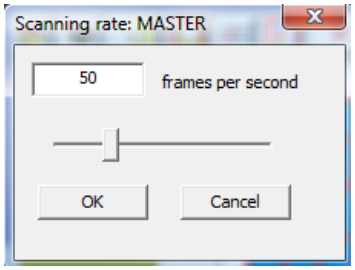
Masks may be defined prior to data collection if they are to be displayed during data collection for online feedback, or automatically exported as part of the MVA file. Please see **section 11** for further details on the MVA file.



- The **Open Mask** icon opens a previously defined mask/sensor
- The **Save Mask** icon saves a mask/sensor definition
- The **Free Drawing** icon provides the ability to draw a mask in any shape.
- The **Rectangle** icon draws a square box around the insole and

selects all sensors within the box.

- The **Sensor on/off** icon turns off/on an individual sensor within the insole
- The **Show all masks** icon shows all masks, or only the selected mask
- The **Delete single mask** icon deletes the currently selected mask
- The **Delete all masks** icon deletes all masks
- The **Change color** icon allows the user to change the color of a mask
- The **Download all masks** icon allows the user to create a new configuration file containing the settings that have been defined here. A typical Windows Save dialog will appear. The user may then select the measurement rate for the new configuration file:

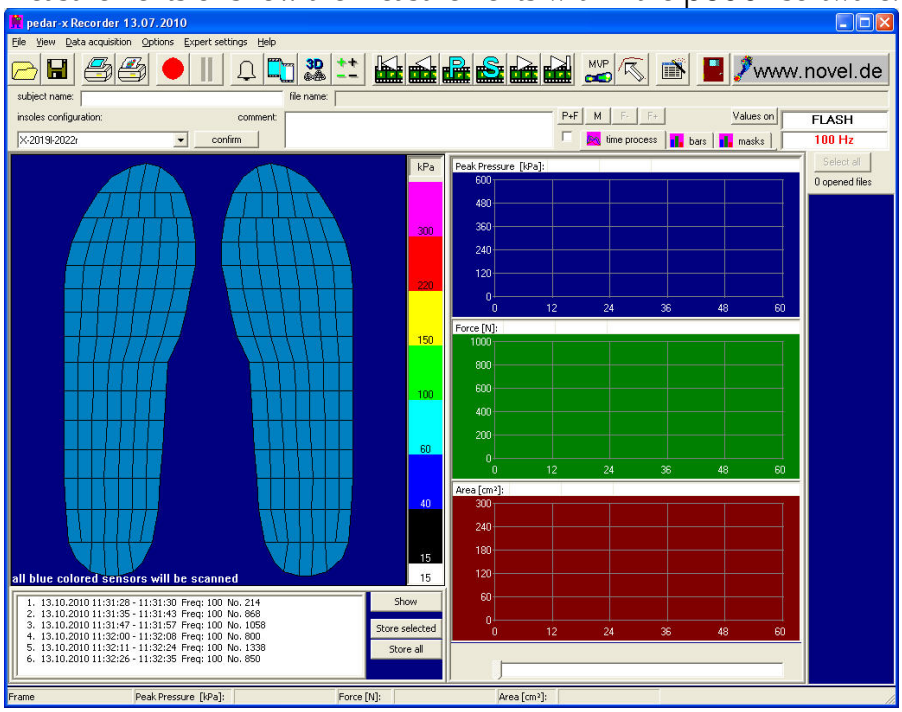


Turning off sensors:
By default the entire left and right insoles are defined and used for collection. To delete either the left or right insoles, select the associated radial button and choose the **Delete single mask** button. No sensors within the insole should be masked to ensure that the insole will not be scanned. If a portion of the insole is to be used, either delete the mask and draw a new mask, or use the **Sensor on/off** button to turn of sensors.

Create a new mask:
To create a new mask select a radial button alongside an empty color  and insert a name for the mask into the text field. Select the appropriate draw tool and draw the mask

g.	Video Time Limit	This menu allows the user to set a maximum recording time for Video.
h.	Video Preview	This menu initiates the video camera and displays a preview window for the video
i.	Configure Flash	**Load current insole configuration to flash:** select this option to download configuration and calibration files to the flash card. This is needed for proper data acquisition with the flash memory. **Erase measurements in flash:** select this option to delete all data from the flash card memory.

j. Read Flash **Read from device:** This menu item will display a Window containing a list of measurements contained on the device. The user may select to **store all** measurements, only **store selected** measurements or **show** the measurements within the pedar software.

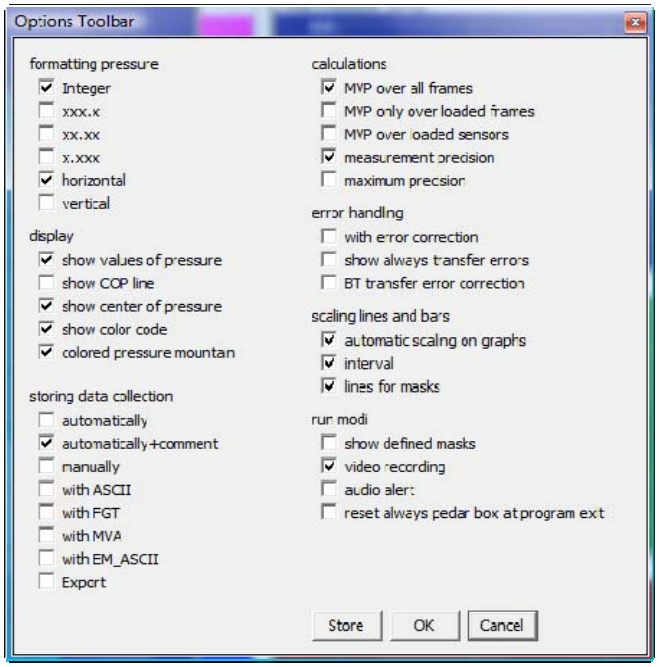


Append new series: This menu item is used after measurements have been read within the software. This will access the memory of the device again for download of additional trials.

Save to FAR File: This menu item saves the entire contents of the flash memory to a file with the extension .FAR. This file can be opened within the pedar software later and individual trials may be downloaded at that time via the same method as described under **Read from device**.

4.2.4 Options Menu

Many of options available within the **Options Menu** may be access by clicking the **Options** button  within the main window toolbar. The options will be described as listed within this dialog box. The options may also be accessed directly from the **Options | Set Options** Menu.



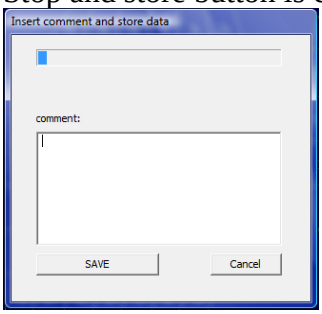
a. Formatting Pressure — Select how many decimal places will display when the pressure values are displayed on the 2D pressure picture. The user may also select whether the values display horizontal or vertical.

b. Display

- **Show values of pressure:** if selected the values of pressure will display on the 2D pressure picture
- **Show COP line:** if selected the COP line for the entire trial will display on the 2D pressure picture
- **Show the center of pressure:** displays a white and black circle of the COP during dynamic playback
- **Show color code:** display the color code block, or turn this option off
- **Colored pressure mountain:** if selected the 3D pressure picture will display in color

c. Storing data collection

- **Automatically:** the data will be stored automatically when the Stop and store is clicked. The data will be stored to the path defined within the Configuration program – typically C:\novel\novfile\data\pedar. If the pedar online program was started through the database a copy of the data will also be returned to the database.
- **Automatically + Comment:** a window will appear when the Stop and store button is clicked:



A comment may be entered regarding what occurred within the trial (i.e. the condition). This comment can be useful later when reviewing the data, or when searching for data within the novel database. As with the setting above, the data will be stored automatically (see above for more details).

- **Manually:** data will remain in the computer's RAM memory until stored. The user must manually go to File |Save | SOL file to save the trial.
- **With ASCII, FGT, MVA, EM_ASCII, Export:** Select these options to automatically export the trial into the exported file formats. These will be saved to the path defined within the Configuration program – typically C:\novel\novfile\ascii\asciiout \pedar. Please see **section 11** for definitions of each file export format.

d. Calculations

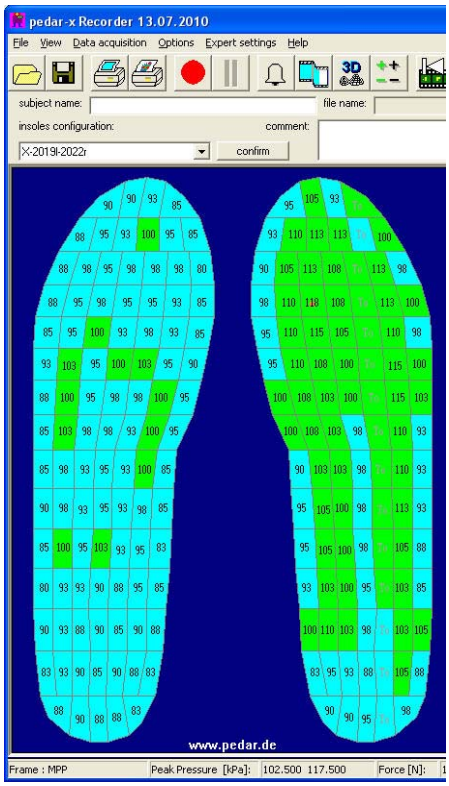
The Mean Value Picture (MVP) can be calculated three ways.

1. **over all frames:** average is calculated including *all* data, even zeros.
2. **only over loaded frames:** entire data *frames* that contain *all* zeros will not be included in the calculation.
3. **over loaded sensors:** *no* zero values will be included in the calculation.

Measurement precision vs. Maximum precision: When pressures are averaged the resolution which results will be higher than the actual resolution of the measurement. The maximum precision will average floating numbers, where measurement precision will only display averaged numbers which are valid with the measurement resolution.

e. Error Handling

- **With error correction:** If an error occurs the software will use the sensor elements around the sensor with the error to approximate the value within the element with the error. If this option is turned on, there is no way to undo the correction. It is recommended that this error be left off unless errors are commonly seen and it is determined that the estimation is appropriate.



In the example above a Timeout error has occurred in a column on the right insole. The sensors around this column are used to estimate the pressures. Please see Section **Point o** below and **Section 9** for additional information on error codes.

- **Show always transfer errors:** If an error occurs, or if using Bluetooth and dropped frames occur, this will display a message regarding the errors. It is recommended that this option ALWAYS be on. Error codes are not stored with the data file. Therefore, if an error occurs, the error code should be reviewed immediately after collection and a note be made. If the BT signal has dropped frames, this option will display the number and percentage of dropped frames which have occurred.
- **With BT transfer error correction:** When using the online + flash mode, this option will automatically fill the dropped frames which occur to the software from the flash memory of the device. Please note that this option is not recommended for larger lapses in data or when the connection via BT is very unreliable. For additional information on BT dropped frames and troubleshooting, please see **Section 8**.

f. Scaling of lines and bars

This menu determines how the time and bar graphs are scaled. The menu item **Options | Scaling of lines/bars** allows you to set these settings, as well as the numbers used of the automatic scaling is turned off.

- **Automatic scaling on graphs:** will change the scale to the maximum pressure, force, area etc which was captured during the trial.

- **Interval:** will auto-scale the time graph to the entire trial.
- **Lines for masks:** This option will display a separate line within the time process window for each of the defined masks. Please see **Section 4.2.3–f** and **Point p** below, for additional information on creating and displaying masks.

g. Run Mode

Show defined masks: This option will display the **Masks** window by default (instead of the time process graphs). Please also see **Section 4.1.3 #5** for further information on this window.

Video Recording: This option turns on the video recording. This option can also be turned on and off by using the Video on/off button in the main toolbar. A  button indicates that video is on; where a  button indicates that video is off. Please also see **Point q** for additional information on video recording.

Audio Alert: This option turns on the audio alert. This option can also be turned on and off by using the Audio on/off button in the main toolbar. A  button indicates that the audio alert is on; where a  button indicates that the audio alert is off. Please also see **Point r** for additional information on the audio alert feature.

Reset pedar box before exit: This option resets the pedar box upon exiting. Please note that the pedar box must still be on and connected to the computer for this option to function properly.

h.  Show 3D/2D/isobar pressure picture

With each mouse click the pressure picture will toggle between the 2D, 3D and isobar displays.

i. Show number of sensors

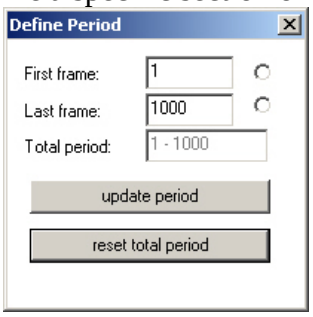
When selected, displays the assigned sensor identification number that corresponds to the ASCII output file.

j. Show area of sensors

When selected, displays the physical area of each sensor in mm^2. The information on this display may be useful when using the ASCII file.

k. Define period

Allows the user to define a specific section of data to view or save.



Select the first and last frame by either typing in a frame number, or by selecting the radial button and then clicking on a point within the time process graph. Once selected, click **update period**. The new data and calculations will display. If the new trial should be saved, use the menu item **File | Save As | SOL file** (see **Section 4.2.1 - f** for additional information. Use the **reset total period** to view the entire data file.

l. Speed of roll over process

Allows the user to change the rate at which the data file is played back.



The dialog will display one frame for the time which is defined and is dependent on the measurement rate that the file was collected. In the example above the software will play back one frame every 20 milliseconds. Therefore, if the data was collected at 100 Hz, or 1 frame every 10 milliseconds, this will display the play back at half speed. In this example a definition of 5 milliseconds would play back the trial at double speed. Click **OK** to apply the change.

m. Simultaneous scrolling

With this option enabled and a second pedar file is opened in a separate window, when the play button is clicked both trials will play back simultaneously. Please note that this does not mean that the trials will be synchronized. This option can also be enabled/disabled within the main toolbar. The  button indicates that simultaneous scrolling is disabled; where a  button indicates that the feature is enabled.

n. Image transparency

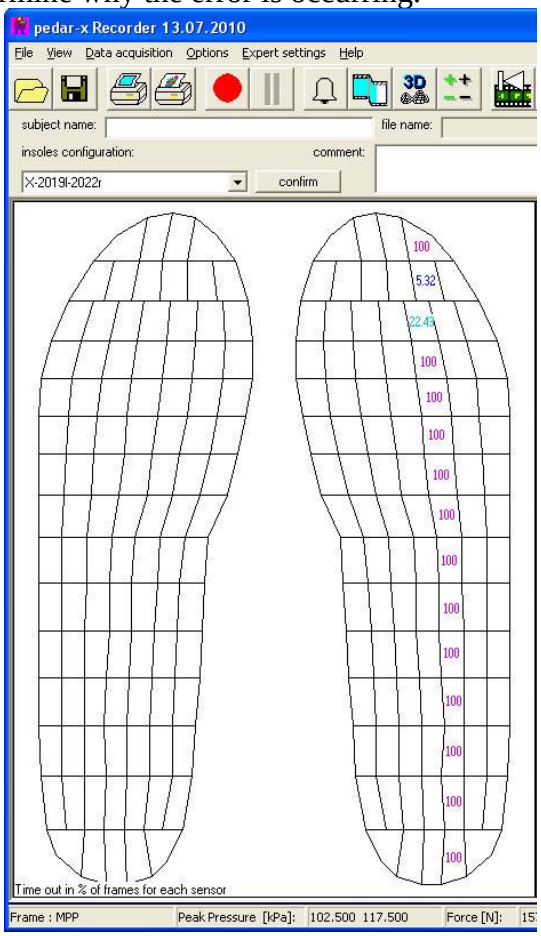
It is now possible to have an image in the background of the Pressure picture window. This image may be selected from the **File | Open Image** menu. The transparency of this image can be modified with this menu item.

o. Error codes

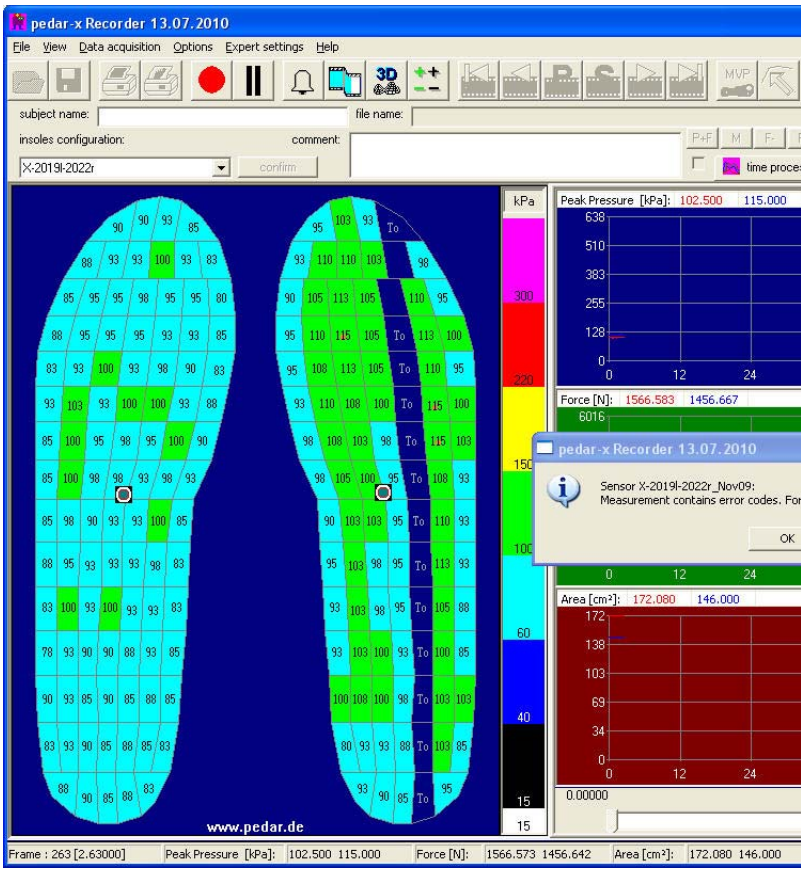
If an error has occurred during data collection, one of the following menu items will become available to view the error. Please note that the error code is NOT saved with the trial, so as soon as the trial is closed, or another measurement is started the error code will be gone. It is important to ensure that the option **Show always last transfer errors** is enabled. This will prompt the user to view the error and make note of it before continuing on to the next measurement.

Once one of the following is chosen the pressure picture window will display in which sensor(s) the errors have occurred and for what percentage of the trial collected. If this is a very low number, the error can likely be ignored. If the time in which the error

occurred is higher, or is occurring quite frequently, steps should be taken to determine why the error is occurring.



- **Converter overload:** This error will occur if the system received data which is above the amplification of the sensor. This error commonly occurs if high loads have been placed on the sensor. For additional information please see calibration **Section 4.4.2.2**.
- **Timeout:** This error indicates that the pedar box did not receive a reading from a sensor(s) when it expected to.



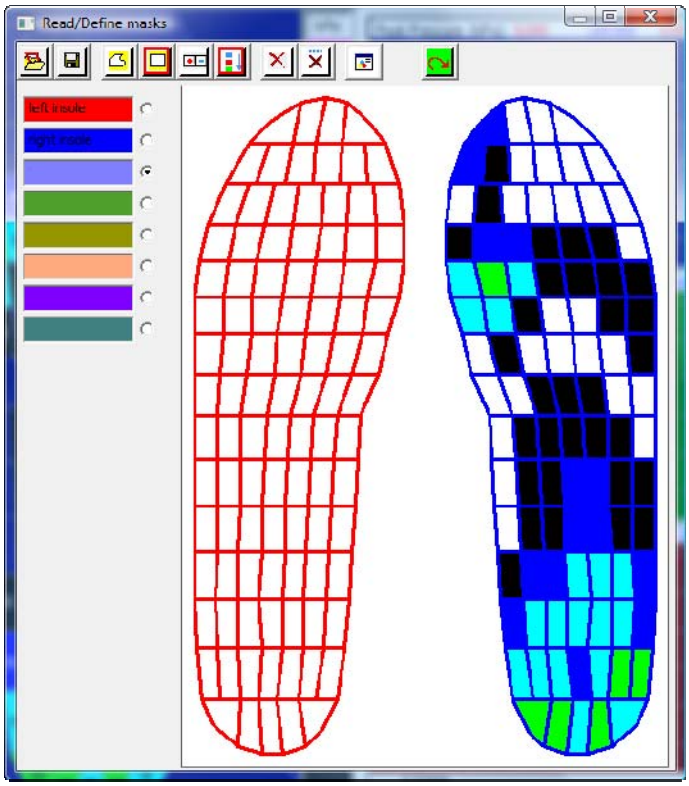
This error will occur if there is insufficient time for the computer to communicate with the pedar-x, or if a sensor element has become damaged. Please see **section 9** for information on testing the pedar insoles for errors.

- **Sensor overload:** This error occurs when one or more sensors have exceeded their maximum capacity.
- **Last transfer error:** This option will display the last measurements error codes. If no error codes were reported the message will state this.

p. Read/define masks

Select this option to define or display a mask file for use with the collected trial(s).

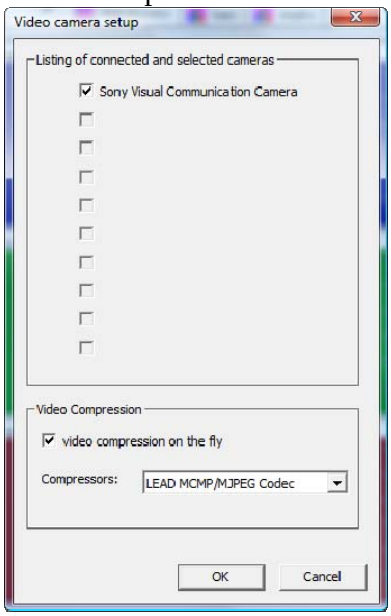
- **Define Masks:** A window similar to the following will appear. All options are the same as that described in **Section 4.2.3-f**. The masks may be applied by clicking the **overlay changed masks** button  .



- **Read masks:** Allows the user to apply a mask to the currently selected file.
- **Read masks for all files of same type:** Allows the user to apply a mask to all currently open files with the same insole setup.

q. Video Camera Setup

The video camera setup dialog allows the user to select and deselect the cameras which are connected to the computer. Only those selected will be captured. The user may also select a video compressor. Please note that codecs may be required to view the collected videos on other computers:

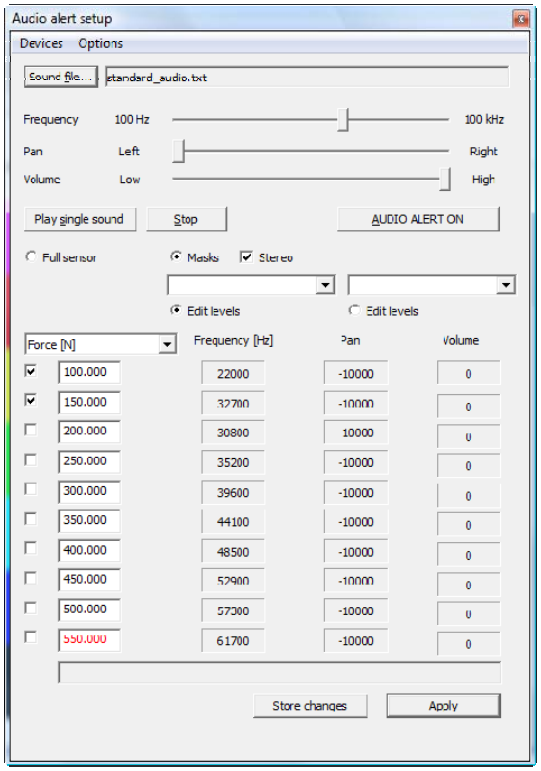


r. Audio Alert Setup

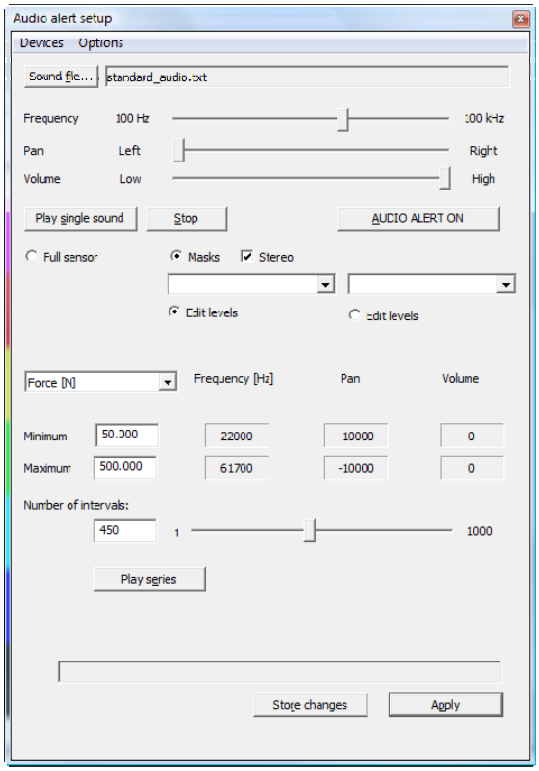
The audio alert function provides the user with the ability to provide audio feedback for data being collected or played back. This option can be useful during biofeedback training sessions. Force, peak pressure, mean pressure and contact area can all be input variables for the determination of the output sound.

- Each variable may be adjusted for the left and right foot independently. This can be done by selecting the **Full sensor** or **Masks** radial button and then the subsequent left or right insole. **Edit levels** should then be selected to edit the level for that particular mask. In some versions of the pedar software you will have the ability to apply the sound to other masks which have been defined along with the left and right insole
- The frequency, pan (left vs. right) and volume may be adjusted for each level selected. To play the sample sound click the **Play single sound** button and then the **Stop** button to stop the sound.
- In some versions of software you will also have the ability to apply a proportional sound, as opposed to discrete sound levels. This option is selected from the **Options** menu.

Discrete Sound:



Proportional Sound:



The proportional sound series may be sampled by clicking the **Play series** button.

- Once the options have been selected the settings may be saved as a new souds file by clicking the **Store changes** button. A previously saved audio alert file may be opened by clicking the **Sound File...** button.
- The audio alert can be turned on by clicking the **AUDIO ALERT ON** button and the **Apply** button will apply the settings to the software for the next data collection.

s. EM_ASCII export setup — This option is used in conjunction with orthotic development. The export directory and default name for the export file can be defined.

t. Pressure range setup — This option allows the user to change the color range settings of a previously collected trial. This change may be applied to only the currently selected file, or all open files.

4.2.5 Expert Settings

Please note that not all options are available for all versions of software. The Basic and Standard versions of software will not be able to access all of these features.

The pedar software uses an insole configuration file for measurement. This configuration file contains the calibration file, as well as information on how the data will be recorded. This includes the measurement rate, the length of the trial which will be recorded, synchronization options, etc. Please see **Section 4.3** for a complete description of the configuration file.

The calibration file, which is contained within the configuration file, describes the relationship between an applied known pressure and the electrical signal from the insoles. This procedure is completed prior to system delivery, but may also be completed by the user with the aid of the trublu calibration device. Please see **Section 4.4** for a complete description of the configuration file.

a.	Edit actual insole configuration	The insole configuration which is selected under the **insoles configuration** drop down list will be opened for editing. Please see **Section 4.3** for a complete description of the configuration file.
b.	Edit insole configuration	An insole configuration can be selected from a typical Windows Open screen for editing. Please see **Section 4.3** for a complete description of the configuration file.
c.	Create New insole configuration	Select to create a new configuration file for a calibration. Please see **Section 4.3** for a complete description of the configuration file.
d.	Settings + calibration Online	Select to complete the calibration process for insoles. The pedar box must be connected to the software prior to selecting this menu item. Please see **Section 4.5** for a complete description of the calibration procedure.
e.	Settings + calibration Offline	Select to review a previously completed calibration. Please see **Section 4.4** for a complete description of the calibration procedure.
f.	Directories	The default directory paths may be changed with this menu item.

4.2.6 Help menu

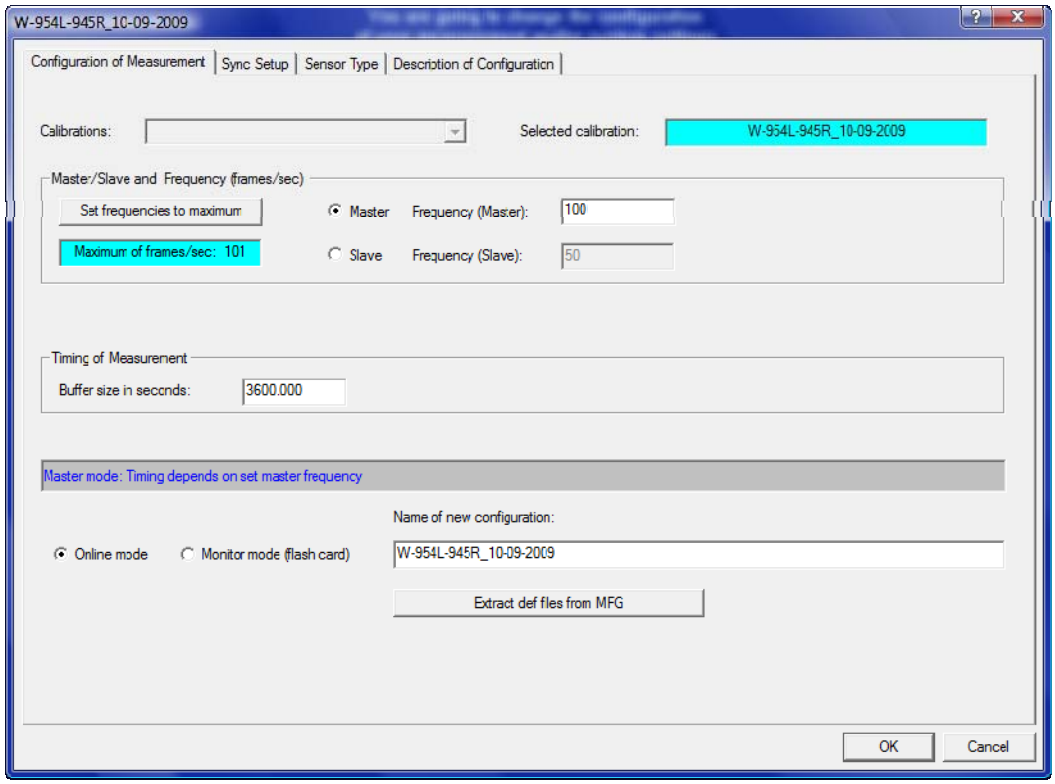
a.	novel (www.novel.de)	Opens the novel.de website in the user's default browser.
b.	About	View software version information.

4.3 The insole configuration file

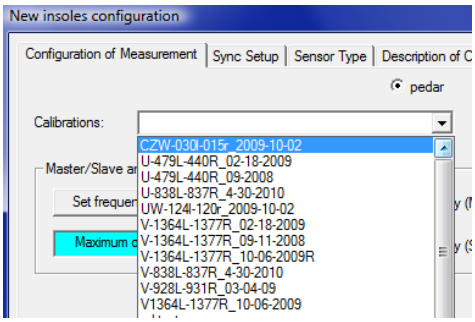
Please note that not all options are available for all versions of software. The Basic and Standard versions of software will not be able to access all of these features.

The pedar software uses an insole configuration file for measurement. This configuration file contains the calibration file, as well as information on how the data will be recorded. This includes the measurement rate, the length of the trial which will be recorded, synchronization options, etc.

The configuration file may be accessed via any of the methods described in **Section 4.3.5-a-c**. Once the configuration file is accessed, the following Window will appear:



If a new configuration is created the same window will appear, but with the option to select the calibration from the **Calibrations** drop down menu:

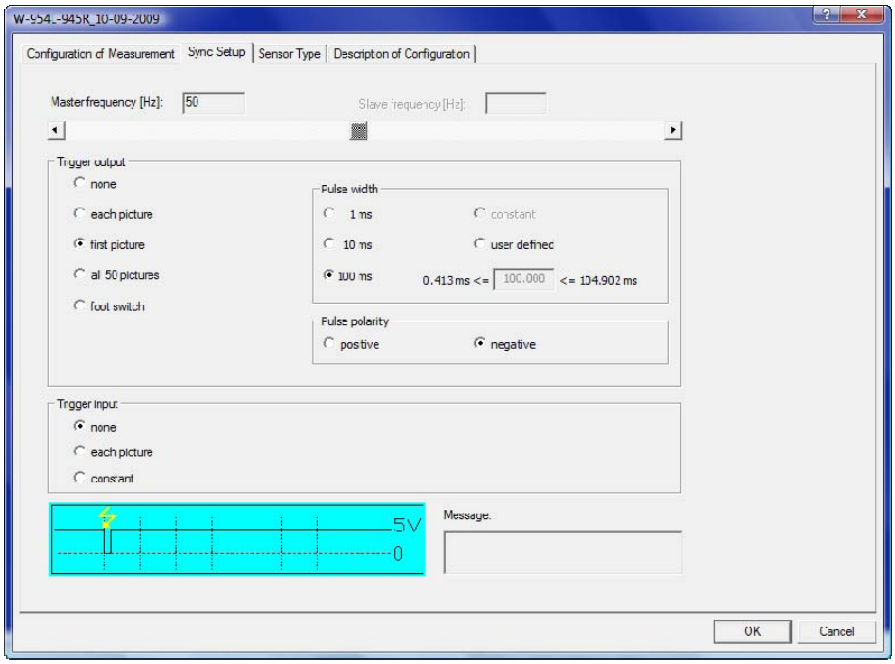


4.3.1 Configuration of Measurement Tab

- **Selected calibration:** will display the calibration file which is associated with this configuration.
- **Master/Slave Frequency (frames/sec):** This section defines the measurement rate of the pedar system. The max sampling rate can be selected by clicking the **Set frequencies to maximum** button. (Please note that typically 100 Hz is used for pedar data collection, not 101 Hz). Alternatively the sampling rate may be entered under the **Master** field. The slave device (if used) can also be entered.
- **Timing of Measurement:** The total collection time may be entered within this field. By default the maximum measurement rate is set to 3600 seconds (one hour). Typically users will stop the data collection manually using the **Stop and store** button, or the system is triggered externally.

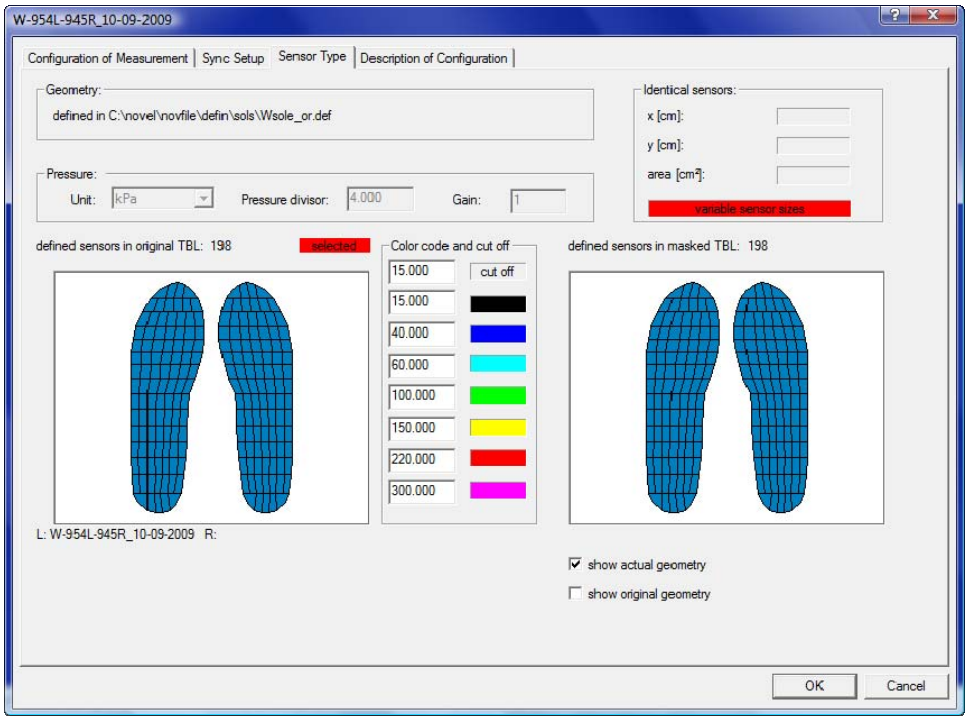
- **Mode:** The configuration must be set to either an online or flash card mode. If online + flash is used, this is considered an online configuration. Please see **Section 2.4.3** for additional information.
- **Name of new configuration:** Enter or change the name of the configuration file here. It is recommended that the configuration file contain the information on the insoles used, the date of calibration and any other important information, such as online vs. flash mode etc.
- **Extract def files from MFG:** This button will extract the calibration files contained within the MFG. This can be useful if the configuration files were transferred to another computer without the calibration files.

4.3.2 Sync Setup



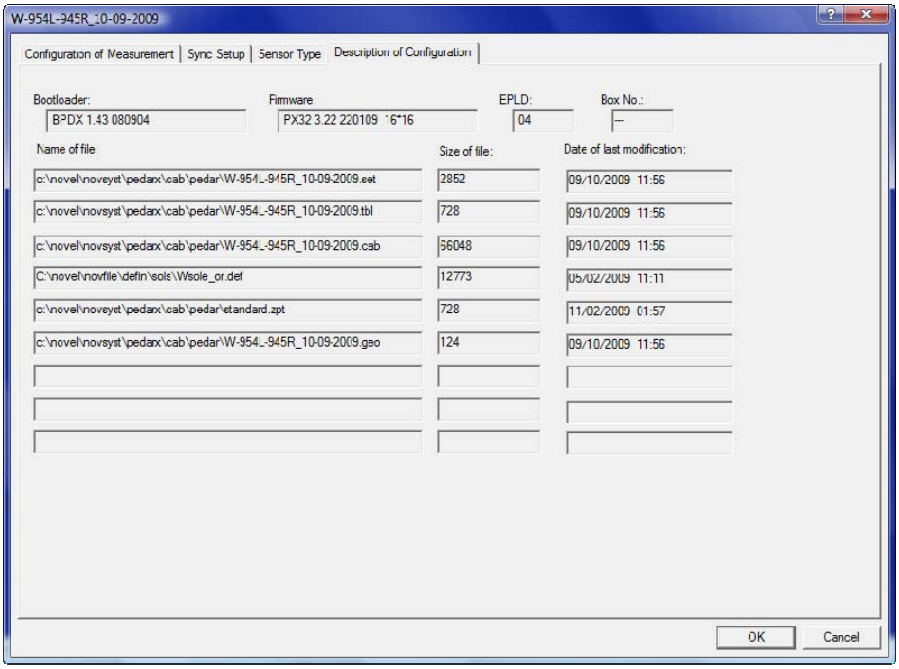
- This window displays the options for a trigger output and trigger input. These options may be used with either the wired or wireless synchronization device (Please note: only certain versions of the wireless sync can use a trigger input – please contact your novel representative if you are unsure if your sync device supports a trigger input).
- For a complete description of the synchronization hardware and software please see **Section 7**.

4.3.3 Sensor type



This window displays the pressure color code and cutoff value and the layout stored in the original and the masked TBL files. The TBL files are created during calibration or masking and are used to determine the maximum sampling rate of the insole pair.

4.3.4 Description of Configuration



This window displays the paths for the set of calibration files. These paths can't be modified and are for reference only.

4.4 Settings and Calibration Software and Calibration Hardware

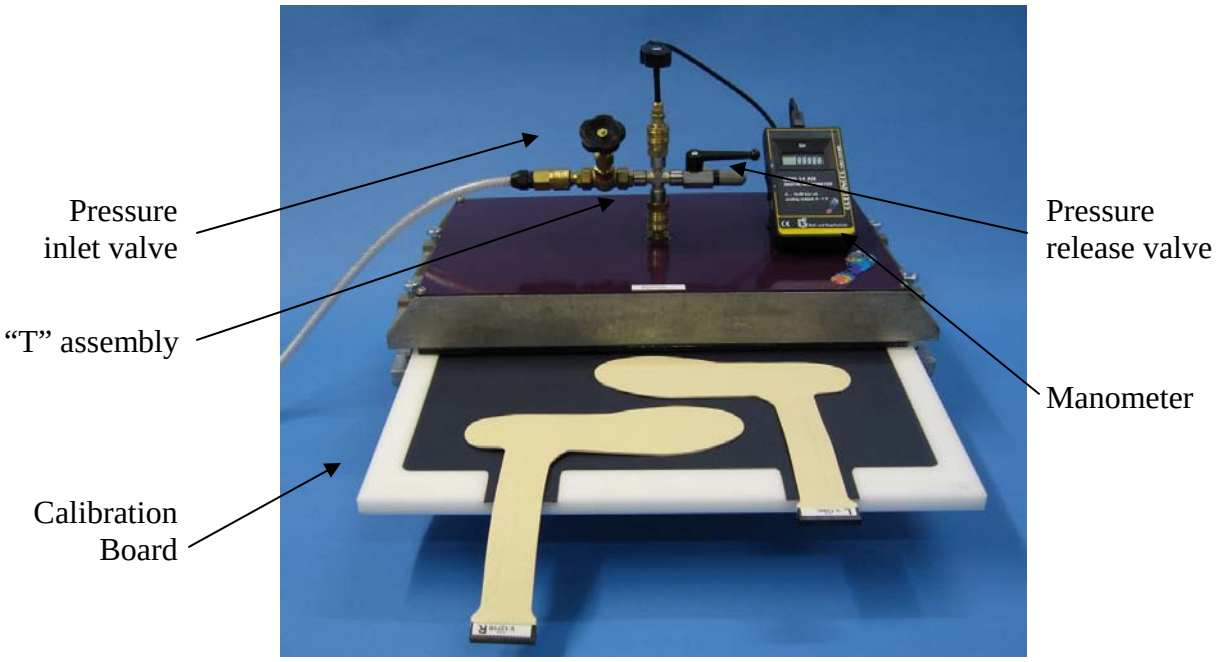
The calibration file describes the relationship between an applied known pressure and the electrical signal from the insoles. This procedure is completed prior to system delivery by novel and the calibration files are installed during installation. This procedure may also be completed by the user with the aid of the trublu calibration device. **Calibration must take place in a trublu calibration device supplied by** novel**.**

IMPORTANT

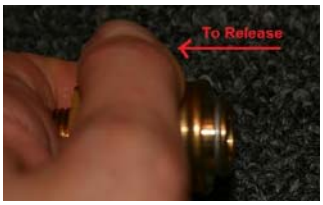
novel recommends checking calibration of each insole approximately every three months by applying known pressures to the insoles(s) in the calibration device and comparing with the pressure obtained from an online measurement. Please see **Section 4.4.2.1 #13** for instructions on checking the calibration.

4.4.1 The Calibration Device

The calibration device has been developed specifically for calibration of pedar insoles and consists of a rubber membrane, housed within a secure unit. Compressed air is fed into the device thus exerting a pressure on the inserted insoles. The rubber membrane ensures a homogeneous pressure distribution.



- **Place the calibration device on a flat, stable surface and secure to prevent it from slipping off the surface.**
- Connect the supplied brass air pressure "T" assembly component and the manometer to the top of the calibration device as shown in the picture above. Please note that all connections utilize a quick release connection. Pull back on the connector to release.

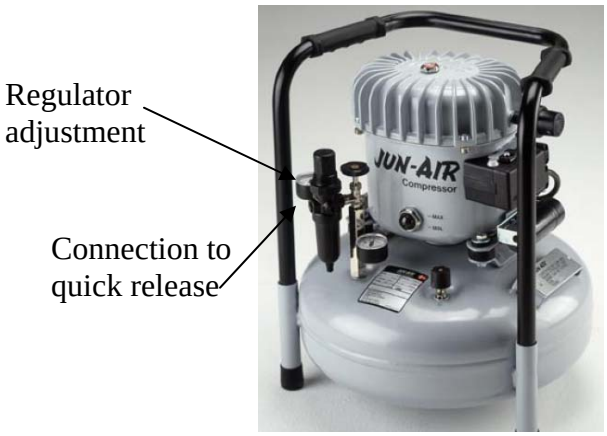


- Connect the air supply (compressor) to the calibration device. The air must flow from the compressor to the calibration device, thus making it possible to read the air pressure on the manometer pressure scale. For the pedar calibration a 90 PSI (6 bar) air supply is required. An extra female fitting (pictured on the left below) is typically provided for connections to the air compressor/supply. This is a standard ¼" NPT connection and can be made utilizing plumber's (Teflon) tape.





- The calibration device must only load pressure with the calibration board pushed completely into the device. Two stoppers on the bottom of the calibration board will indicate when the board is in the proper position
- To allow pressure into the calibration device close the release valve by turning it 90 degrees clockwise. SLOWLY open the inlet valve by turning the knob counterclockwise. The manometer should begin to read pressure. Ensure that you can obtain the required 6+ bar for calibration.
- High levels of pressure should not be left inside the calibration device for long periods of time, as this can cause slow leaks in the bladder system. Ensure that the steps performed with calibration are performed systematically and quickly.
- Commonly novel will provide a Jun-air air compressor with the calibration device. This is a low noise compressor and is designed for laboratory/office applications. The female quick release picture above should be connected to the regulator on the air compressor, as pictured below. The air compressor should be regulated to just above 6 bar. Please refer to the compressor manual for complete instructions.



IMPORTANT

It is extremely important to close the inlet valve from the compressor (clockwise) and open the outlet valve (parallel to the outlet pipe) when not using the calibration device. If this is not performed, it is possible that air can leak from the compressor into the calibration device over time and rupture the rubber bladder.

IMPORTANT

The manometer is calibrated in bar.

1 bar = 10 N/cm^2 = 100 kPa

4.4.2 Settings + calibration online

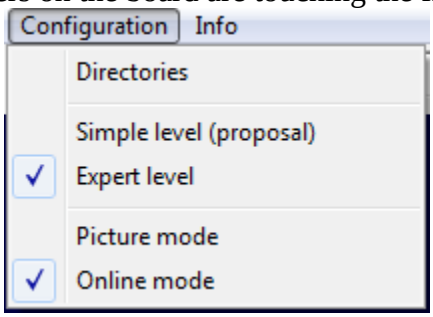
Select this option to complete the calibration process for insoles. The computer must be connected to the pedar device to select this option. The process includes setting the amplification of each sensor element within the matrix (the "Settings" procedure) and then performing a multi-point calibration. The final step of the calibration procedure is checking the calibration within the pedar online software.

There is no need to use this program during daily routine use of the system. Calibration should be repeated or checked approximately every three months.

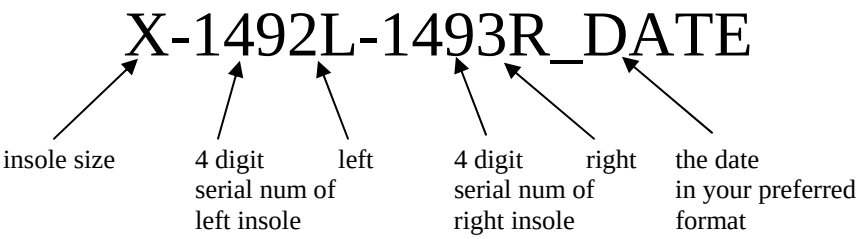
4.4.2.1 Calibrating Insoles (Simple Level)

1. Select the menu item **Expert Settings | Settings + Calibration Online** from the pedar online software.
2. Connect the pedar system and insoles as described in **Section 2.4.4**.
3. Place the insoles within the calibration device.

a. The insoles should be placed within the calibration device as shown in the first picture in **Section 4.4.1** above.
b. The insole has two sides; one side shows a square pattern of the leads and the other the "s" shaped sensors (see picture below). Place the leads side UP on the calibration board. This means that the left insole will end up on the right side of the board and the right insole of the left side. In this configuration the pedar insole label (showing the serial number such as X.1234L) should be UP and visible.
c. Ensure that the insoles are properly spaced within the device. They should not be touching each other and should be about 1 cm from the edge of the board.
d. Push the calibration board into the device. Ensure that it is pushed in completely in and the two stoppers on the board are touching the front of the device.



4. Within the Calibration software go to the menu **File | NEW calibration** or select the  button from the toolbar. It is recommended that a new calibration be performed with each calibration.
5. A Windows Save As window will appear. Give a name to the calibration file. The calibration file should be named like the following:

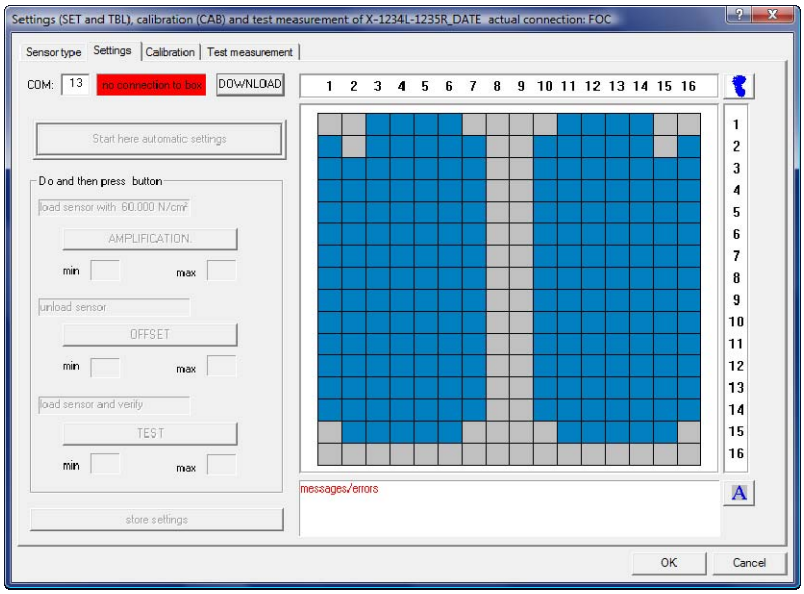


It is very important that the size and width of the insole is indicated properly. This is done by starting the calibration file name with the <u>first letter</u> that is shown on the pedar insole label, as for instance the X. For some insole types there is a second letter, which must be used as the second letter in the calibration file name:

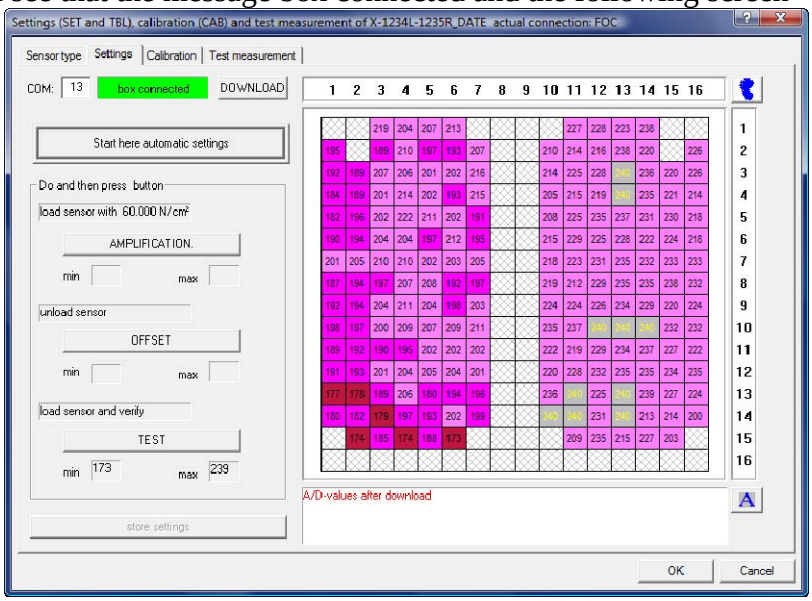
- A wide insole will also have a "W" associated with the serial number, such as XW.1234 L
- An extra wide insole will have an "E", such as XE.4321 R.
- An insole with a long lead will have a "S" such as XS.1672 L.

Click **Save**.

6. The following screen will appear

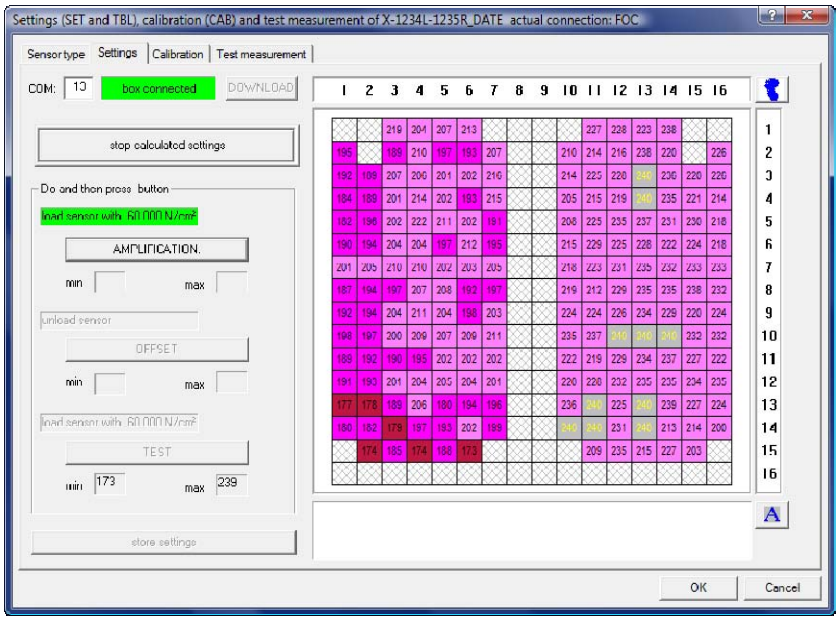


7. Verify the system is connected to the specified COM port and click **DOWNLOAD**.
8. You will then see that the message box connected and the following screen



The numbers and colors shown after the Download button is clicked are not important. Any messages (error codes, warnings, etc) regarding the progress of calibration can be found in the message field in the bottom right of the window.

9. Click **Start here automatic settings**. You will receive a message within the message window on the bottom right of the screen which says “Please wait: preparing automatic settings”. The software will then indicate with a green window what should occur. The first step will be to load the sensors with the maximum load, which is 60 N/cm^2, or 6 bar.



a. The software will then indicate with a green window what should occur. The first step will be to load the sensors with the maximum load, which is 60 N/cm^2, or 6 bar. To ensure that amplification of the sensors occur properly at high loads, it is recommended that you load the insoles slightly higher than the amplification number mentioned above, i.e. 0.1 bar higher.
   i. Load the sensors in the calibration device to the level indicated. Follow the instructions for loading pressure described in **Section 4.4.1**. Please note that the manometer will display in bar and the software is, by default, set to display in N/cm^2. Please make this conversion (a factor of 10, accordingly.
   ii. Click the **AMPLIFICATION** button.
   iii. The min and max values under Amplification should read between 235 and 245. If a value is greater than 240 it will be highlighted in yellow. This is not an issue.

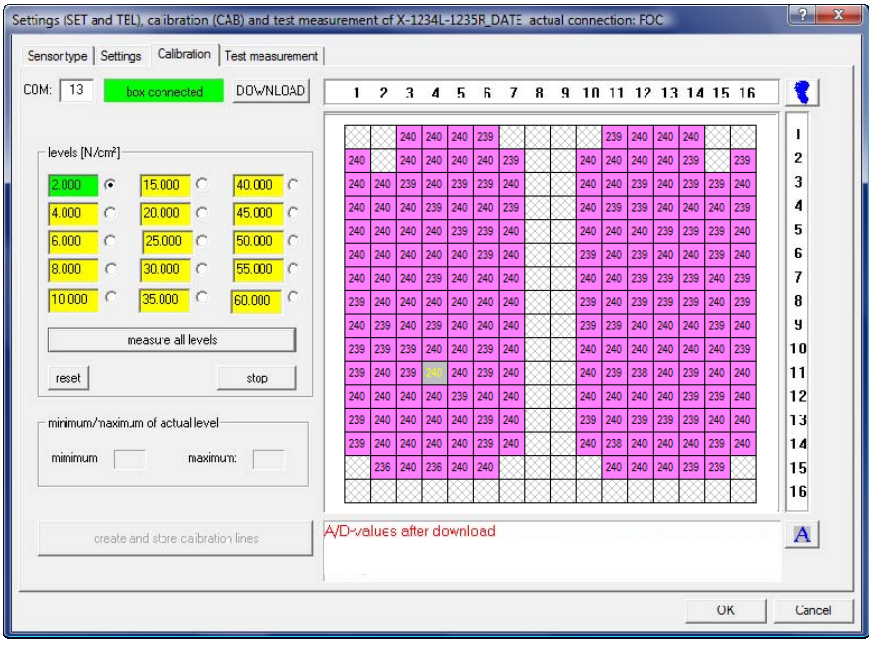
b. The **unload sensor** field will turn green.
   i. Release the pressure by opening the release valve and release all pressure.
   ii. Click the **OFFSET** button. The software will wait 10 seconds and then capture two unloaded states. Do NOT load the system with any pressure during this time.
   iii. The min and max values should read between 25 and 35.

c. The **load sensor with 60 N/cm^2** field will turn green.
   i. Load the sensors in the calibration device to the level indicated (60 N/cm^2).
   ii. Click the **TEST** button.
   iii. The min and max values under Test should read between 235 and 245. If some sensors read a gray > sign, this is OK.
   iv. Unload the calibration device.

d. Click **store settings** to save the settings. If any errors have occurred they will be listed under the messages/errors box and the next step will not be able to be

performed. Please refer to the Expert Mode described in **Section 4.4.2.2** for further information.

10. Click on the CALIBRATION tab. A window similar to the following will appear:



The first level, 2 N/cm2, will be green. Click **measure all levels**.

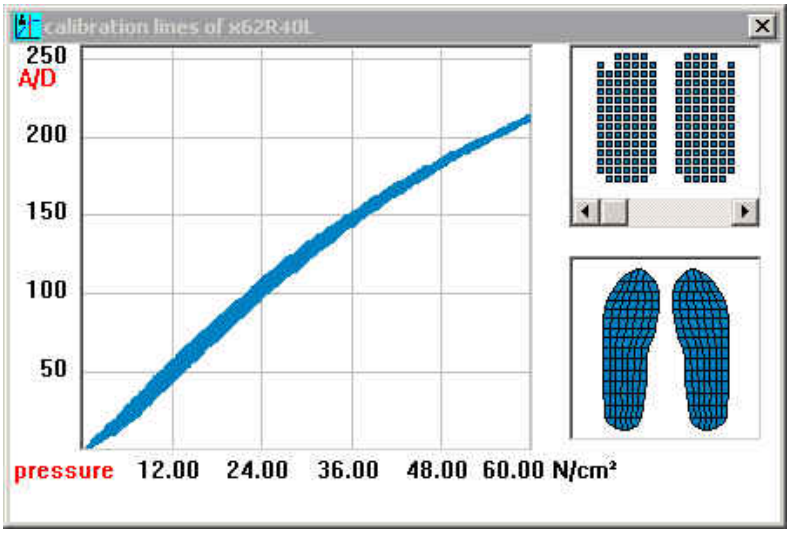
a. A message will appear to unload the insoles. Ensure that no pressure is in the calibration device and click **OK**. Please note that this step is similar to the unload requested when the pedar online software requests that the insoles be unloaded.
b. The software will begin to read the data in real-time from the pedar device. Apply the specified pressure (the value will be green). Once the sensors in the calibration device are loaded to the specified level, press the space bar or Enter key to acquire the A/D values corresponding to the pressure level.
c. As soon as a level is taken the next level will turn green. You should also hear a "ding" if the volume is turn on. Increase the pressure placed on the insoles to the next level and click the space bar. Continue the process until the last level has been acquired.

NOTES:

- The pressure should be increased to each level; you should not go over a level and come back down. The process should occur smoothly and not take a long period of time (less than a few minutes), so as not to introduce hysteresis on the sensors. If a level is over shot, release the pressure to below the level and come back up, but this can cause a slight change in curvature of the lines. It is recommended that the process be restarted:
  - **Stop** button: click to stop the calibration process.
  - **Reset** button: click to reset the calibration process.
  - **Zero** button: re-zero the sensors (no pressure should be applied).
  - **Measure all level** button: click to begin the calibration process.

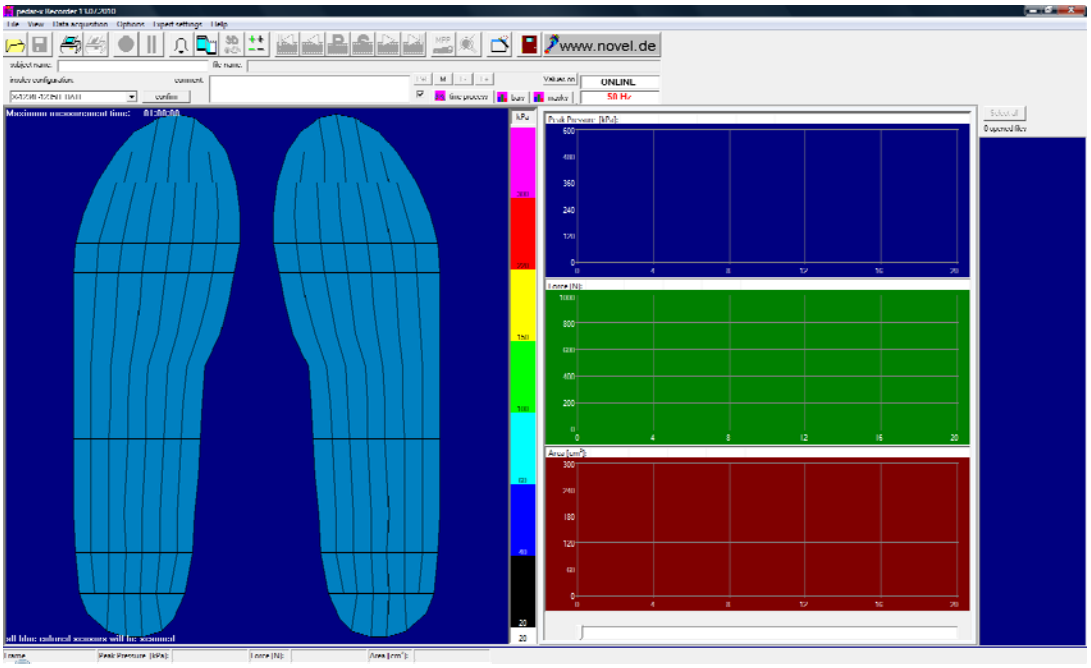
- Minimum and maximum levels: this displays the min and max value that occurred at each pressure level. These values may be important if having problems with the calibration. Errors concerning these values will be displayed in the messages/errors box. Please contact your novel representative for assistance.
- Font icon  : use to specify the font of messages.
- The foot icon  in the upper right corner is used to switch the view of the sensor layout between the electronic layout and the physical layout.

11. When the calibration is over click **create and store calibration lines**. A window similar to the following will appear.



Close the calibration lines window.

12. Click the **OK** button within the calibration window. A message will appear asking if you would like to create a new configuration file. Click **YES**. A new configuration file will be created for use within the pedar online software. The configuration file will now be selected and displayed in the pedar online software:



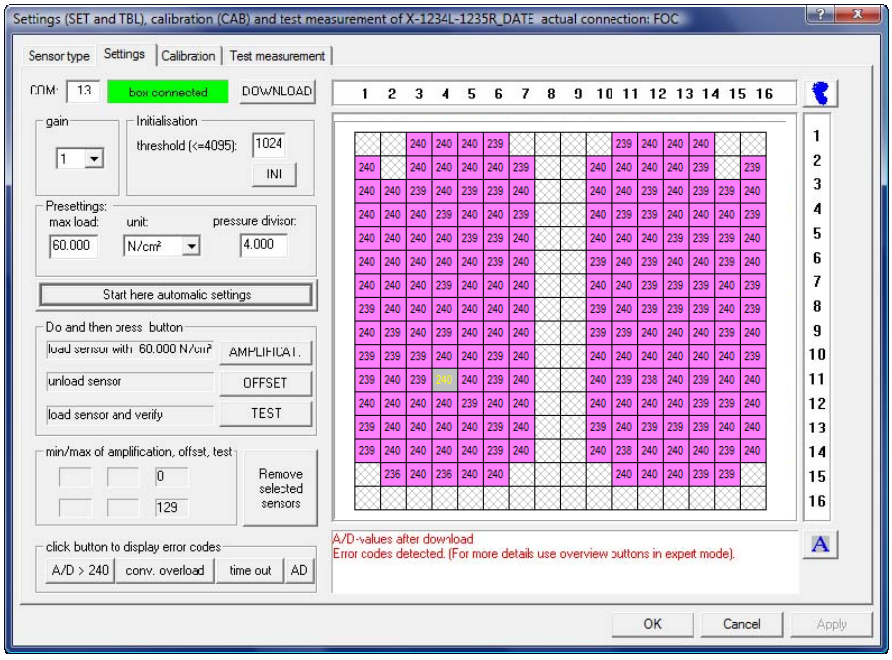
13. The insoles will now be tested within the calibration device to ensure that the calibration is accurate. This process is also the process that should occur when testing whether or not the insoles need recalibrated.
    a. Choose the appropriate **insole configuration** file from the drop down menu, if not displaying already.
    b. Click the **Confirm** button to confirm the use of this insole configuration.
    c. Click **OK** to unload the left and right insole. Ensure that no pressure is within the calibration device.
    d. Click the **Record** button and a trial will begin.
    e. Click the **Values on** button
    f. Load the sensors to any pressure level; choose something lower than 1 bar to begin with.
    g. Read the **Mean Pressure** value for the left and right insole. This value should read similar to that of the manometer. Also ensure that the **Peak Pressure** value is not far off from the mean pressure value.
    NOTE: Keep in mind that the values displayed in the online software are most likely kPa and the manometer displays in bar. These have a factor difference of 100, i.e. 100 kPa = 1 bar.
    h. Check additional pressure levels up to the maximum calibrated value (600 kPa / 6 bar / 60 N/cm^2) in the same manner. When you have finished, unload the sensors and click the stop and store button. If you would like you may save this trial, though it is not required.
    i. NOTE: The new configuration file will default to the configuration file settings in the default configuration. It may be necessary to modify the measurement frequency, sync settings, color code or cutoff etc. This should be done prior to the first data collection. Please see **Section 4.3** for additional information.

4.4.2.2 Calibrating insoles (Expert Level)

This method of calibration is recommended for Expert users, or for those recommended to do so by their novel representative, and for users who would like to perform the following:

- Calibrate one insole at a time
- Remove/turn off sensors from the calibration
- Change the default maximum load for calibration (600 kPa / 6 bar / 60 N/cm^2) to another level.

Steps 1-5 described in **Section 4.4.2.1** should be followed to set up the calibration. In general the same steps for the remainder of calibration should be the same; however the display is slightly different:



Please follow **Steps 6-13** in that section aside from the following changes:

- **Step 9-d** - click the Apply button instead of the store button. A message will appear asking to save the new settings, click Yes.
- **Step 11** – it is not required to click create lines and store. As soon as the last level of pressure is taken (presumable 6 bar) the lines will be created. Once the lines are reviewed and the created lines window is closed the user must click **Apply** to create the calibration lines. A message will appear asking if the lines should be created, click **Yes**.
- As with the simple level the user should then move to **Step 12** and click **OK** and the configuration file will be created.

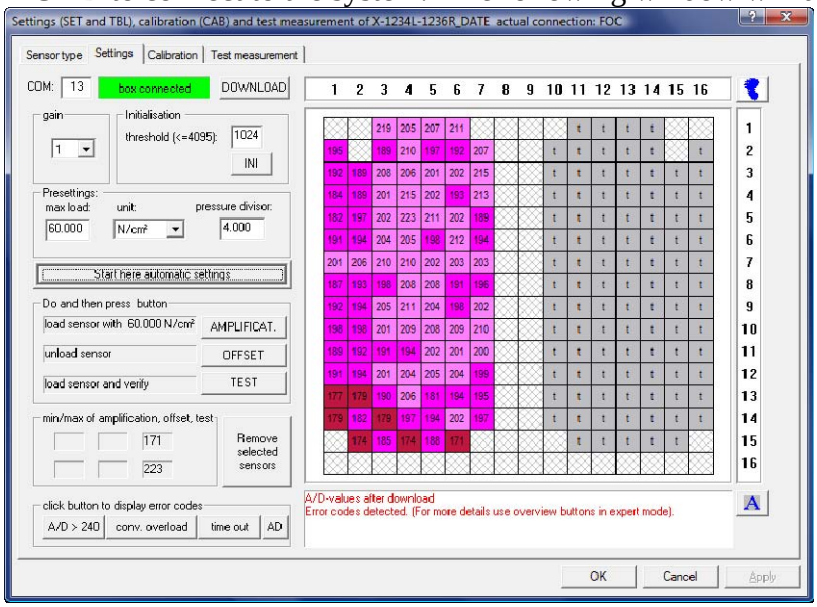
Continue with **Step 13** to test the success of the calibration within the pedar online software.

4.4.2.2.1 Calibrating one insole at a time

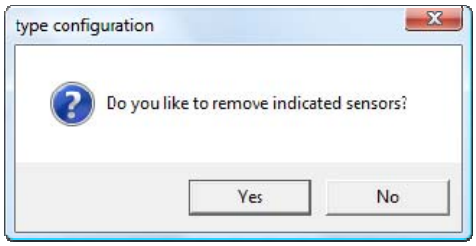
This method of calibration is used when:

- Only one insole is required for data collection
- Planning to combine insoles of different sizes
- Only one insole fits inside of the calibration device.

1. Connect the system as described in **Section 2.4.4**, however, leave one of the insoles disconnected, i.e. calibrate the left insole only, so leave the right insole disconnected.
2. Follow **Steps 1-6** described in **Section 4.4.2.1** to set up the system for calibration.
3. Click **DOWNLOAD** to connect to the system. The following window will appear:



4. Click the **time out** button. A message will appear asking if the user would like to remove the indicated sensors. Click **Yes**.



NOTE: It may be possible that some sensors on the disconnected side actually receive a reading. If so, these sensors will need to manually be removed. Please perform the process described in **Section 4.4.2.2.2** below after #7 in this section.

5. A message will appear in the message window that the sensors have been removed and to start the settings again.



6. Click **DOWNLOAD** again. The settings will be downloaded without the right insole.
7. If only one sensor will be used for data collection, please proceed to the next steps. If calibrating insoles for combination with another insole, or because the insoles do not fit

within the calibration device, connect the other sensor to the empty insole connector at this time.

8. Complete **Steps 9-11** as described in **Sections 4.4.2.1 and 4.4.2.2**
   a. If the user plans to use just one insole for data collection, please complete **Steps 12 and 13** as well
   b. If you are combining insoles later, do NOT create the configuration file. Once OK is clicked in **Step 12**, click **NO**.
9. Repeat **Steps 1-9** of this section for the other insole. To complete the combination of two insoles please refer to **Section 4.4.2.3**.

4.4.2.2.2 Removing or turning off sensors

This menu can be useful when a sensor has become damaged, but calibration needs to be performed. If an error occurs during the calibration procedure, calibration lines cannot be created. This option may also be used to reduce the number of sensors in the matrix for an increase in sampling rate. However, the pedar online menu item **Data Acquisition | Define sensors to be scanned** is more commonly used for this (see **Section 4.2.3 – f**).

1. Follow **Steps 1-6** described in **Section 4.4.2.1** to set up the system for calibration
2. Click **DOWNLOAD** to connect the system to the pedar
3. If a **t** appears in one of the sensors, or this sensor has been identified previously or during the Settings procedure as problematic it may be necessary to remove it:



   a. Click on sensors to be removed. The sensor will turn gray with an '**x**' in it.
   b. When all sensors to be removed are selected, click the **remove selected sensors** button.
   c. A message will appear in the message window that the sensors have been removed and to start the settings again.
   d. Click **DOWNLOAD** again.
   NOTE: it is best to remove sensors *before* completing the settings procedure, so the procedure will not have to be repeated.
4. Complete **Steps 9-13** as described in **Sections 4.4.2.1 and 4.4.2.2**

4.4.2.2.3 Changing the maximum load of the insoles

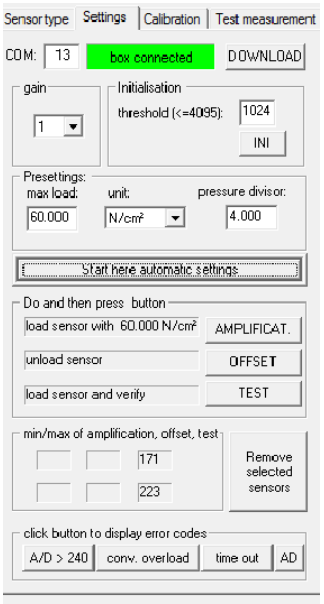
The default maximum load for the insoles is 600 kPa / 6 bar / 60 N/cm^2. The insoles have been designed for measurement within the 20 – 600 kPa range. They are most accurate in the mid-range of measurement where most data will occur during regular walking and running. Because the pedar-x system has a fixed gain amplification, changing the maximum load will change the measurement resolution. However, because the sensors are not perfectly linear, decreasing or increasing the maximum load will not double or halve your measurement resolution. Please consult your novel representative to determine the proper load, or alternative sensor types for your application.

If it has been determined that an alternative load is appropriate the following steps should be performed:

1. Follow **Steps 1-6** described in **Section 4.4.2.1** to set up the system for calibration
2. Click **DOWNLOAD** to connect to the pedar system
3. Within the Expert Mode under **max load** enter the desired calibration level for the pedar insoles. Again, the default listed will be 60 N/cm^2. Change this to the desired maximum load. You may also change the units of measurement. The pressure divisor will be automatically calculated.
4. Complete **Steps 9-13**. The level which will be obtained for the Settings procedure and the levels of calibration for the step-wise Calibration procedure will be altered automatically. Follow these steps accordingly.

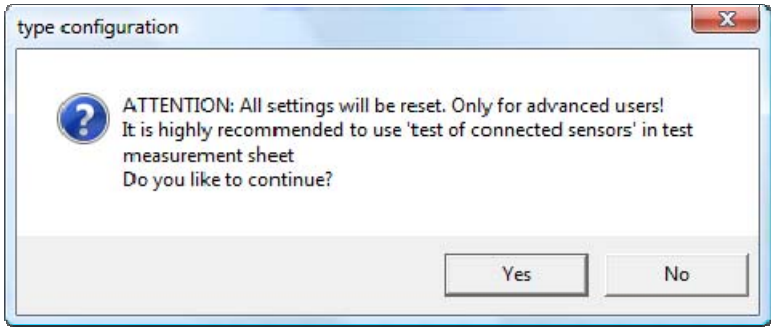
4.4.2.2.4 Additional tools of the Expert Level Calibration

1. Settings Tab:

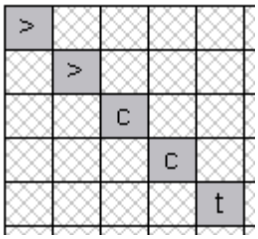


a. Gain: The gain of the system can be changed if amplification can't be reached with a gain setting of "1" (the default value). This may occur when smaller insoles, such as children's size insole N-T or even the adult U insole. It may also be needed if the insoles are calibrated to a maximum pressure lower than the standard 600 kPa. It is not recommended that the insoles are calibrated lower than 300 kPa. To discuss alternative insole types, please contact your novel representative.

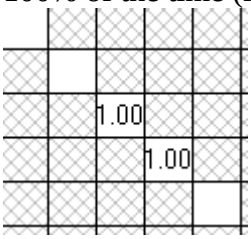
b. **INI button:** Click this button to reset all values to the initial settings. This feature is typically not used, as all predefined settings are stored with the insoles. The following message will appear:



c. When the **Test** button is clicked, or when the **Download** button is first clicked one of the following error codes may display: **>**, **c**, **t**



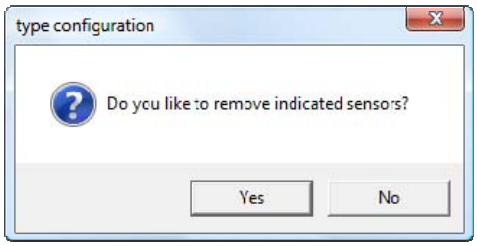
Click the following buttons to display the error codes. The data window will display the amount of time in which the error occurred. A value of 1.00 equals 100% of the time (i.e. a value of 0.41 = 41% of the time).



i. **A/D >240, symbol is ">":** Any sensors that exceeded an AD value above 240 will be automatically set to 240. This message is only a warning and does not affect the precision of the calibration. This error code is common during the **Test** mode when the insoles are loaded to the maximum level.

ii. **conv. Overload, symbol is "c":** Any sensor which reads higher than the amplification previously performed will display. This error code should be avoided and the amplification process should be repeated if it occurs.

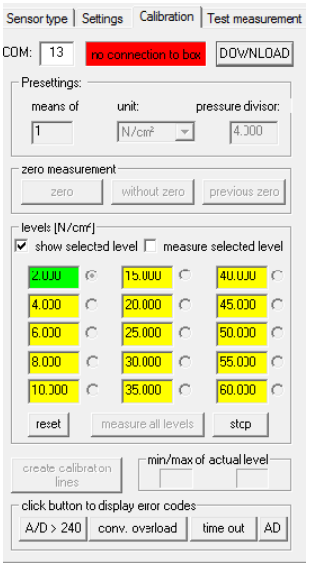
iii. **Time out, symbol is "t":** Any sensor which is not being read properly will display. When the button is clicked a message will appear asking to remove the selected sensors:



The sensor may be removed. See **Section 4.4.2.2.2** for additional information.

iv. AD: This button will switch back the display of the data from

2. Calibration Tab:



a. **zero:** use this setting to repeat a zero. This may be done if the calibration procedure was stopped reset to begin again.
b. **without zero:** do not use during calibration process.
c. **previous zero**: do not use during calibration process.
d. **measure selected level:** if a level was not correct when measuring all levels, a single level can be re-measured using this feature. This process may not produce the same results as completing the level during a step wise calibration. It may be necessary to repeat the entire step-wise procedure. **See Section 4.4.2.1 #10** for additional information.
    i. Select this option.
    ii. Select the level to be repeated.
    iii. Load insole to selected level.
    iv. Hit the enter key to make measurement.
    v. Click 'create calibration lines' to view changes.
e. **reset:** Use this option to restart the process of measuring all levels to form calibration lines. Values are set to zero so the process can be repeated.
f. **stop:** Use this option to end the measure process during any portion of the process.

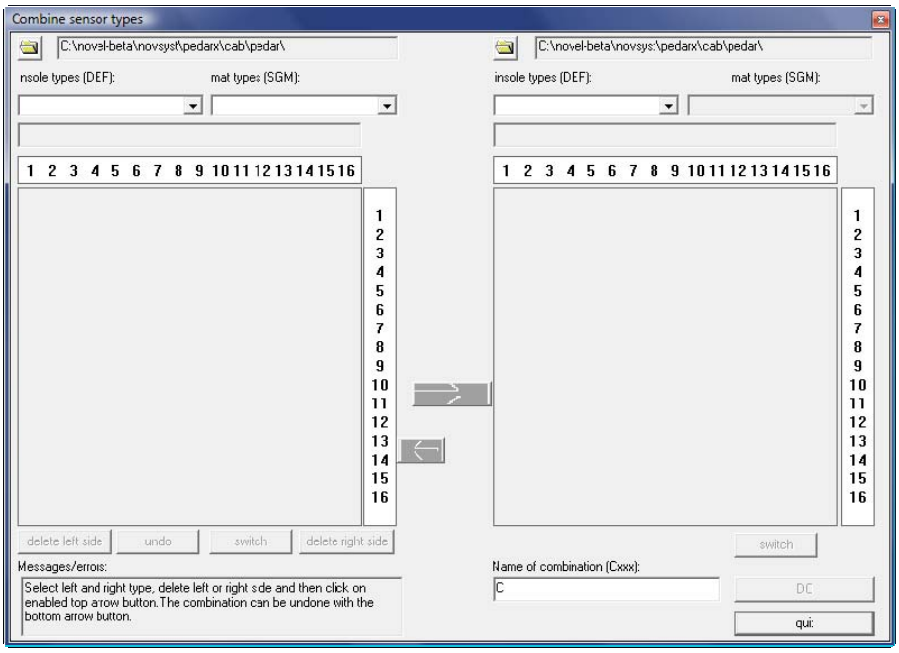
g. **min/max of actual level:** displays the minimum and maximum A/D values at each pressure level.

4.4.2.3 Combine insoles

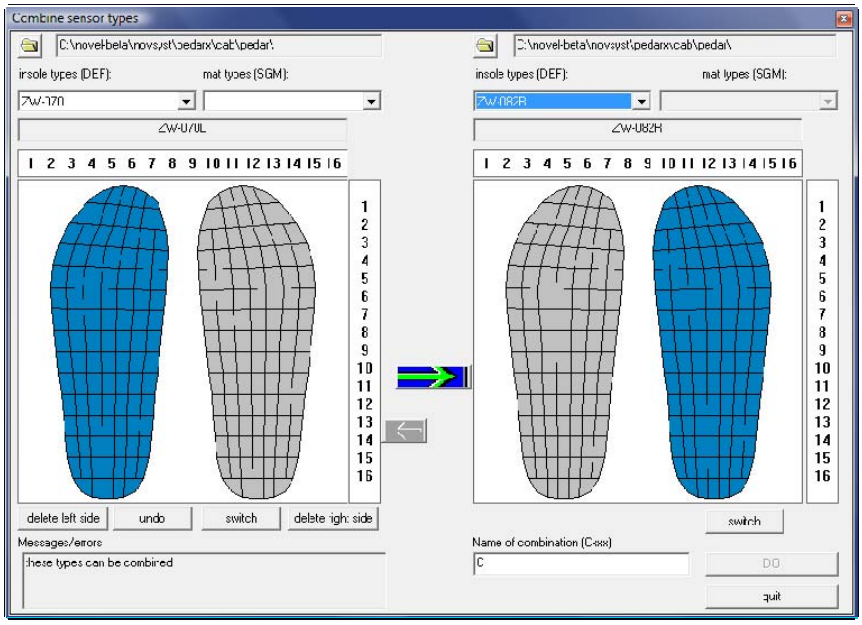
This menu item is used to combine the calibration files from two different individual calibrations. The insoles must have been calibrated previously as individual insoles. Please see **Section 4.4.2.2.1** for additional information on calibrating one insole. It is possible to combine the calibration files of two different insole sizes, which may be helpful with subjects possessing two different sized feet. It is also possible to combine two insoles which were calibrated individually because they did not fit within the calibration device.

NOTE: This process does NOT work to use two pair of calibration files and select just the right or left information from each file. The insoles must have been calibrated separately first.

1. Within the pedar settings calibration program click **File | Combine sensor type** or click the  button. The following window will appear.



2. In the **insole types (DEF)** pull down menu on the left side of the window, choose the calibration file for the left insole.
3. In the **insole types (DEF)** pull down menu on the right side of the window, choose the calibration file for the right insole. A window will appear similar to the following:



4. Click the blue/green arrow located between the sensor windows to display both insoles on the right side.
5. The **name of combination** field located below the right set of insoles will automatically fill in with a name for the combined sensor file. The name will start with the insole type. Adjust the name (i.e. enter the date and remove the extra insole type indicator) if desired.
6. Click the **Do** button. The combined file will be created. A message will appear when the conversion is complete.
7. A second message will open asking to create a configuration file. Click **Yes** to create the configuration file. The configuration file will then open within the pedar online program and may be tested according to **Step 13** in **Section 4.4.2.1**.
8. Click **quit** to exit the program if a configuration file was not created.

4.4.2.4 Menu Options

1. File

a.	NEW Calibration 	Select to create a new calibration. See **Section 4.4.2.1**
b.	REFRESH Calibration 	Select to modify an existing calibration
c.	Combine Sensor Types 	Select to combine single insole calibrations from two different insole sizes or insoles calibrated separately becuase of their large size. See Section **4.4.2.3**.
d.	Show calibration lines 	Select to view calibration lines which were previously created
e.	Print screen	Select to print a copy of the screen
f.	Capture rectangle and print	Use to select a portion of the screen to be printed. 1. Select option from menu. 2. Set origin with the RIGHT mouse button.

3. Move the cursor to the end point and click the LEFT mouse button.
4. A print setup screen will appear. Choose options and then print.

g.	Exit	Select to exit the program

2. Configuration

a.	Directories	Use this option to set default paths for files
b.	Simple/Expert Level	Select either the simple or expert user level for calibration. The calibration procedure for the Simple level may be found in **Section 4.4.2.1** and for the expert level in **Section 4.4.2.2**.
c.	Picture/Online Mode	The **Online mode** transfers data to the computer faster than the **Picture Mode.** The Online Mode should be used by default. If the computer does not display the data within the calibration windows correctly, the Picture mode can be used.

3. Info

a.	About	Displays the software version information

#####

5 THE pedar-x STEP ANALYSIS PROGRAM

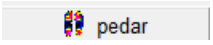
This program is used to perform a step analysis in which the pedar-x measurements' steps can be examined, post-processed and then saved. The pressure pictures for a selected step can be viewed (in 2D or 3D) as well as the maximum pressure picture (MPP) and the mean value picture (MVP) for the entire data sequence. The force and pressure time profiles for all foot strikes can be observed together with gait timing parameters (step time, swing time, double support time etc) for each foot strike.

The step analysis program can be very useful when undesired steps exist within the pedar data collection file. This may occur due to the "starting and stopping" effect (i.e. the subject needs time to get to a steady walking state), the subject "turned-around", the subject tripped or simply additional steps were recorded than what will be analyzed. This also may be useful when other activities aside from walking or running occurred, such as cutting, stair climbing etc. Depending on the analysis not all steps may be required.

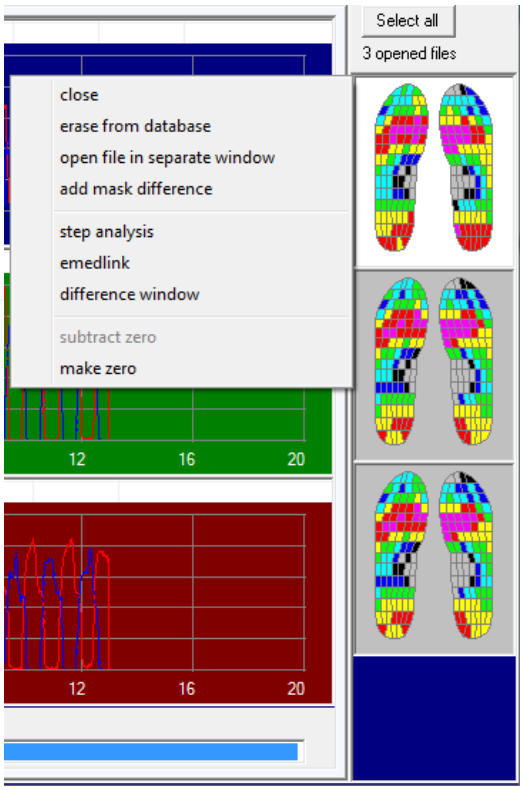
The general steps for deleting steps are:

1. Opening the data file
2. Reviewing the data
3. Deleting the undesired steps
4. Saving the data as a .COL file for further analysis

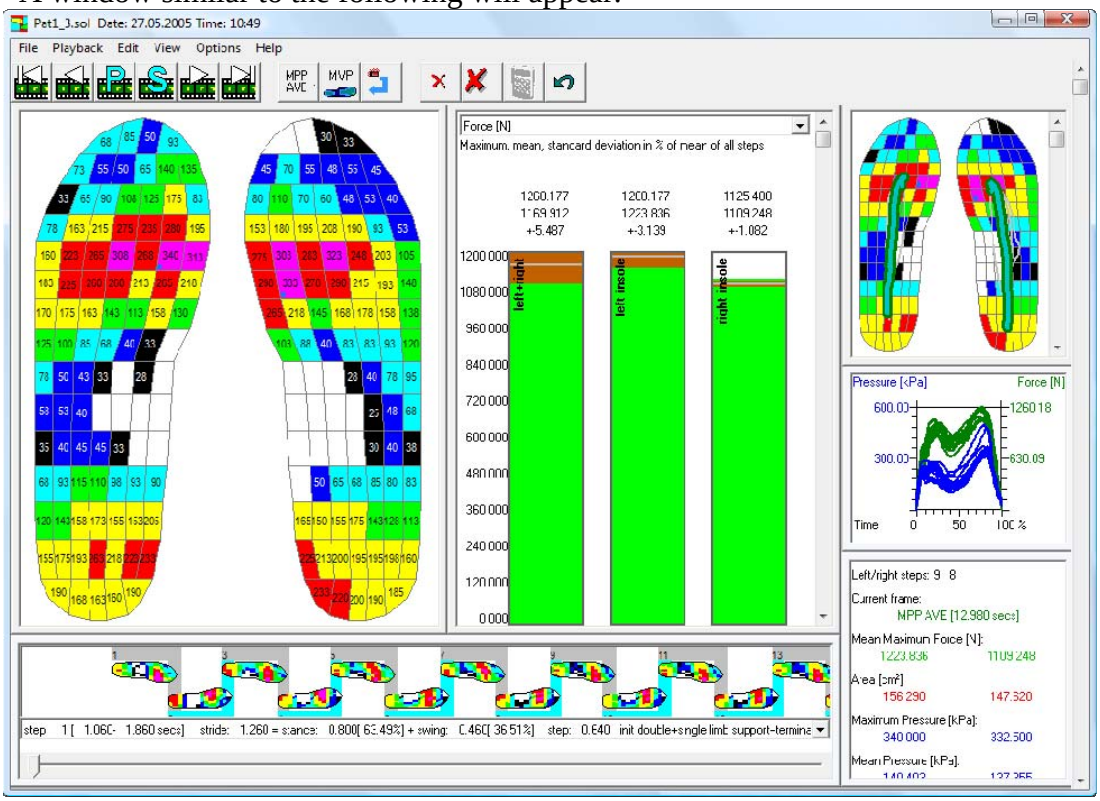
5.1 Viewing the data file in step analysis

1. The step analysis program can be opened via database (see **Section 3c**) or from the pedar online program. If the novel database was used to collect the pedar data it is recommended to open the files in step analysis through the database, as any files which are saved with then be automatically returned to the database. If the pedar measurements are not opened in step analysis through the database, a warning will appear stating that they will not be returned to the database.
2. To open the pedar file from the database, select the trial(s) from the list of pedar data and then click the Run Application  button and then the 

button and then click the  button to begin the step analysis program. Please refer to the database medical manual for complete instructions.
3. To open the pedar file from the pedar online program, select the measurement from the list on right side of the screen with the left mouse button. Please click with the right mouse button on the selected measurement picture. The following selection list will appear.



4. Select the **step analysis** option with the left mouse button.
5. A window similar to the following will appear.



6. Several different screen configurations are possible – this is just one example. Each Window is described in detail in **Section 5.1.2 – 4a**. This arrangement shows:
    a. **Insoles:** Top left window - shows the Average Maximum Pressure Picture (MPP AVE – average picture of the MPP for each step) with the respective pressure values.
    b. **Bars:** Middle window – shows the force, pressure, area, FTI, PTI, Stance or Stride time in bar graph format. The maximum, mean and standard deviation of the variable for the left+right, left and right insole are displayed.
    c. **Gait lines:** Top right window – shows the 2D pressure display with center of pressure lines. The green line is the average of all of the COP lines and the red represents the selected step's COP line.
    d. **Pressure- and force-time curves:** shows the pressure and force time curves for each step in the file.
    e. **Steps window:** A MPP representation of each step within the file. A pull-down list in the bottom of this window will describe timing information for the selected step.
    f. **Values:** Displays the values of peak pressure, maximum force, contact area, mean pressure and time, as well as the configuration and comments.
7. The toolbar at the top of the window is used to navigate through the data file. Specific icons are described in **Section 5.1.2**.

5.2 Menu options

1. File

a.	Save as	• COL file: if any changes were made to the data file (i.e. steps were removed), the file will be saved as a COL file format. If the software was started from the database the COL file will be returned to the database. This file can then be used for further analysis within the novel scientific programs. • GTC file: All timing information and gait phase characteristics can be saved in this format. The GTC file can then be opened in a program such as Excel.
b.	Save MPP AVE	• MPP file: Saves the MPP AVE information in a file format that is read by the step analysis and pedar online program. • MPA file (ASCII): Saves the MPP AVE information in an ASCII file format.
c.	Close	Closes the windows
d.	Print	• **1:1 left values:** prints a 1:1 picture of the left MPP AVE insole with values • **1:1 right values:** prints a 1:1 picture of the right MPP AVE insole with values • **1:1 left values + color:** prints a 1:1 picture of the left MPP AVE insole with values and color

- **1:1 right values + color:** prints a 1:1 picture of the right MPP AVE insole with values and color
- **Graphics window:** prints a copy of the graphic windows

e. Exit — Exits the program

f. Exit and save configuration — Exits the program and saves any changes to the layout of the windows

2. Playback

a.  Play — Play back the data file.

b.  Stop — Stop playing the data file.

c.  First frame — Move to first frame of data file.

d.  Previous frame — Move one frame back in data file.

e.  Next frame — Move one frame forward in data file.

f.  Last frame — Move to last frame in data file.

g. speed of roll over — Allows the user to change the rate at which the data file is played back.



The dialog will display one frame for the time which is defined and is dependent on the measurement rate that the file was collected. In the example above the software will play back one frame every 20 milliseconds. Therefore, if the data was collected at 100 Hz, or 1 frame every 10 milliseconds, this will display the play back at half speed. In this example a definition of 5 milliseconds would play back the trial at double speed. Click the **X** to apply the change.

h.  MPP AVE — View the Average Maximum Pressure Picture (MMP AVE) of the file.

i.  MVP — View the Mean Value Picture (MVP) of the file.

3. Edit

a.  Delete single steps — Deletes one step from the data file. This option allows "bad" steps to be removed from the file, or just from the display of the data within this program. When this button is selected and a step is clicked within the steps window, the step will be removed. Clicking the step again will add the step back to the data file. Data is not recalculated until the

Calculate button is clicked (see below).

b.  Delete all steps — Select to remove all steps in the file, or just from the view of the file. This menu option can be helpful when only one step is required. After deleting all steps the **Delete Single Steps** button will be activated and the step can be selected

c.  Calculate — Select to apply changes to the calculations. Updated calculations will take into effect any removed steps.

d.  Undo — Select to undo ALL changes applied to the data file.

e. To difference picture — Opens the difference program and allows the user to select two different files. See the novel tools manual for additional information.

4. View

a. Step Analysis — Opens the settings and descriptions for the step analysis being performed. Numbers/options in blue can be modified to change the analysis being performed.



- **First/last steps excluded:** by default the first and last steps within the file will be deleted. If this option is not desired, uncheck this checkbox.
- **Force threshold:** this number is used to determine the recognition of a step. The blue number is a percentage of the maximum force. The black number to the left is the actual value in N of this percentage. All data below the force threshold will be set to 0. This percentage can be changed if there is a high offset in the data and steps are not being recognized.
- **Min step length (s) + frames:** this options sets the minimum step length in terms of time. 10 milliseconds + a defined number of frames can be used.
- **Results:** this section displays the results of the step analysis. This is similar to the data displayed in the **Bars**

and **Numbers** windows (please see below for additional information).

- **Velocity:** The velocity that the subject traveled can be calculated if a known distance was traveled.
- **Renew/Reset/Cancel:** Click the **Renew** button to apply the changes, **Reset** button to reset to the default settings and **Cancel** button to cancel all changes.

b. Windows

- **Insoles:** displays the 2D pressure distribution for each foot. Pressure values/colors for each sensor of the insole are represented here.
- **Color Code:** displays the color code relating the pressure values to a color. Also shows the units of pressure.
- **Values:** displays information relative to each frame of data.
    - **Left/right steps:** Displays the number of steps for the left and right foot. Only displayed when MPP AVE is displayed.
    - **Range:** Displays the number of data frames represented by the selected step
    - **Current frame:** displays the current frame number and the corresponding time.
    - **Force (N):** displays total force for the left (in the left column) and right (in the right column) foot.
    - **Area (cm^2):** Displays the total contact area.
    - **Maximum Pressure (kPa):** Displays the maximum pressure that occurred in any one sensor for the displayed frame
    - **Mean Pressure (kPa):** Displays the average pressure over the frame shown.
    - **Time per frame (sec):** Displays the time duration of each frame. This value is dependent on the sampling rate.
    - **Insole type:** Displays the calibration file used to collect the data file.
    - **Comments:** Displays any comments saved with the trial
- **Time Curves:** displays force verses time (green) and pressure verses time (blue) curves for each step. A line may be selected and the corresponding foot strike will be highlighted. This is helpful when identifying an extreme outlier step.
- **Pressure Mountain:** Displays 3D representation of the pressure map. The color can be turned on/off and the height scale can be adjusted.
- **Isobar:** Displays a 'smoothed' view of the 2D pressure picture.
- **Steps:** Displays a MPP representation of each step within the file. The display also shows the stance and double

support time as represented by the gray and blue bars. The highlighted step is indicated in red. A pull-down list in the bottom of this window will describe timing information for the selected step. The scroll bar at the bottom can be used to scroll through the steps.

- **Gait Lines:** Displays a 2D representation of the pressure map with the Center of Pressure lines also displayed. The blue COP line is the average COP line and the red COP line is the COP line for the selected step. All other COP lines are displayed in gray.
- **Bars:** Displays the force, pressure, area, FTI, PTI, Stance or Stride time in bar graph format. The maximum, mean and standard deviation of the variable for the left+right, left and right insole are displayed. The gray horizontal bar in each bar graph represents the average value.
- **Title bar:** this option allows each window to appear with or without the title of the window.
- **Store current windows configuration:** this option allows the user to arrange all the desired windows on the screen and then save the layout for later use. A save as window will appear. The layout can later be chosen from the **File | Open | Configuration** menu.
- **Reset default:** use this option to reset the default screen configuration.

c.	MPP AVE	View the Average Maximum Pressure Picture (MMP AVE) of the file.
d.	MVP	View the Mean Value Picture (MVP) of the file.
e.	Gait line	Choose this option to view the gaitline on the 2D pressure picture.
f.	Show MPP steps	If selected, the steps in the **steps** window will be displayed as MPP's.
g.	Show MVP steps	If selected, the steps in the **steps** window will be displayed as MVP's.
h.	Rotate	Use to rotate the pressure picture 90 degrees counter clockwise. Continue to choose this option to rotate another 90 degrees.

5. Options

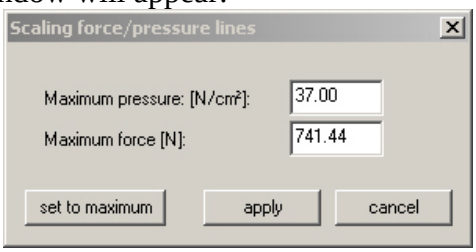
a.	Mode	• **Step mode:** data are displayed as a MPP only for each step. • **Frame Mode:** each frame of data within the steps is displayed.
b.	MVP AVE/MVP Precision	**Measurement precision vs. Maximum precision:** When pressures are averaged the resolution which results will be higher than the actual resolution of the measurement. The maximum precision will average floating numbers, where measurement precision will only display averaged numbers

which are valid with the measurement resolution.

c. MVP calculation
- **MVP over all frames:** all frames and all zeros are included in the calculation.
- **MVP only over loaded frames:** completely unloaded frames are not included in the calculation.
- **MVP over loaded sensors:** no zeros are included in the calculation

NOTE: Choose between these options to determine how the mean pressures are calculated. Each option addresses whether zeros are included or not in the mean calculation.

d. Scaling of pressure/ force lines
Allows the force/pressure-time curves to be scaled. The following window will appear:



The maximum Y coordinates of the graphs can be set. Type a value into the appropriate box and click apply.

e. Step size in pixel
Provides the user with the ability to change the size of the steps in the **step** window.

f. Format
Select to choose how the fonts are displayed for the program text and the pressure values.
- **Font: text:** choose the font for the text display in the windows
- **Font: pressure values:** choose the font for the pressure values in graphics window.
- **Values: (4 options):** These options allow the pressure values in the graphics window to be displayed with a different numbers of decimal points.

g. Show number of sensor
When selected, displays the assigned sensor identification number that corresponds to the ASCII output file.

h. Show area of sensors
When selected, displays the physical area of each sensor in mm^2. The information on this display may be useful when using the ASCII file.

i. Masks
Values for masks can be displayed within the **Bars** window.
- **Overlay:** opens a Windows open dialog in which a pedar mask (.msk) file can be applied to the data.
- **Show All:** displays the masks on the 2D pressure picture and includes a list of the masks. Displays the data selected for the masks within the bar graphs on the right.
- **Standard:** displays the standard left and right mask only. Displays the data selected for the masks within the bar graphs on the right.

- **Individual:** displays the masks within the bar graphs on the right

6. Help
   a. novel — Opens the www.novel.de homepage in your default browser
   b. About — Displays the version number of the software.

#####

6 THE pedar-x EMEDLINK PROGRAM

As with the pedar step analysis program, this program is used to perform an analysis in which the pedar-x measurements' steps can be examined, post-processed and then saved. The 2D pressure pictures and the force curves are displayed within the emedlink program. The emedlink program also provides the ability to save the steps within a pedar file into individual steps with a homogenous sensor grid. Some estimation exists with the force when saving individual steps and this is displayed. Step (.STP) files are used as emed files within the novel scientific and analysis software.

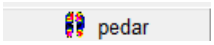
The step analysis program can be very useful when undesired steps exist within the pedar data collection file and the force profile of the sensor is desired to determine this. The program is also useful when an individual frame, or selected periods are to be excluded. Finally, the program is used to create step (.STP) files for further analysis as well. This also may be useful when other activities aside from walking or running occurred, such as cutting, stair climbing etc are collected.

The general steps for deleting steps are:

5. Opening the data file
6. Reviewing the data
7. Deleting the undesired steps or frames
8. Saving the data as a .COL file for further analysis

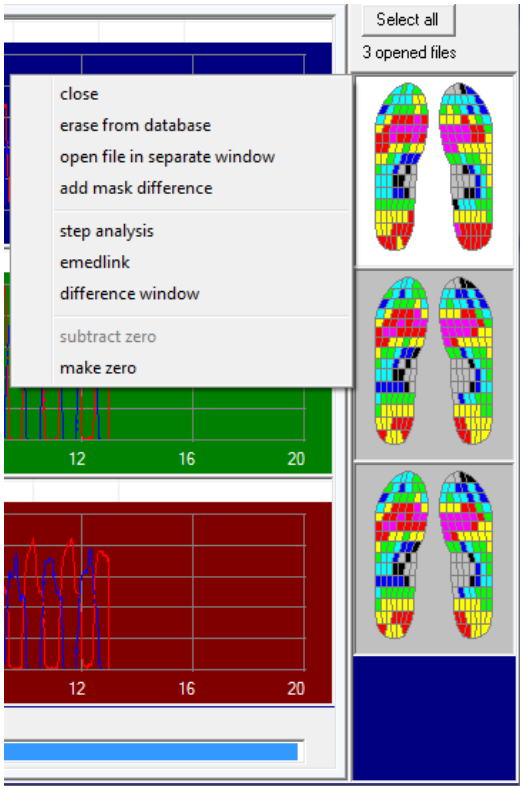
In general, the menu options for this program are very similar to the pedar step analysis program.

6.1 Viewing the data file in emedlink

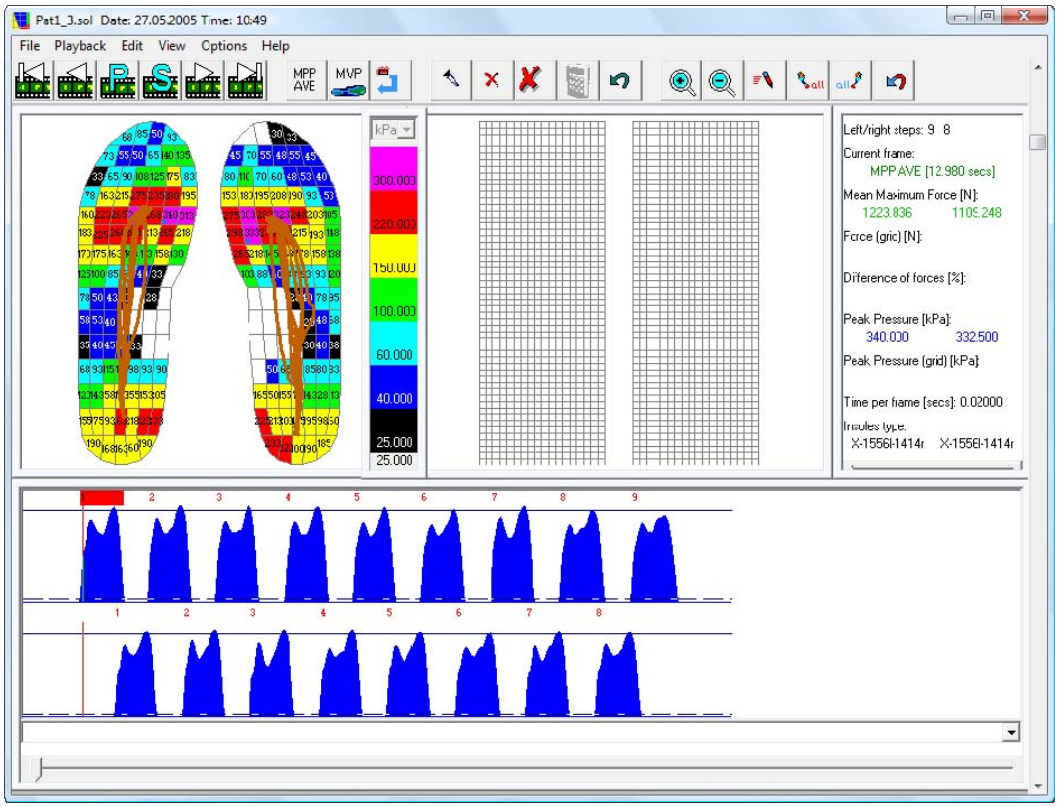
1. The emedlink program can be opened via database or from the pedar online program. If the novel database was used to collect the pedar data it is recommended to open the files in emedlink through the database, as any files which are saved with then be automatically returned to the database. If the pedar measurements are not opened in emedlink through the database, a warning will appear stating that they will not be returned to the database.
2. To open the pedar file from the database, select the trial(s) from the list of pedar data and then click the Run Application  button and then the 

button and then click the  button to begin the emedlink program. Please refer to the database medical manual for complete instructions.
3. To open the pedar file from the pedar online program, select the measurement from the list on right side of the screen with the left mouse button. Please click with the right

mouse button on the selected measurement picture. The following selection list will appear.



4. Select the **emedlink** option with the left mouse button.
5. A window similar to the following will appear.



6. The following windows will appear
    a. **Graphics:** displays the 2D pressure picture with pressure values and gaitline (if enabled).
    b. Color Code: displays the color code, pressure value for the colors and cutoff for the pressure picture.
    c. Grid: displays the pressure values in a homogeneous 4 sensor/cm^2 grid.
    d. Values:
        i. **Left/right steps:** Displays the number of steps for the left and right foot. Only displayed when MPP AVE is displayed.
        ii. **Range:** Displays the number of data frames represented by the selected step
        iii. **Current frame:** displays the current frame number and the corresponding time.
        iv. **Force (N):** displays total force for the left (in the left column) and right (in the right column) foot.
        v. **Force (grid) (N):** displays the force value for the grid file
        vi. **Difference of forces (%):** displays the percentage difference in the force from the original file compared to the file with the homogeneous grid. The maximum difference is also listed.
        vii. **Peak Pressure (kPa):** Displays the maximum pressure that occurred in any one sensor for the displayed frame
        viii. **Peak Pressure (grid) (kPa):** Displays the maximum pressure that occurred in any one sensor for the displayed frame in the grid file
        ix. **Time per frame (sec):** Displays the time duration of each frame. This value is dependent on the sampling rate.

x. **Insole type:** Displays the calibration file used to collect the data file.
xi. **Comments:** Displays any comments saved with the trial

e. Force curves: displays a force curve for each step. A selected step is indicated in red. Also listed are the gait timing parameters (step time, swing time, etc).

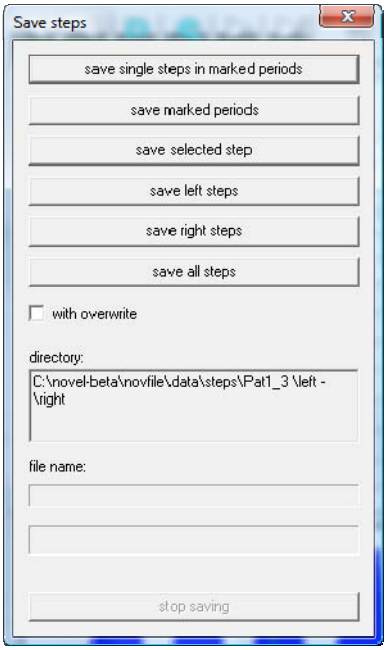
6.2 Menu options

Please refer to the step analysis instructions, **Section 5.2**, for complete listing of all features. Just those features which are only included within the emedlink program will be listed here. Not all features within the step analysis program are available within the emedlink program.

1. File

Save as

- **COL file:** if any changes were made to the data file (i.e. steps were removed), the file will be saved as a COL file format. If the software was started from the database the COL file will be returned to the database. This file can then be used for further analysis within the novel scientific programs. Please note that if the marked period options are used, only these marked periods will be saved within the COL file. See **#3** in this section for instructions on marking periods.
- **STP file:** Each individual step will be saved as a .stp file. If the software was started from the database the STP files will be returned to the database as emed files. The following window will appear. See **#3** in this section for instructions on marking periods.



- **Save single steps in marked periods:** will save the individual steps within the marked period
- **Save marked periods:** will save the marked period as

one .stp file (even if multiple steps exist).

- **Save selected step:** will save the step which is selected with the red bar within the force curves window
- **Save left steps:** will save all left steps as individual .stp files
- **Save right steps:** will save all right steps as individual .stp files
- **Save all steps:** will save all steps as individual .stp files.

2. Playback: please see **Section 5.2 #2**.

3. Edit:

a. 	Mark period	Allows the user to select a time period by clicking and dragging. A time period can be selected independently for the left and right steps. If all steps are selected using this option will deselect a desired period.
b. 	Mark total left	Selects all left steps.
c. 	Mark total right	Selects all right steps.
d. 	Undo marked periods	Deselects all marked period.
e. 	Delete Single Frame	Deletes a single frame from the data file.

4. View: please see **Section 5.2 #4.**

5. Options: please see **Section 5.2 #5.**

6. Help: please see **Section 5.2 #6.**

#####

7 PEDAR-X SYNC BOXES AND SOFTWARE

This section describes how to set up the synchronization of the pedar-X device with other devices such as video, EMG or motion capture systems. Generally there are two methods of synchronization:

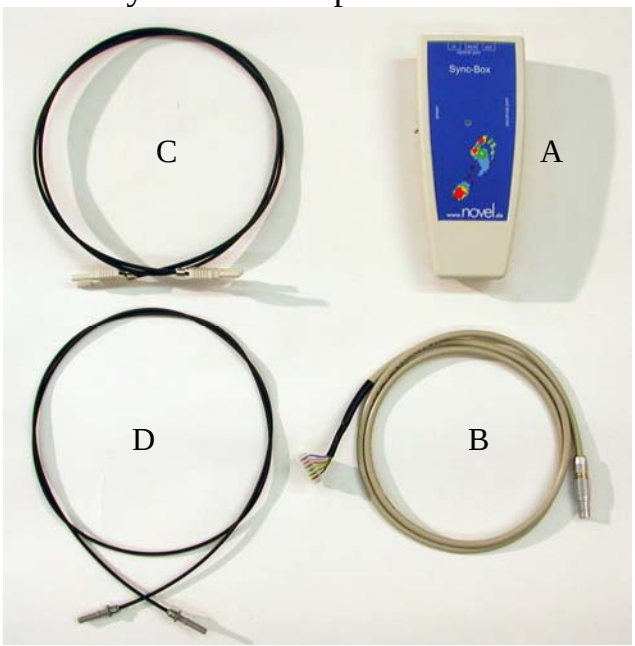
1) Two (or more) devices run with their own clock. The pedar-X sends or receives a signal that is recorded/sent by the other devices (all devices run as "master").
2) One device sends clock pluses (the master); the other devices ("slaves") run with the clock speed defined by the master.

If one uses more than two devices methods 1) and 2) can be combined. The pedar-X sync box can either send or receive a signal, therefore making triggering an easy way to "sync" data streams from two different devices. Typically this is done by syncing the start of a trial. The two data streams are then exported for simultaneous analysis later. The pedar-X sync device can send a 5 V trigger to another device which can start that device's measurement, or be recorded as part the device's data stream. Similarly the pedar-X sync device can receive a signal triggering the data collection to start within the pedar-X online software.

The modes of synchronization described in this manual are available in all operating modes of the pedar-X (**online**, **online+flash**, **flash memory only**). When you use the last two modes (**online+flash**, **flash memory only**) the configuration must be stored in the flash memory of the pedar-X. For descriptions of each of these modes, please refer to **Section 2.4.3**.

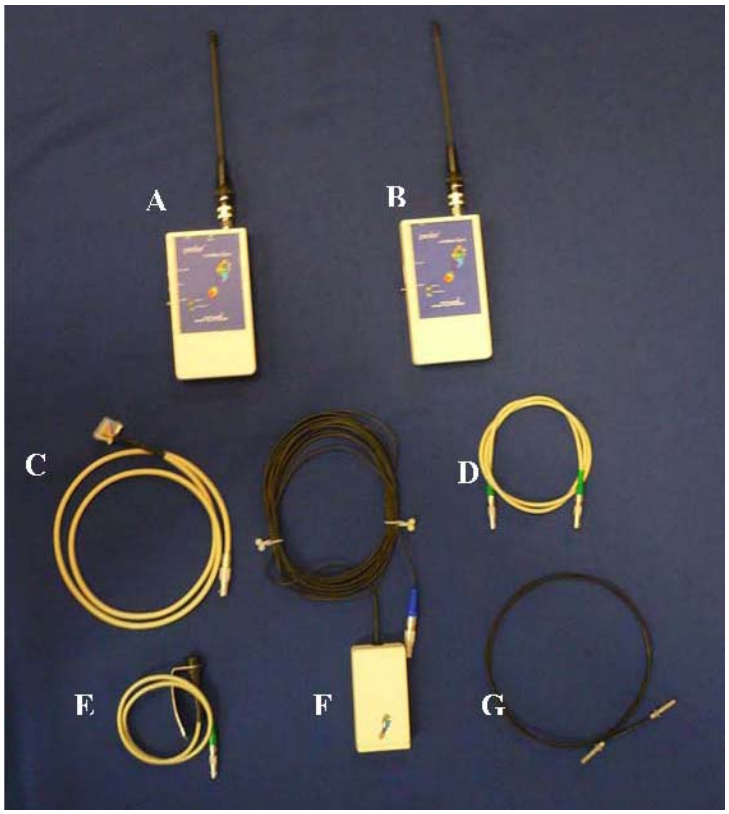
7.1 pedar-X Sync Box Components and Connections

7.1.1 Wired Sync Box Components



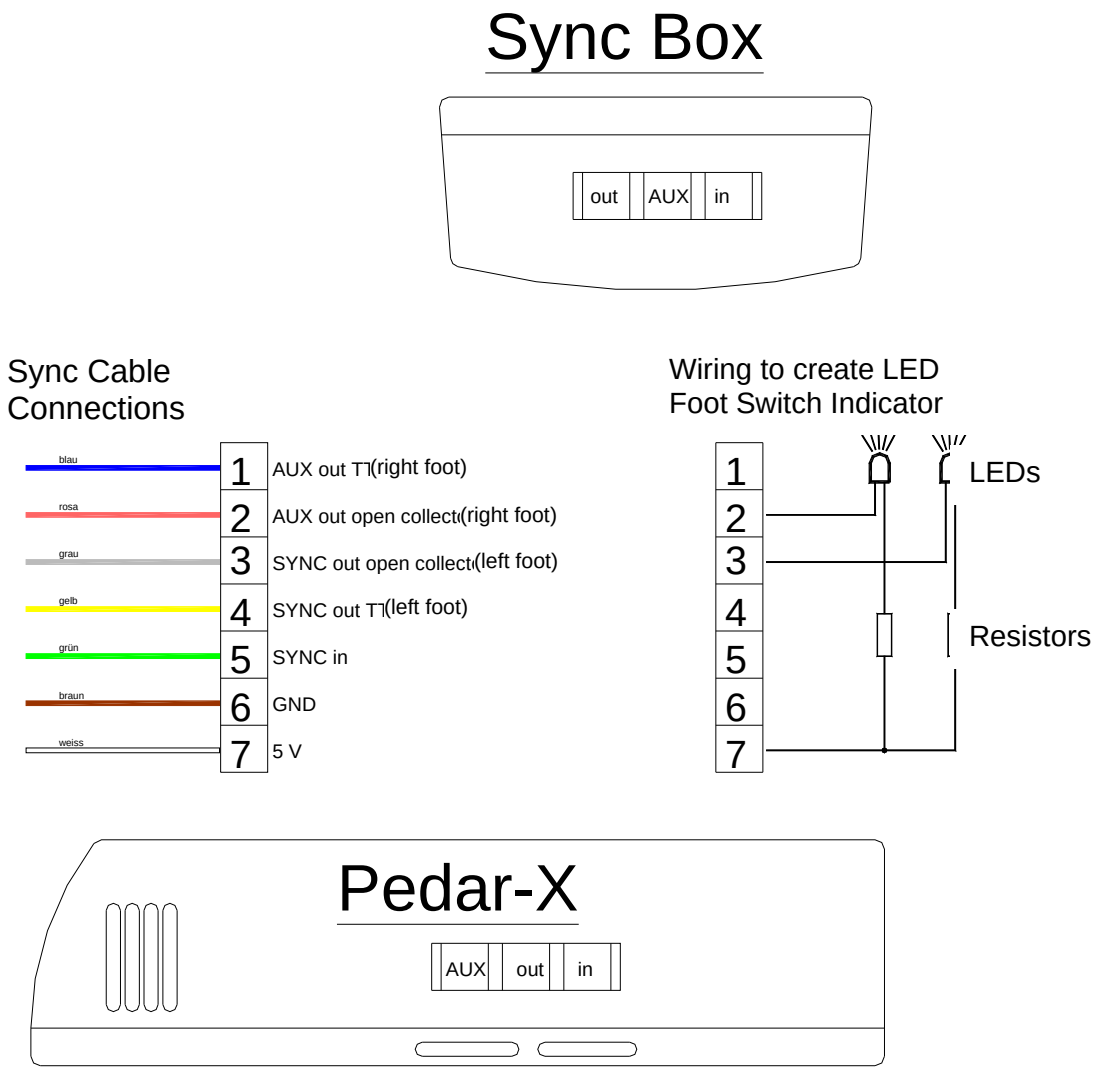
A) The wired synchronization unit
B) Electrical Signal Cable
C) Sync Out Fiber Optic Cable (2 pins)
D) Sync In Fiber Optic Cable (1 Pin)

7.1.2 Wireless Sync Unit Components



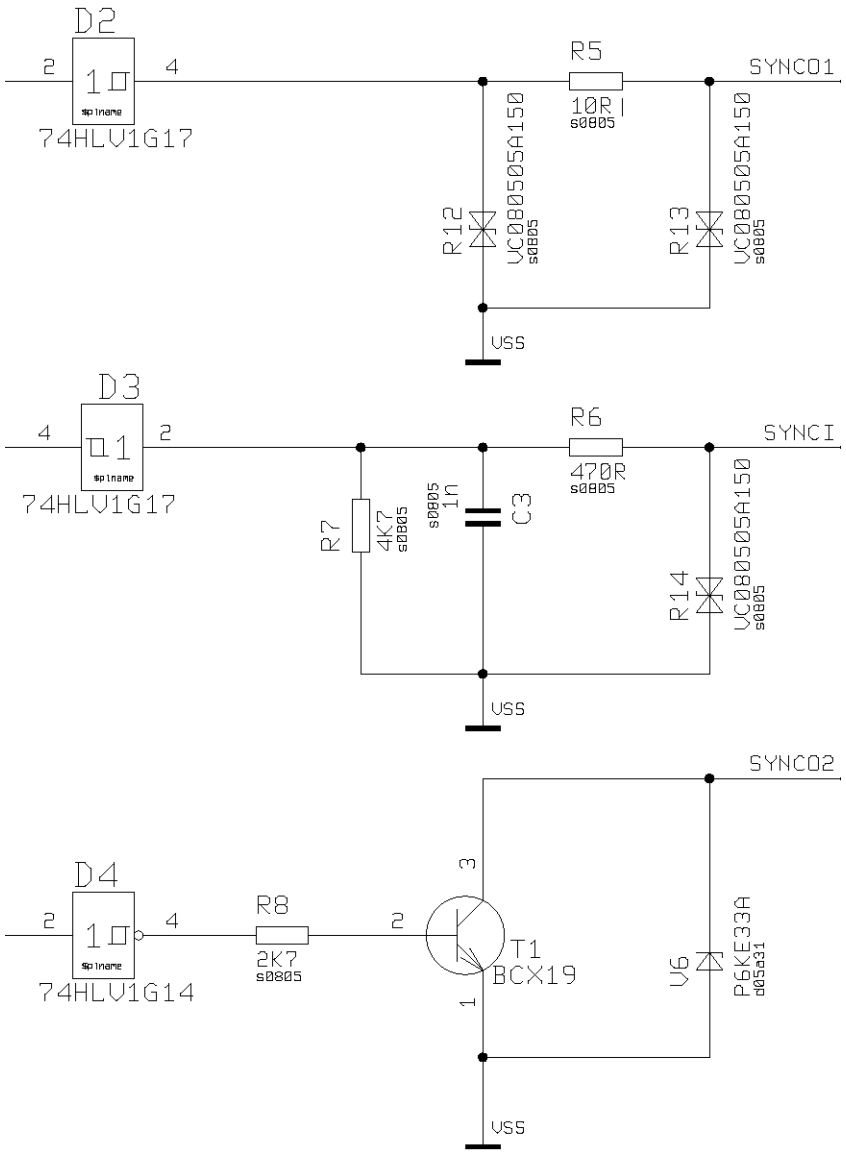
A) Wireless synchronization unit: PC Side
B) Wireless synchronization unit: pedar Side
C) Electrical Signal Cable
D) Sync Unit Trigger Cable
E) Trigger Cable
F) LED Flash Unit
G) Sync Fiber Optic Cable (1 Pin)

7.1.3 Connections of the Sync Boxes and pedar Box



7.1.4 Schematic diagram of the electrical ports of the Sync Box

SYNC01: TTL output
SYNCI: input
SYNC02: Open collector output



7.1.5 Connecting the wired sync box to the pedar device



Connections:

(1) For a sync out connection:
   (a) For a sync out TTL signal: connect the 1 pin FOC cable to the sync out port on the pedar box and the sync box
   (b) For a sync out open connector in combination with an AUX connection (i.e. the foot switch option): connect the 2 pin FOC cable to the sync out and AUX ports on the pedar box and the sync box
(2) For a sync in connection connect the 1 pin FOC cable to the sync in port on the pedar box and sync box.
(3) Connect the electrical signal cable to the wired sync box.
(4) Utilize the pinouts described in **Section 7.1.3** to capture the electrical signal

7.1.6 Connecting the wireless sync box to the pedar device

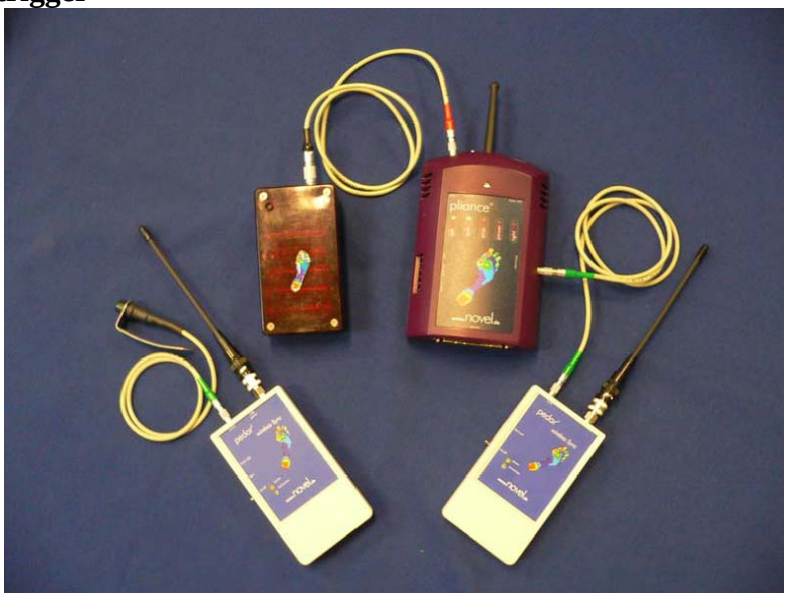
7.1.6.1 Electrical signal output/input



Connections:

(1) For a sync out connection:
   (a) For a sync out TTL signal: connect the 1 pin FOC cable to the sync out port on the pedar box and to the pedar side wireless sync box
   (b) For a sync out open connector in combination with a sync in connection: connect the 2 pin FOC cable to the sync out and sync in ports on the pedar box and the pedar side wireless sync box
(2) For a sync in connection connect the 1 pin FOC cable to the sync in port on the pedar box and pedar side wireless sync box.
(3) Connect the electrical signal cable to the PC side wireless sync box.
(4) Utilize the pinouts described in **Section 5.1.3** to capture the electrical signal

7.1.6.2 Remote trigger



Connections:

(1) Connect the sync unit trigger cable to the pedar side wireless sync box and to the start/stop port on the pedar box.
(2) Connect the start/stop cable to the switch port on the PC side wireless sync box.
(3) This setup may be used in combination with the electrical signal sync out option described in the previous section.

7.1.6.3 LED indicator

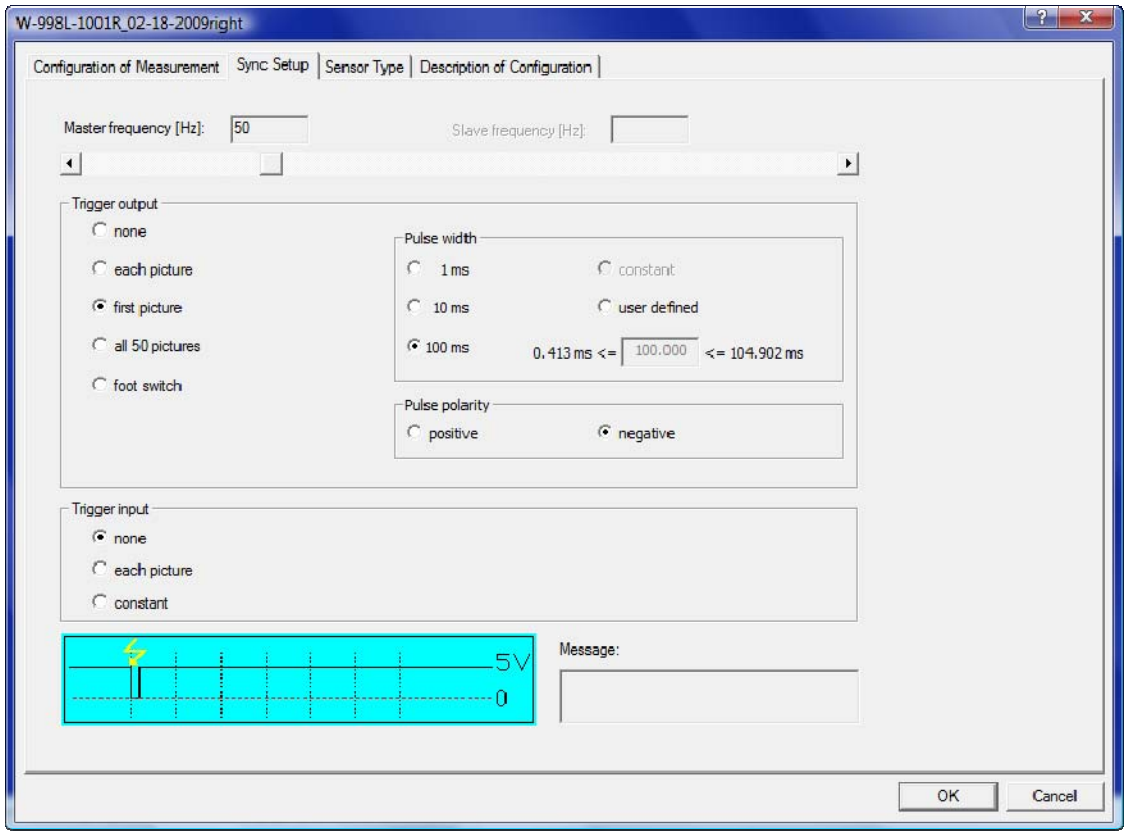


Connections:

(1) Connect the 1 pin FOC cable to the sync out port on the pedar box and to the pedar side wireless sync box.
(2) Connect the flash LED to the Flash LED port on the PC side wireless sync box.
(3) This option may be used in combination with the electrical signal sync out option described in the previous section.

7.2 The Sync Setup Tab in the pedar online Software

The settings for the synchronization signal are found within the insole configuration file. The configuration file can be accessed from the **Expert Settings | Edit insole configuration** or **Edit actual configuration** menu. Please see **Section 4.3** for additional information on the insole configuration file.



7.2.1 Master and Slave Frequency

The master frequency will typically have been set under the **Configuration of Measurement** tab (see **Section 4.3.1**). This setting can be adjusted here as well by moving the slide bar.

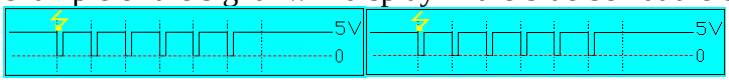
The slave frequency is used with the trigger input and an **each picture** setting. The pedar device will then wait to receive a signal from the "master" device.

7.2.2 Setting the Trigger Output

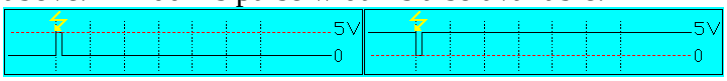
There are four options for a trigger output with the sync system:

(1) Each picture: With this option a trigger output will be sent with each frame of data collection from the pedar device. The pulse width must be defined: 1 ms, 10 ms, constant or user defined length. The pulse polarity must also be set, either positive or negative (rising or falling). This should be set according to the

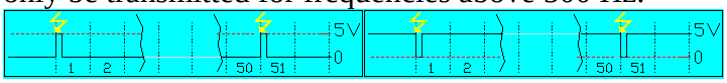
requirements and measurement rate of the device which is being triggered. An example of the signal will display in the blue box at the bottom of the screen:



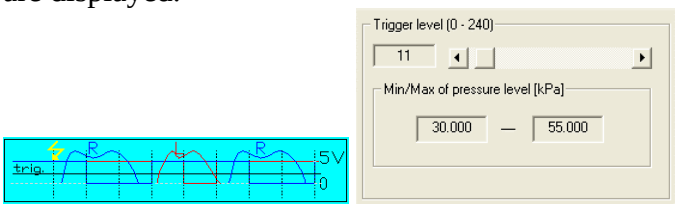
(2) First picture: With this option a trigger output signal is sent with the first frame of data collected by the pedar. The pulse width must be defined, as described above. A 100 ms pulse width is also available.



(3) **All 50 pictures:** With this option a trigger output signal will be sent with every 50th picture, i.e. 1st, 50th, 100th etc frames. The polarity and pulse width are defined as with the **each picture** option. NOTE: in this mode a constant pulse can only be transmitted for frequencies above 500 Hz.



(4) **Foot switch:** With this option trigger output signals are provided for the left and right foot contact, providing information similar to that of a foot switch. The trigger level to obtain a valid "step" must be set. The trigger level is not related to the calibrated pressure values, but to the raw data of the sensors (A/D values). Therefore the minimum and maximum pressure values that can trigger the contact are displayed.



7.2.3 Setting the trigger input

There are two options for a trigger input with the sync system. The pedar is either triggered on the rising (positive) or falling (negative) edge of the signal. The duration is not important, as long as it is not below 1 ms:

(1) **Each picture:** This option is used when the pedar device runs in the "slave" mode. With each pulse (either a positive or negative polarity), the pedar will take a measurement. The frequency of measurement must be set as the "slave" frequency. Ensure that the frequency is not above the maximum frequency for the sensor layout.

(2) **Constant:** With the constant option the pedar will record with a constant input, either positive or negative polarity. The pedar will collect at the frequency set by the master frequency. To begin recording the **Measure** button must be clicked within the pedar online software. The device will wait for the trigger pulse. The **stop and store** button must be clicked to stop the measurement.

(a) **Edge-triggered:** Select this option to trigger both the start and stop of the measurement with a pulse, either positive or negative polarity. The advantage of this option is that only two pulses are needed to control the start and stop of the measurement.

NOTE: The trigger output options available with the trigger input change depending on the settings which are chosen.

############

8 TROUBLESHOOTING BLUETOOTH CONNECTION

8.1 General Bluetooth Information

The pedar-x system transmits wirelessly to the computer with the pedar online software via Bluetooth (BT) transmission. A BT dongle is provided to all novel customers which is rated to transmit for 100 meters. Many laptops also have a BT connection, though this should not be used, and possibly disabled, so as not to interfere with the pedar BT transmission. Please see **Section 2.4.6** for information on installing the BT dongle and connecting the pedar-x box to the computer via BT.

Bluetooth transmission uses "line-of-sight", which means that they pedar box must be able to "see" the BT dongle connected to the computer. However, it does not need direct line of sight, but rather can use the walls and ceilings to bounce the signal off of. It is generally a good idea to have the BT dongle up and out in the open as best as possible. You may want to utilize the extension cable provided with the dongle or a longer extension cable if your data collection computer is surrounded by walls, etc. It is also important that the BT dongle's antennae point in the same direction that the pedar box's antennae points. For example, if you are using the standard pedar belt, the pedar box will sit vertically on the subject. Therefore ensure that your BT antennae points vertically as well.

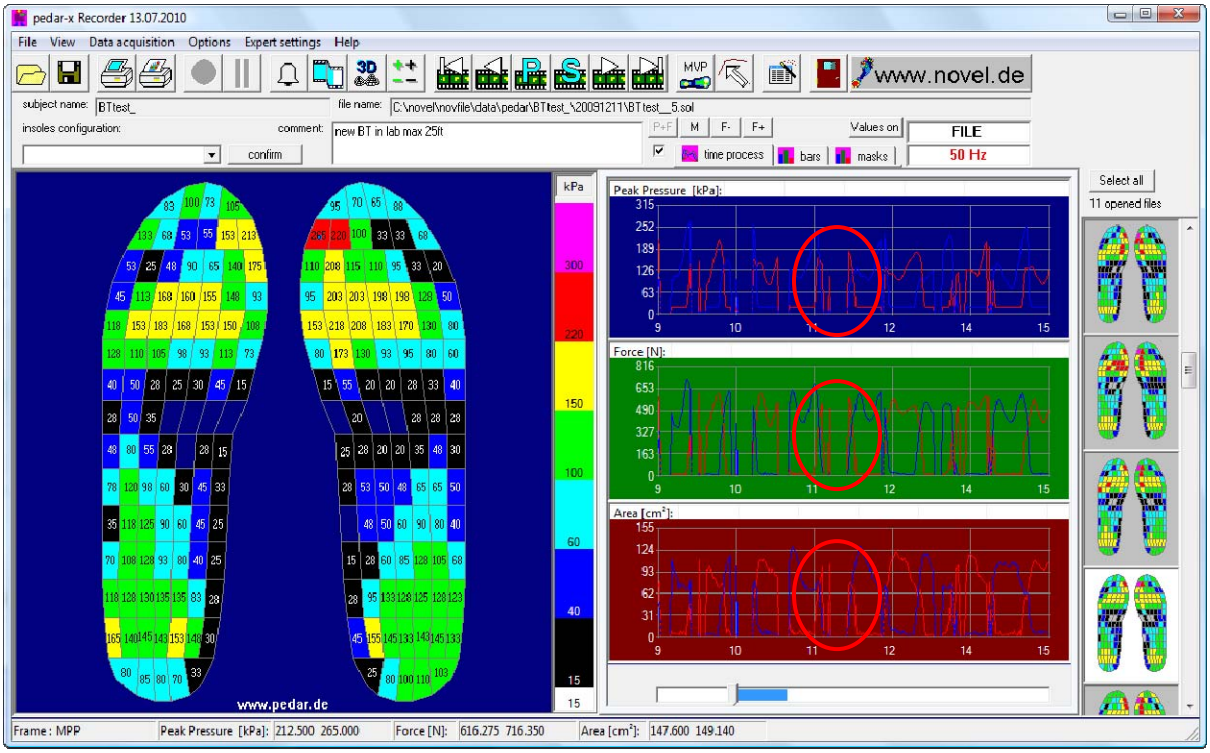
The quality of the Bluetooth transmission is highly dependent on the environment and the number of other signals which are also present. The following are items which can interfere with the BT transmission. It is recommended that, when possible, these items are removed from the field for data collection:

- Other Bluetooth devices such as
  - Mice, keyboards, headsets
  - Cell phones
  - Other computers transmitting BT
- Wireless Internet routers
- Media devices which transmit a Internet POC

To see a listing of other BT devices which are in the area, please see **Section 2.4.6**.

8.2 Possible Solutions for Dropped Frames

If dropped frames are occurring regularly within the BT transmission, you will see the pressure/force/area values drop to “0” when pressure should be displayed, as shown in the following screen capture:

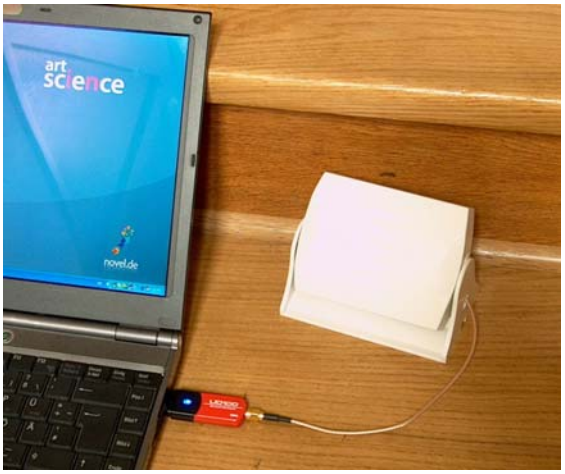


To try to resolve this issue consider the following:

1) Determine what other wireless devices (please see the previous section on devices which may cause interference) are within the data collection space and try to remove these from the space.
2) If a wireless connection is present on the computer you are collecting data, disable this for data collection.
3) Try relocating the BT dongle to a space which is more out in the open utilizing an extension cable if necessary
4) If you are transmitting outdoors, or if the subject’s body is often blocking the antenna/computer line of sight, consider using a pedar removable antenna (please see picture below). This antenna allows the user to place the antennae on the shoulder, top of the head or another location which will always have a line of sight to the BT dongle. If you do not have a removable antenna, please contact your novel representative for information on obtaining such a device.



5) novel also offers a stronger long range patch antenna which can significantly boost the BT signal for both outdoor and indoor activities in which the space is very open, or has many other wireless signals. Please contact your novel representative for information on obtaining such a device.



6) Consider using the **online+flash** mode for data collection. With this mode the data will also be captured to the flash memory and can be downloaded after collection if dropped frames have occurred. For more information on this mode of data collection please see **Section 2.4.3** for additional information.
7) Turn on the BT error correction option. This option must be selected prior to data collection under the Options tab (please see **Section 4.2.4-e**)
When using the **online + flash** mode, this option will automatically fill the dropped frames which occur to the software from the flash memory of the device.

8.3 If Bluetooth COM Ports are Not Created Automatically

Problem Description:

One of the following issues may arise when connecting the pedar-x box to a Windows XP SP 2 or 3 or Windows Vista computer via a Bluetooth Connection:

- After completing the **Add New Bluetooth Device** wizard there are no COM ports created. Typically an outgoing and incoming COM port will be listed after completing the Add New BT Device wizard. COM ports can also be found under the COM ports tab.
- When adding a COM port, you receive the message **access is denied**

Solution:

One of two issues could be the problem:

1) The first issue can be read about in the following Microsoft Knowledge Base article:
   http://support.microsoft.com/kb/894035/

   a) To work around this issue, follow these steps:
      1. Open the Microsoft Management Console (MMC) snap-in for Services. To do this in Windows, click Start, type **services.msc** in the **Start Search** box, and then click **Services** in the **Programs** list. Windows Vista: If you are prompted for an administrator password or for a confirmation, type the password, or click **Continue**.
         To open the Microsoft Management Console (MMC) snap-in for Services in Windows XP, click **Start**, click **Run**, type **services.msc**, and then click **OK**.
      2. Double-click the **Bluetooth Support** service.
      3. If the Bluetooth Support service is stopped, click **Start**.
      4. On the **Startup type** list, click **Automatic**.
      5. Click the **Log On** tab.
      6. Click **Local System account**.
      7. Click **OK**.
      8. If you prompted to restart the computer, click **Yes**

   b) After completing these steps, remove and re-add the pedar-x device within Bluetooth Devices. An outgoing COM port should be created. Test the connection with the pedar-x Online software.

2) If the above process does not work, the second issue can be read about in the following Microsoft Knowledge Base article:
   http://support.microsoft.com/kb/892891/en-us

   a) To work around this issue, follow these steps:

      1. Click **Start**, click **Run**, type **cmd**, and then click **OK**.
      2. At the command prompt, type
         esentutl /g %windir%\security\database\secedit.sdb
         You should receive output that is similar to the following output:

```
Microsoft(R) Windows(TM) Database Utilities
Version 5.1
Copyright (C) Microsoft Corporation. All Rights Reserved.
Initiating INTEGRITY mode...
        Database: C:\WINDOWS\security\database\secedit.sdb
```

```
  Temp. Database: TEMPINTEG1704.EDB
Checking database integrity.
                    Scanning Status (% complete)

          0    10   20   30   40   50   60   70   80   90  100
          |----|----|----|----|----|----|----|----|----|----|
          ..................................................
Integrity check successful.
Operation completed successfully in 1.592 seconds.
```

If the Windows security database is corrupted, the output will contain the following line:

```
This operation may find that this database is corrupt.
```

To work around this problem, follow these steps:

1. If both the Edb.log file and the Edb.chk file are in the %windir%\security directory, follow these steps to repair the security database:
   a. Click **Start**, click **Run**, type **cmd**, and then click **OK**.
   b. At the command prompt, type **esentutl /r edb /l %windir%\security /s %windir%\security**
2. If either the Edb.log file or the Edb.chk file is in the C:\WINDOWS\security directory, or if neither file exists in the directory, follow these steps to repair the security database: Click **Start**, click **Run**, type **cmd**, and then click **OK**.
   a. At the command prompt, type **esentutl /p %windir%\security\Database\secedit.sdb**
   b. Click **OK** when you receive the following message:
      You should only run Repair on damaged or corrupted databases. Repair will not apply information in the transaction log files on the database, and may cause information to be lost. Do you wish to proceed?
3. After you repair the security database, follow these steps to modify the registry so that you can access the registry key that is required to configure the Bluetooth device:
   a. Create the C:\bt.inf file that has the following text.

```
[version]
signature="$CHICAGO$"
revision=1
DriverVer=07/01/2001,5.1.2600.1106
[Registry Keys]
"MACHINE\SYSTEM\CurrentControlSet\Services\BTHPORT",0,"D:AR(A;CI;GRGW;;;PU)(A;CI;GRGW;;;LS)"
```

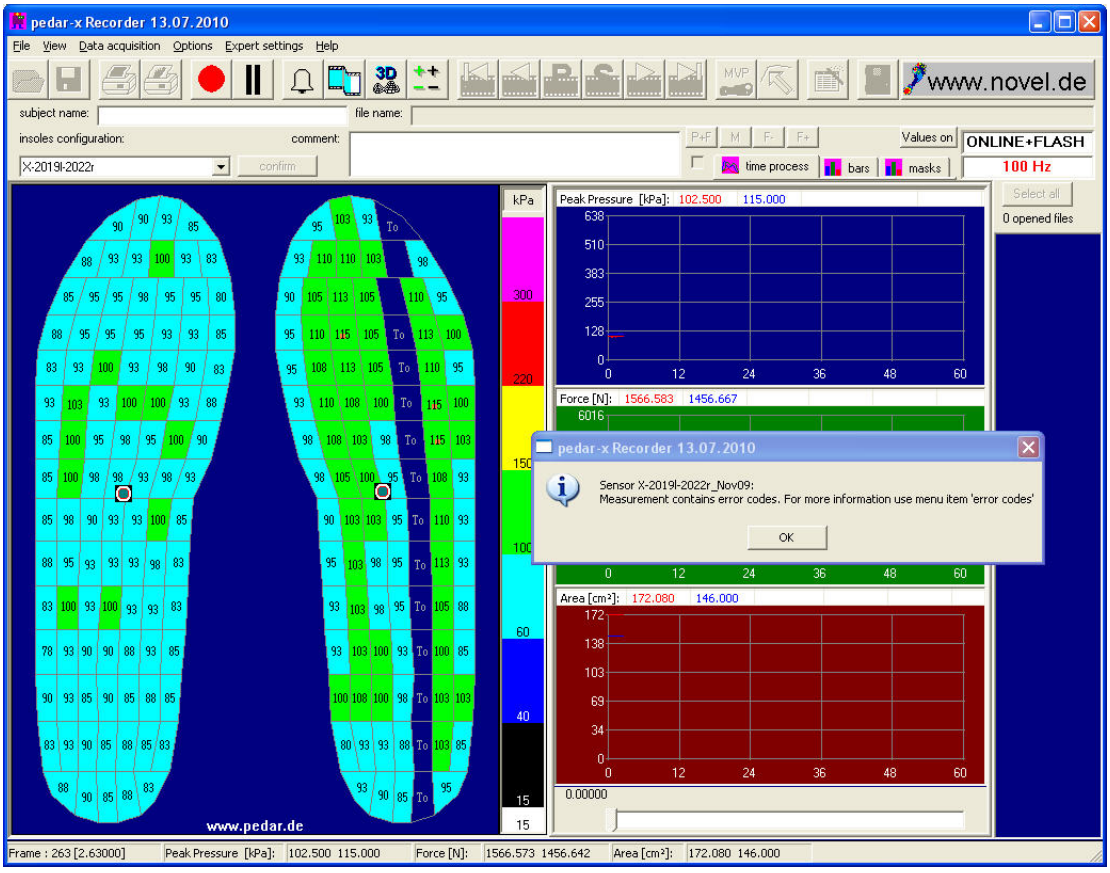
   b. Click **Start**, click **Run**, type **cmd**, and then click **OK**.
   c. At the command prompt, type **secedit /configure /cfg C:\bt.inf /db C:\bt.sdb**
   d. Delete the C:\bt.inf file and the C:\bt.sdb file.

b) After completing these steps, remove and re-add the pedar-x device within Bluetooth Devices. An outgoing COM port should be created. Test the connection with the pedar-x Online software.

9 TESTING INSOLES AND CABLES FOR ERRORS

If a frequent timeout error code is being read by the pedar online software, it may be useful to check where the error is coming from. This process can be helpful in reporting the information to your novel representative for further investigation or repair of the insole.

A typical error may occur in one sensor, one row, one column or multiple locations on the insole. Once an error occurs you will notice a TO in the sensor element and should receive the error code warning message, if you have this option turned on (see **Section 4.2.4 – e**).



The insoles may have an error in just one section or multiple sections. The following process will help to narrow down where the error is occurring, and therefore the possible cause of the error.

First, it may be helpful to ask these questions:

1) Does the error occur on only one pair of insoles, or multiple pairs of insoles?
2) If the error occurs on multiple pairs, is it every pair?
3) If so, the issue could be an issue with the insole cable, not the pedar insole itself. Please see below for additional information.

9.1 Testing the insole for errors

1) Connect the pedar device as described in **Section 2.4.4**. Connect the insoles in which the errors are occurring.
2) Lay the system flat on a table.
3) Open the pedar online software. In menu Options – Error Codes please uncheck the option With error correction. Choose the insole relevant configuration file, click the *Confirm* button, and when the message appears "Unload left insole" click *Cancel*. Begin a recording. – A measurement will start, showing the basic sensor values without the deduction of zero values.

   See **Section 4.1.1** for additional information on collecting data within the pedar online software.
4) Starting from the insole connector, "wiggle" the insole back and forth, not to the point of bending.
5) Observe the pressure data which is displaying in the 2D pressure picture within the pedar online software. Determine if the movement you are making causes the error to occur/stop.
6) Continue this process moving down the lead of the insole. Typically an error in the lead will cause more than one sensor (i.e. an entire row or column) to show a TO.
7) Continue the "wiggle" process from the heel to the toes. Again, make note if a certain location causes the error to occur, or stop occurring.
8) Once the entire insole is checked, stop the recording.

If the error occurs close to the insole connector, this could because too much stress is being placed on the insole lead where it is being attached to the ankle of the subject. Ensure that the cable is attached proximally to the calf well and that the connector does not "flap" around as the subject is moving.

If the error occurs at the juncture of the insole lead and measuring insole, this could be due to the fact that there is not enough "slack" being placed in the insole lead when it is being attached to the ankle of the subject. Please see **Section 2.4.7** for information on correct attachment.

If the error occurs within the heel or toes of the insoles, it is possible that the insole is not being fit within the shoes properly, or that the heel cup is too deep for the insoles. Please check each of these scenarios with each subject.

9.2 Testing the insole cable for errors

1) Inspect the insole cable. Look at the connector which connects to the pedar box. Are there any pins bent? This can occur if the cable is not pushed straight in and pulled straight out (i.e. if there is a lot or torque placed on the cable during the connection process). Also inspect the insole connector to determine if any of these pins are bent. If pins are bent, please contact your novel representative for repair instructions. If not, please continue to step #2.
2) Connect the pedar device as described in **Section 2.4.4**. Connect any pair of insoles.
3) Lay the system flat on a table.

4) Open the pedar online software. In menu Options – Error Codes please uncheck the option With error correction. Choose the insole relevant configuration file, click the *Confirm* button, and when the message appers “Unload left insole” click *Cancel.* Begin a recording. – A measurement will start, showing the basic sensor values without the deduction of zero values.

   See **Section 4.1.1** for additional information on collecting data within the pedar online software.
5) Starting from the connector which attaches to the pedar box, “wiggle” the cable for which the insole is receiving errors (either right or left) back and forth to the point of bending, but not crimping the cable.
6) Observe the pressure data which is displaying in the 2D pressure picture within the pedar online software. Determine if the movement you are making causes the error to occur/stop.
7) Continue this process moving down the entire length of the cable. Typically an error in the cable will cause more than one sensor (i.e. an entire row or column) to show a TO.
8) If the error is occurring on the other pedar insole as well, repeat this process for that side’s cable.
9) Once the entire cable is checked, stop the recording.
10) Repeat such a test with other insole pairs. If exactly the same columns or lines are instable or missing, it indicates a damage of the cable.

Once you have determined that the cable is faulty, please contact your novel representative for repair instructions.

10 DEFINITION OF VARIABLES

Maximum Pressure Picture (MPP)
Displays the highest pressure that was reached for each sensor at any time during the data collection period. This is therefore an artificial picture that provides a summary of the highest pressures that were reached over the entire surface of the insole – it does not represent any one step or event. This picture is displayed within the pedar online, step analysis and emedlink software programs.

Average Maximum Pressure Picture (MPP AVE)
Displays the average of all the MPP's in a data file. This picture is displayed within the step analysis and emedlink software programs.

Mean Value Picture (MVP)

over all frames:
Displays the mean over all frames of the data collection process is calculated (even if the insole was completely unloaded during a frame) for each sensor within the insole.
***without unloaded frames*:**
Displays the average pressure recorded by each sensor during the contact phases of the data collection period, calculated by summing the pressures for each sensor and dividing by the number of frames that the sensor is loaded.
***without unloaded sensors*:**
For each sensor the mean is calculated over the frames during the data collection period when the sensor was loaded, i.e. all zero values are ignored.

Force (N)

for the MPP:
The maximum force is shown. This value will depend on how the MPP was calculated. In the pedar-x online software, it is the maximum force for the entire data collection. For one step, as in the step analysis and emedlink software programs, it is the maximum force of that step
for the MVP:
The mean force is shown. This value depends on the MVP calculation. If the MVP was calculated over all frames, then the force will be the mean over all frames. If the MVP was calculated without unloaded sensors, then the force will be the mean of the steps only.
for any other picture:
The force is calculated as the sum of forces experienced by each sensor loaded at that time.

Area (cm^2)

Area is calculated as the sum of the area of all loaded sensors shown in the picture or frame.

Peak Pressure (kPa)

The maximum pressure (sometimes referred to as maximum pressure) is the highest pressure experienced by any one sensor in the picture/frame.

Mean Pressure (kPa)

The average pressure is calculated for each picture/frame by summing the pressures in all sensors and dividing by the number of sensors (99).

Force-time Integral (N*s)

This value is calculated once for each trial within the pedar online software, or for each step within the step analysis and emedlink software programs. It is the summation of the force at each frame multiplied by the time per frame, or more simply, the area under the force-time curve.

Pressure-time Integral (kPa*s)

This value is calculated once for each trial within the pedar online software, or for each step within the step analysis and emedlink software programs. It is the summation of the peak pressure at each frame multiplied by the time per frame, or more simply, the area under the peak pressure-time curve.

Stance (sec)

The amount of time from heel strike to toe off (i.e. the amount of time the foot is making contact with the ground) for one step of the left or right foot. Available within the step analysis and emedlink software programs

Swing (sec)

The amount of time from toe off to heel strike (i.e. the amount of time the foot is not making contact with the ground) for one step of the left or right foot. Available within the step analysis and emedlink software programs

Stride (sec)

The amount of time from heel strike of one step to heel strike of the next step (i.e. stance time + swing time). Available within the step analysis and emedlink software programs

#####

11 DEFINITION OF FILE TYPES

11.1 pedar Data Files

By default, the data is stored in the folder C:\novel\novfile\data\pedar\. This directory can be changed from the **Expert Settings** menu of from the **Configuration** program.

- ***.sol** – pedar data file. This file can be opened in the pedar online software, step analysis and emedlink software programs, as well as many of the novel scientific software programs (please refer to these manuals directly for additional information).
- ***.col** – corrected pedar data file. This file was created from an original *.sol file and has been post-processed within either of the step analysis or emedlink software programs. This file can be opened in the pedar online software, step analysis and emedlink software programs, as well as many of the novel scientific software programs (please refer to these manuals directly for additional information).

11.2 pedar Calibration and Configuration Files

By default, the calibration files are located at C:\novel\NOVSYST\pedarx\cab\pedar and the configuration files are located at C:\novel\NOVSYST\pedarx\mfg. These directories can be changed from the **Expert Settings** menu of from the **Configuration** program. These files can only be accessed from the pedar online software. All six calibration files must be located within the defined directory in order for a calibration to be accessed and used.

- **Calibration Files:**
  - ***.def:** Contains information regarding the values at each step used during the calibration process.
  - ***.cab:** Contains the calibration curves for each sensor.
  - ***.tbl:** Contains information regarding the arrangement of sensors and which sensors are activated.
  - ***.set:** Contains information regarding the pressure range of the mat and the span of the range across the A/D values.
  - ***.geo:** Contains information regarding the shape and size of the sensors.
  - ***.chl:** Not implemented with the pedar-x system.
- **Configuration File:**
  - ***.mfg:** Contains the calibration file, as well as the information on the configuration of measurement, synchronization settings and sensor layout. Please see **Section 4.3** for additional information.

11.3 pedar Exported files

These files can be exported and created from the .sol or .col files. They are tab delimited files which can be open in programs such as Excel, Notepad, etc. These files can be saved manually from the pedar online software program, or exported automatically when the file is saved

immediately after data collection. Please see **Section 4.2.1 -f and 4.2.4 -c** for additional information

- ***.asc:** This is the raw ascii data from the pedar data collection file. It contains the individual pressure values for each of the 99 sensor elements for the left and right insole throughout time. A header is also included. A portion of an example output is shown here:

file name: date/time 27.05.05 10.48
sensor type: x-1556l-1414r
total time time per f scanning rate [Hz]: 50
pressure values in kPa

mask definition:
left insole: 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,5
right insole: 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,

time[secs	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0.02	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.04	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.06	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.08	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.18	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.24	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.26	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.28	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.32	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.34	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.38	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.42	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.44	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.46	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.48	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0.52	0	27.5	32.5	27.5	30	0	27.5	0	40	0	32.5	0	0	0	0
0.54	87.5	85	90	90	92.5	87.5	90	85	112.5	75	75	60	47.5	47.5	45
0.56	147.5	130	130	135	140	160	157.5	160	197.5	135	127.5	102.5	112.5	92.5	100
0.58	165	137.5	137.5	145	155	195	187.5	187.5	227.5	160	150	122.5	155	120	122.5
0.6	162.5	132.5	135	140	150	197.5	190	187.5	230	162.5	152.5	120	162.5	122.5	127.5
0.62	157.5	132.5	130	137.5	147.5	197.5	190	190	230	162.5	152.5	120	167.5	125	132.5
0.64	165	135	135	140	150	207.5	200	200	240	172.5	157.5	127.5	182.5	137.5	145
0.66	172.5	145	140	147.5	155	215	207.5	207.5	250	180	165	132.5	195	145	152.5
0.68	172.5	140	140	147.5	155	215	207.5	210	250	182.5	165	132.5	197.5	147.5	155
0.7	170	140	137.5	145	155	212.5	207.5	205	247.5	180	162.5	132.5	195	147.5	152.5
0.72	162.5	135	132.5	137.5	150	205	197.5	197.5	240	170	160	127.5	190	145	150
0.74	152.5	127.5	122.5	127.5	140	192.5	187.5	187.5	227.5	162.5	150	120	177.5	135	140
0.76	137.5	117.5	112.5	115	132.5	177.5	172.5	175	212.5	150	140	112.5	167.5	122.5	132.5
0.78	122.5	105	102.5	107.5	122.5	160	160	162.5	200	137.5	127.5	102.5	155	112.5	120
0.8	112.5	100	97.5	97.5	112.5	150	150	152.5	187.5	125	120	92.5	145	102.5	112.5
0.82	102.5	92.5	92.5	92.5	102.5	140	140	145	180	120	110	87.5	135	100	107.5
0.84	97.5	85	87.5	87.5	97.5	132.5	132.5	135	167.5	110	102.5	80	125	92.5	100
0.86	90	80	80	80	90	122.5	122.5	122.5	157.5	100	97.5	70	120	85	92.5
0.88	80	72.5	72.5	72.5	82.5	110	112.5	112.5	147.5	90	90	65	110	77.5	87.5
0.9	67.5	65	65	65	72.5	100	102.5	102.5	132.5	82.5	80	55	95	72.5	80
0.92	60	55	55	55	65	87.5	92.5	87.5	120	70	70	47.5	85	67.5	70
0.94	45	45	50	45	52.5	75	82.5	75	102.5	55	60	40	72.5	60	60
0.96	40	40	40	40	40	60	70	60	87.5	45	50	32.5	60	47.5	55
0.98	27.5	32.5	32.5	27.5	30	47.5	55	45	72.5	32.5	40	27.5	40	35	40
1	0	27.5	27.5	0	25	32.5	45	32.5	55	0	32.5	0	25	30	35
1.02	0	0	0	0	0	27.5	32.5	0	45	0	27.5	0	0	25	25

- ***.fgt:** This exported file contains the force and gaitline (center of pressure) for the left and right insole throughout time. A header is also included. A portion of an example output is shown here:

file name:	date/time 27.05.05 10.48					
sensor type: x-1556l-1414r						
force time integral[N*s]: 6874.022 6426.975						
pressure time integral [kPa*s]: 1733.100 1630.500						
total time	time per f	scanning rate [Hz]: 50				
	left insole			right insole		
time[secs	force[N]	x[mm]	y[mm]	force[N]	x[mm]	y[mm]
0.02	39.118	50.83	187.55	873.55	56.16	74.23
0.04	14.273	20.65	203.89	895.385	56.78	79.6
0.06	4.62	21	240	897.625	57.4	85.34
0.08	4.62	21	240	878.992	57.7	90.84
0.1	4.62	21	240	866.537	57.74	97.71
0.12	4.62	21	240	846.713	57.75	105.1
0.14	4.62	21	240	842.583	56.93	114.79
0.16	4.62	21	240	837.677	56.47	124.24
0.18	4.62	21	240	847.778	55.79	135.32
0.2	4.62	21	240	840.53	54.8	146.45
0.22	4.62	21	240	839.78	53.95	156.11
0.24	4.62	21	240	810.718	52.44	167.61
0.26	4.62	21	240	820.275	50.57	174.28
0.28	4.62	21	240	846.28	49.17	177.47
0.3	4.62	21	240	865.92	47.5	180.74
0.32	4.62	21	240	883.157	45.92	183.2
0.34	4.62	21	240	915.475	44.95	184.35
0.36	0	0	0	943.09	44.28	185.2
0.38	8.92	49.44	186.49	975.063	43.6	186.39
0.4	4.62	21	240	1012.12	43.03	187.37
0.42	4.62	21	240	1050.473	42.46	188.23
0.44	0	0	0	1085.443	41.9	188.96
0.46	0	0	0	1128.167	41.25	189.37
0.48	0	0	0	1160.932	40.72	189.77
0.5	4.62	21	240	1184.975	40.13	190.06
0.52	42.5	53.5	45.41	1198.49	39.4	190.26
0.54	268.078	54.01	46.33	1191.5	38.73	190.22
0.56	478.337	53.15	43.86	1158.983	38.2	189.61
0.58	614.695	53.52	50.78	1090.787	37.82	188.6
0.6	669.885	52.92	59.6	962.597	37.91	187.09
0.62	720.617	52.52	65.9	763.997	38.68	186.03
0.64	818.237	51.84	72.29	551.943	39.61	184.92
0.66	898.97	51.86	75.98	340.383	39.68	183.14
0.68	938.958	52.07	78.19	171.96	37.31	179.99
0.7	980.352	52.27	80.76	33.647	27.33	175.81
0.72	1000.165	52.56	83.64	9.158	11.92	171.31
0.74	1006.34	52.72	88.52	4.325	10	179
0.76	993.532	52.78	92.26	0	0	0

- ***.mva:** This exported file contains data throughout time for the masked parameters. (For more information on creating masks please see **Section 4.2.4 – p**. The parameters included are force, peak pressure, mean pressure and standard deviation of pressure. A header is also included. A portion of an example output is shown here:

file name: date/time 27.05.05 10.48
sensor type: x-1556l-1414r
total time time per f scanning rate [Hz]: 50
pressure values in kPa

mask definition:
left insole: 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,
right insole: 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,6
left heel: 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,
right heel: 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,
left forefoot: 41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,77,78,79,80,81,82,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98
right forefoot: 41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,9
medial : 1,1,2,2,6,6,7,7,8,8,13,13,14,14,15,15,20,20,21,21,22,22,27,27,28,28,29,29,34,34,35,35,36,36,41,41,42,42,43,43,48,48,49,49,50,50,55,55,56,56,57,57,62,62,63,63,64,64,69,69,70,7
lateral: 3,3,4,4,5,5,9,9,10,10,11,11,12,12,16,16,17,17,18,18,19,19,23,23,24,24,25,25,26,26,30,30,31,31,32,32,33,33,37,37,38,38,39,39,40,40,44,44,45,45,46,46,47,47,51,52,52,53,53,54,54

	left insole				right insole				left heel				right heel			
time[secs	force[N]	max press	mean pre:	+-% of me	force[N]	max press	mean pre:	+-% of me	force[N]	max press	mean pre:	+-% of me	force[N]	max press	mean pre:	+-% of me
0.02	39.118	32.5	2.273	323.94	873.55	190	50.764	102.27	0	0	0	0	654.338	190	93.651	58.72
0.04	14.273	27.5	0.829	559.77	895.385	180	52.033	92.78	0	0	0	0	642.768	180	91.995	54.83
0.06	4.62	27.5	0.268	1019.4	897.625	167.5	52.163	84.01	0	0	0	0	609.818	167.5	87.279	49.76
0.08	4.62	27.5	0.268	1019.4	878.992	150	51.08	75.8	0	0	0	0	565.503	150	80.936	45.65
0.1	4.62	27.5	0.268	1019.4	866.537	135	50.357	67.51	0	0	0	0	518.145	135	74.158	42.03
0.12	4.62	27.5	0.268	1019.4	846.713	127.5	49.205	60.7	0	0	0	0	462.505	117.5	66.195	39.39
0.14	4.62	27.5	0.268	1019.4	842.583	135	48.965	55.94	0	0	0	0	402.09	102.5	57.548	38.58
0.16	4.62	27.5	0.268	1019.4	837.677	145	48.68	57.39	0	0	0	0	342.065	102.5	48.957	40.07
0.18	4.62	27.5	0.268	1019.4	847.778	152.5	49.266	63.1	0	0	0	0	278.547	100	39.867	45.61
0.2	4.62	27.5	0.268	1019.4	840.53	160	48.845	73.29	0	0	0	0	208.417	92.5	29.829	57.75
0.22	4.62	27.5	0.268	1019.4	839.78	167.5	48.802	82.44	0	0	0	0	150.518	82.5	21.543	75.14
0.24	4.62	27.5	0.268	1019.4	810.718	175	47.113	94.51	0	0	0	0	72.882	70	10.431	152.19
0.26	4.62	27.5	0.268	1019.4	820.275	175	47.668	100.19	0	0	0	0	37.505	55	5.368	229.86
0.28	4.62	27.5	0.268	1019.4	846.28	187.5	49.179	102.74	0	0	0	0	25.775	42.5	3.689	255.77
0.3	4.62	27.5	0.268	1019.4	865.92	200	50.321	106.45	0	0	0	0	10.07	30	1.441	424.13
0.32	4.62	27.5	0.268	1019.4	883.157	215	51.322	110.14	0	0	0	0	0	0	0	0
0.34	4.62	27.5	0.268	1019.4	915.475	232.5	53.201	111.41	0	0	0	0	0	0	0	0
0.36	0	0	0	0	943.09	245	54.805	113.28	0	0	0	0	0	0	0	0
0.38	8.92	27.5	0.518	706.39	975.063	262.5	56.663	114.99	0	0	0	0	0	0	0	0
0.4	4.62	27.5	0.268	1019.4	1012.12	280	58.817	116.01	0	0	0	0	0	0	0	0
0.42	4.62	27.5	0.268	1019.4	1050.473	300	61.046	117.88	0	0	0	0	0	0	0	0
0.44	0	0	0	0	1085.443	320	63.078	120.08	0	0	0	0	0	0	0	0
0.46	0	0	0	0	1128.167	340	65.561	121.01	0	0	0	0	0	0	0	0
0.48	0	0	0	0	1160.932	360	67.465	122.64	0	0	0	0	0	0	0	0
0.5	4.62	27.5	0.268	1019.4	1184.975	380	68.862	123.63	0	0	0	0	0	0	0	0
0.52	42.5	40	2.47	327.48	1198.49	392.5	69.647	124.34	37.88	40	5.421	200.55	0	0	0	0
0.54	268.078	112.5	15.579	167.69	1191.5	392.5	69.241	123.95	240.42	112.5	34.41	85.37	0	0	0	0
0.56	478.337	197.5	27.797	161.71	1158.983	367.5	67.351	121.59	449.8	197.5	64.377	75.7	0	0	0	0
0.58	614.695	227.5	35.721	145.59	1090.787	315	63.388	116.69	557.85	227.5	79.841	67.6	0	0	0	0
0.6	669.885	230	38.929	131.7	962.597	290	55.939	110.55	573.023	230	82.013	65.43	0	0	0	0
0.62	720.617	230	41.877	120.83	763.997	232.5	44.398	102.5	588.758	230	84.265	63.25	0	0	0	0
0.64	818.237	240	47.55	110.24	551.943	160	32.075	102.29	636.597	240	91.112	62.14	0	0	0	0
0.66	898.97	250	52.241	104.11	340.383	112.5	19.78	119.42	681.07	250	97.477	61.61	0	0	0	0
0.68	938.958	250	54.565	99.2	171.96	77.5	9.993	157.8	696.27	250	99.652	59.14	0	0	0	0
0.7	980.352	247.5	56.971	93.06	33.647	45	1.955	389.38	712.065	247.5	101.913	57.7	0	0	0	0
0.72	1000.165	240	58.122	86.67	9.158	27.5	0.532	687.7	711.792	240	101.874	54.86	0	0	0	0
0.74	1006.34	227.5	58.481	79.53	4.325	25	0.251	989.64	689.662	227.5	98.707	51.17	0	0	0	0
0.76	993.532	212.5	57.737	72.96	0	0	0	0	658.508	212.5	94.248	49.4	0	0	0	0
0.78	963.557	200	55.995	67.58	0	0	0	0	620.313	200	88.781	46.49	0	0	0	0
0.8	936.013	187.5	54.394	63.81	0	0	0	0	579.792	187.5	82.982	44.6	0	0	0	0
0.82	919.335	180	53.425	60.2	0	0	0	0	558.167	180	79.887	43.95	0	0	0	0
0.84	910.395	167.5	52.905	56.07	0	0	0	0	528.607	167.5	75.656	42.82	0	0	0	0
0.86	916.905	157.5	53.284	50.92	0	0	0	0	502.128	157.5	71.866	40.16	0	0	0	0
0.88	944.48	147.5	54.886	46.21	0	0	0	0	475.207	147.5	68.013	38.07	0	0	0	0
0.9	959.51	132.5	55.76	44.02	0	0	0	0	440.965	132.5	63.112	35.48	0	0	0	0

- ***.gtc:** This file is exported from the step analysis program only. It contains the gait timing information such as stance time, swing time, stride time etc. A header is also included. An example output is shown here:

file name:	left insole	right insole: X-1556l-1414r													
total time	time per f	scanning r	distance[r	velocity[m/min]: 0.000											
step analysis:															
left insole:															
	mean star	min stanc	max stance: 0.920 secs												
	mean max force +/- deviation in % of mean: 1218.509 [N] +/- 2.376 %														
	threshold (% of mean max force): 60.925 N = 5.00 %														
right insole:															
	mean star	min stanc	max stance: 0.880 secs												
	mean max force +/- deviation in % of mean: 1097.923 [N] +/- 1.849 %														
	threshold (% of mean max force): 54.896 N = 5.00 %														
left steps:	right steps: 8														
measurement precision															
step_no	foot	max. force	mean forc	max press	start	start in %	stride	step	stance	stance in %	swing	swing in %	init doubl	single lim	terminate
2	left	1255.655	933.682	375	1.84	14.65	1.28	0.66	0.8	62.5	0.48	37.5	10.94	40.63	10.94
4	left	1227.373	909.028	320	3.12	24.84	1.26	0.62	0.78	61.9	0.48	38.1	11.11	38.1	12.7
6	left	1238.078	884.232	340	4.38	34.87	1.28	0.64	0.8	62.5	0.48	37.5	12.5	37.5	12.5
8	left	1232.377	899.665	340	5.66	45.06	1.24	0.62	0.76	61.29	0.48	38.71	12.9	37.1	11.29
10	left	1189.517	884.2	305	6.9	54.94	1.28	0.64	0.8	62.5	0.48	37.5	12.5	37.5	12.5
12	left	1207.568	910.45	315	8.18	65.13	1.3	0.62	0.8	61.54	0.5	38.46	12.31	35.38	13.85
14	left	1237.047	912.91	315	9.48	75.48	1.38	0.7	0.86	62.32	0.52	37.68	13.04	37.68	11.59
16	left	1160.455	825.71	330	10.86	86.46	0	1.72	0.92	0	0	0	0	0	0
1	right	1108.762	795.335	497.5	1.2	9.55	1.3	0.64	0.78	60	0.52	40	0	36.92	10.77
3	right	1126.382	820.963	360	2.5	19.9	1.24	0.62	0.76	61.29	0.48	38.71	11.29	38.71	11.29
5	right	1129.377	803.855	355	3.74	29.78	1.28	0.64	0.8	62.5	0.48	37.5	12.5	37.5	12.5
7	right	1091.675	790.745	330	5.02	39.97	1.26	0.64	0.8	63.49	0.46	36.51	12.7	38.1	12.7
9	right	1081.525	805.218	330	6.28	50	1.26	0.62	0.78	61.9	0.48	38.1	11.11	38.1	12.7
11	right	1078.012	797.972	320	7.54	60.03	1.26	0.64	0.8	63.49	0.46	36.51	12.7	38.1	12.7
13	right	1095.6	772.477	330	8.8	70.06	1.38	0.68	0.86	62.32	0.52	37.68	13.04	36.23	13.04
15	right	1072.047	799.42	320	10.18	81.05	0	0.68	0.88	0	0	0	0	0	0

11.4 Other pedar Files

- ***.sgm:** This file is the sensor geometry file. It is located at C:\novel\novfile\defin\SOLS. This file is used in definition of the insole size and layout.
- ***.msk:** This is the mask file which is created during the Read/Define Masks process or during the Define sensors to be scanned process. Please see **Sections 4.2.4 – p and 4.2.3 – f** for additional information.
- ***.far:** This file contains a group of measurements which were collected to the flash memory of the pedar device. This file may be opened using the menu item File | Open FAR file (see **Section 4.2.1-c**). The file is created from the menu item Data Acquisition | Read Flash | Save to FAR file (see **Section 4.2.3-j**).
- ***.mst:** This file is created when using the PDA option with the pedar device. It can be read with the menu item File | Open PDA file (see **Section 4.2.1-d**).

#####

12 INSOLE SIZING CHART AND MEASUREMENT CONVERSIONS

Pressure Measurements

1 N/cm^2 = 0.1 Bar
10 kPa
74.006 mmHg
1.4504 Psi

Shoe Sizes

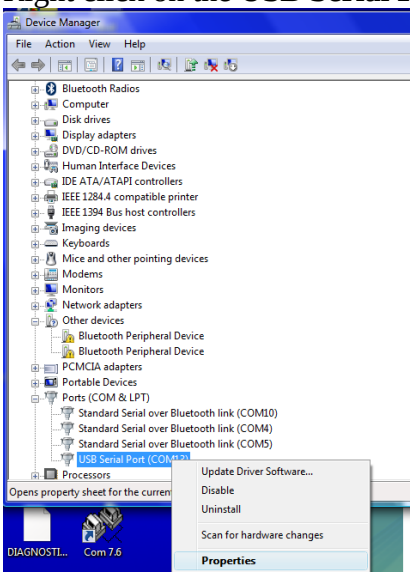
pedar-x insoles	European	United States		
		Children	Women	Men
N	22/23	5.5/6.5		
O	24/25	7/8		
P	26/27	8.5/10		
Q	28/29	10.5/11.5		
R	30/31	12/13		
S	32/33	1/1.5		
T	34/35	2/3		
U	36/37	3.5/5	5.5/6.5	
V	38/39	5.5/6	7/8.5	6/7
W	40/41		9/9.5	7.5/8.5
X	42/43		10/11	9/10
Y	44/45			10.5/11
Z	46/47			11.5/12.5
A	49			14-15

#####

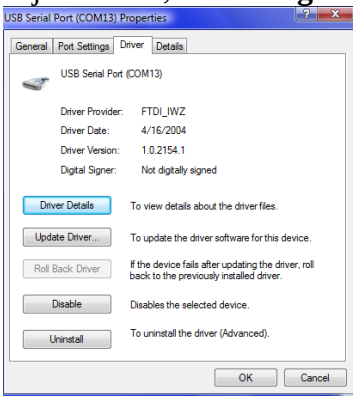
13 ISSUES WITH THE FIBER OPTIC CABLE DRIVER INSTALLATION

If the Fiber Optic Cable drivers installed automatically without your control, or you are not making a connection with the pedar online software, please follow these steps to check that the proper drivers have installed.

1. Check which driver is installed
    a. plug in the USB converter
    b. Go to Device Manager
    c. Choose **Ports COM&LPT**
    d. Right click on the **USB Serial Port** and choose **Properties**



    e. Under the **Driver** tab the Driver provider should be **FTDI_IWZ**. If it is not, and is just **FTDI**, the **wrong** drivers are installed.



    f. Close out of these properties

2. Browse to C:\novel\NOVSYST\driver\usb

3. Run the program **FTdiunin.exe**.
    a. It will first ask you to unplug the USB converter
    b. Click Continue

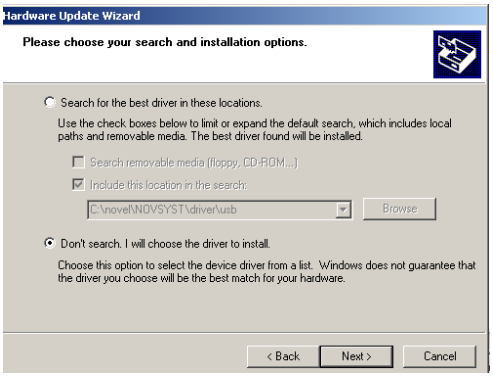
i. Possible error: the program may ask to be in a different location. If so copy the necessary file to the location requested and re-run the program

c. Click **Finish** once successfully uninstalled

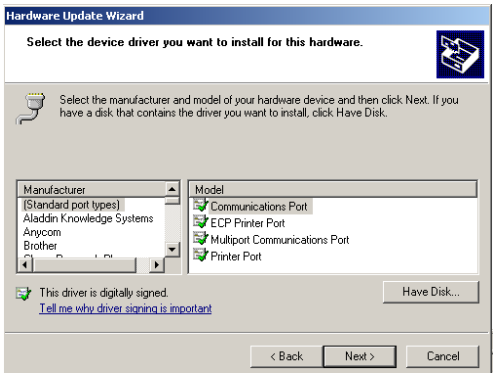
4. Plug back in the USB converter
   a. Follow the steps outlined in **Section 2.4.5.1**.
5. Once this driver is installed go back to the device manager and check for the FTDI_IWZ is now listed
6. Note the COM port listed
7. Open the pedar online software and test the connection. Please see **Sections 4.1.1 and 4.2.3 – a** for additional information.

If the above process still does not install the proper driver, it may be required to manually install the driver via this process:

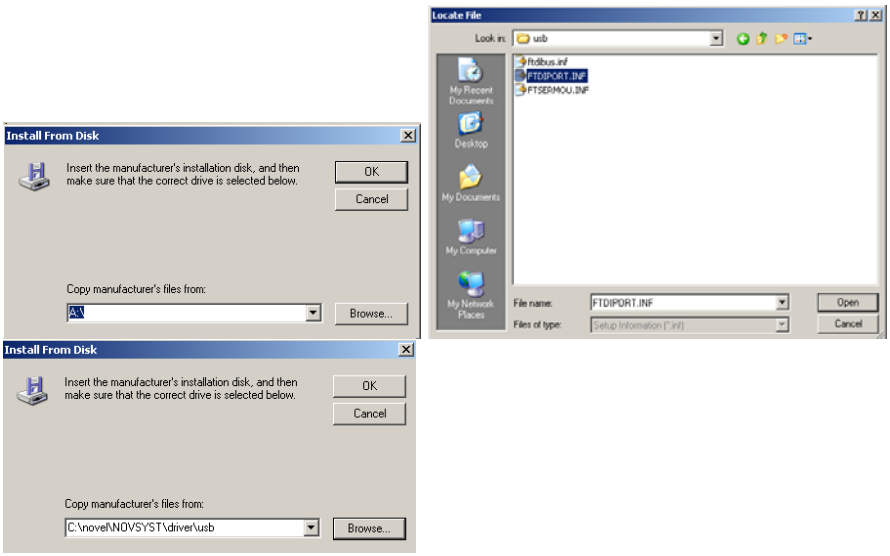
1. Follow step #1 above to check which driver is installed.
2. Choose **Update Driver**
3. The **Found New Hardware Wizard** will appear. This time choose **Don't search, I will choose the driver to install**



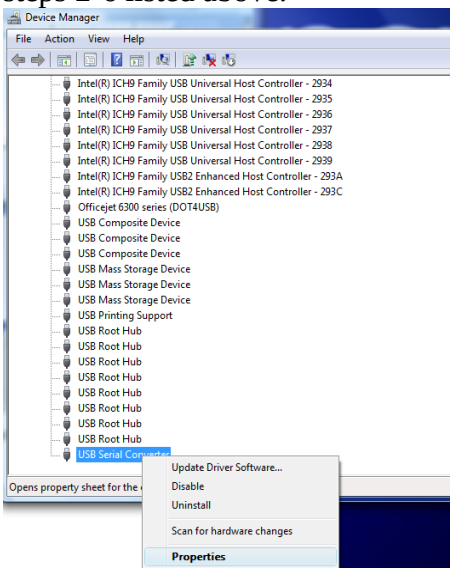
4. Choose **Have Disk**



5. Browse to the C:\novel\NOVSYST\driver\usb directory and choose the **FTDIPORT.inf** file. Click **Open**, then **OK**, and then click **Next**.



6. Click **Finish** to complete the installation.
7. It may be necessary to complete this process for the second driver as well. If so, choose the **USB Serial Converter** driver under the **Universal Serial Bus Controllers** and follow steps 2-6 listed above.



8. Open the pedar online software and test the connection. Please see **Sections 4.1.1 and 4.2.3 – a** for additional information.

#######